The Geeqie User Manual

John Ellis

The Geeqie User Manual

John Ellis
Copyright © 2004 John Ellis
Copyright © 2008-2024 The Geeqie Team

Abstract

Geeqie is an image viewer for Linux and compatible systems.

Feedback

Comments and general help: <geeqie@freelists.org>

Bugs and Feature Requests: Geeqie Issues [https://github.com/BestImageViewer/geeqie/issues]

Website: geeqie.org [https://www.geeqie.org]

Revision History
Revision 0.1 2004-01-01
John Ellis

Gqview Project
Revision 0.2 2016-05-30
Colin Clark

The Geeqie Team
Revision 0.3 2024-09-16
Colin Clark

The Geeqie Team

Table of Contents

1. Introduction 1
    Features 1
        Basic Features 1
        Advanced Features 1
2. Main Window 3
    File Pane 3
        List view 3
        Changing the file list font 4
        Icon view 4
        File Filter 4
        Multiple selection 5
        Sorting 5
        Context menu 5
        Drag and Drop 6
    Collections 6
        Viewing a collection in the files pane 7
    Folder pane 7
        Menu bar 8
        Tool bar 8
        Path entry 8
        Folder list 9
        List view 9
        Tree view 9
        Sorting 9
        Context menu 10
        Drag and drop 11
    Image Pane 11
        Image display 12
        Image Overlay 12
        Image panning and manipulation 12
        Navigation 13
        Context menu 13
        Drag and Drop 14
        Displaying images side by side 14
    Navigation 14
        Moving around in the file system 14
        Moving around within a folder 15
        Going home 15
        Direct path entry 15
        Stepping through images sequentially 15
    Menus 15
        File Menu 15
        Go menu 16
        Select menu 17
        Edit menu 18
        Plugins menu 20
        View menu 20
        Windows menu 22
        Help menu 22
        Tear off menus 23
        Changing the main window menu bar keyboard shortcuts 23
    Status Bar 23
        Progress Bar 23
        Sort method 24
        File list 24

Image Details 24
Zoom and Scroll 24
Buttons 25
Layout 25
Selecting the location of each pane 25
Changing the size of a pane 26
3. Sidebars 27
Info Sidebar 27
Adding a metadata key pane 27
Histogram 27
Title 27
Description 27
Keywords 27
Keyword Autocompletion 29
List panes 29
Map Display 32
Sort Manager Sidebar 32
Using destination shortcuts 32
Folder view 33
Collection view 34
4. Stand-alone Windows 36
Image Window 36
Navigation 36
Drag and Drop 36
Exif Window 36
Pan View 37
Select a folder 37
Choose a view mode 37
How each image should be represented 38
Find 38
5. Search and Select 39
Caching Data For Searches 39
Search Window 39
Search location 39
Search parameters 40
Results list 42
Status bar 42
Drag and Drop 43
Finding Duplicates 43
Adding files to be compared 43
Comparison methods 43
Results list 44
Exporting results for scripts 45
Status bar 45
Thumbnails 46
Ignore Orientation 46
Sort 46
Ignore matches within the same folder 46
Compare two file sets 46
Drag and Drop 47
Image Data Window 47
Marking Images 47
6. Plugins 48
7. File Management 49
Copy and Move 49
Copy and clipboard commands 49
The Copy/Move dialog 50
The error dialog 51

The overwrite dialog 51
Rename 52
Rename single file dialog 52
Rename multiple files dialog 52
Delete 53
Confirm delete dialog 54
Safe Delete 54
Cache and Data Maintenance 54
Geeqie thumbnail and sim. cache 54
Shared thumbnail cache 55
Create thumbnails 55
Create file similarity cache 55
Metadata 55
Clean and Update Caches 56
Command line program 56
8. Color Management 58
9. Image Presentation 59
Slide Show 59
Starting a slide show 59
Stopping a slide show 59
Slide show status 59
Pausing a slide show 59
Selecting the images to include in a slide show 59
Recursive slide show 60
Repeating a collection 60
Slide show sequence navigation 60
Configuring a slide show 60
Full Screen 60
Turning on full screen 60
Turning off full screen 61
Configuring full screen 61
10. Printing 62
Print options 62
11. Preferences 63
General Options 63
Color scheme 63
Thumbnails 63
Star Rating 66
Slide show 66
Image loading and caching 66
Menu style 67
Expand menu/toolbar 67
Hide Selectable Bars 67
Show predefined keyword tree 67
Timezone Database 67
On-line help search 68
Image Options 68
Zoom 68
Fit Image To Window 68
Tile Size 69
Appearance 69
OverlayScreen Display 69
Overlay Screen Display 69
Window Options 71
State 71
Size 72
Full Screen 72
Keyboard Options 72

File Filters Options 73
Filters 73
Grouping sidecar extensions 73
File types 73
Metadata 74
Metadata writing process 75
Step 1 Options: 75
Steps 2 and 3 Option: 76
Miscellaneous 76
Auto-save options 77
Spelling Checks 77
Pre-load metadata 77
Metadata Write Sequence Flowchart 77
Keywords 77
Keyword Search 78
Keyword List 78
Color management options 78
Stereo image management 78
Behavior Options 78
Delete 78
Behavior 80
Navigation 82
Debugging 82
Toolbar 82
Advanced 83
External preview extraction 83
Thread Pools 84
12. Additional Preferences 85
Window Layout 85
Plugins Configuration 85
Plugins Configuration Dialog 85
Adding new plugin 86
Modifying an existing plugin 86
Deleting a plugin 86
Convert image 86
Symlink grouped files 87
Special plugins 87
Geeqie desktop file keys 87
Legacy Configuration File Options 88
13. Reference 90
Command Line Options 90
Independent Instances 94
Remote Keyboard Actions 94
Thumbnails 94
Format 95
Location 95
Size 95
Metadata Processing 95
Metadata templates 95
Metadata handling 96
Options that influence the process 96
Lua Extensions 97
Requirements 97
How to use Lua 97
Geeqie Lua built-in functions 97
Examples 97
Warning 98
Lua API 98

Configuration Files and Locations 98
File date types 99
XMP, Exif and IPTC 99
Geeqie pre-formatted tags 100
Info sidebar panes 102
Exif IPTC alternative keys 103
Collection file format 104
Supported File Formats 104
RAW image previews 105
Similarity Algorithms 105
Standard Algorithm 105
Alternate Algorithm: Perceptual Hash (pHash) 106
Additional pixbuf loaders 106
JP2/JPC/JPX/J2K/JPF 106
Standard Plugins 106
Apply the orientation to image content 107
Camera import 107
Exif datetime to file 107
Export jpeg 107
Open With 107
Random image 107
Resize image 108
Symlink 108
Tethered photography 108
UTC and Daylight Saving Time (DST) 109
Decoding Latitude and Longitude 109
Standards 109
Perl Compatible Regular Expressions 110
14. Useful Software 111
Entangle 111
Photini 111
Unison 111
Keyword Merge 111
Keyword Marks 111
Install Geeqie from sources on Debian 111
Generic install script 111
Recently used Collections 111
Decoder for .sim files 112
15. Frequently Asked Questions 113
Move to Trash failed 113
How do I fix move to trash failed? 113
Image viewing 113
Limiting enlargement of images 113
Command line 113
How do I open an image in an existing Geeqie application ? 113
Setting the slide show delay 114
Miscellaneous 114
How can I save and restore multiple Geeqie windows 114
I want to be able to edit images with Geeqie 114
Exif Auto Rotation 114
How do the Exif auto-rotation commands work? 114
Dark Mode 115
How to set Geeqie to use a dark mode 115
16. Licence 116
17. Credits 117
18. GNU Free Documentation License 118

List of Tables

2.1. 3
2.2. Tool bar 8
2.3. Context menu 10
3.1. Pre-formatted tags 30
11.1. Overlay Screen Display templates 69
11.2. Overlay Screen Display template examples 71
13.1. Pre-formatted tags 100
13.2. Fixed panes 102
13.3. Location and GPS pane 102
13.4. Copyright pane 102
13.5. Exif pane 102
13.6. Conversion table 103
13.7. Resize image options 108

Chapter 1. Introduction

Geeqie is an image viewer that makes navigation of images easy. Images can be viewed as thumbnails, in multiple windows or in full screen mode.

Geeqie is an image manager, with complete copy, move, rename, and delete functions available at the press of a button.

Extensive image collections can be easily handled using a combination of Geeqie's flexible Keyword facility and powerful Image searches.

Image collections can be refined with a duplicate image finder.

Geeqie is ideal for making presentations. A slide-show feature is built in, and Geeqie can be remotely controlled either via command line scripts or an infra-red device. Information such as f-stop, shutter speed and descriptive text can be overlaid onto the image display.

There is no need to import images into Geeqie - simply point Geeqie to a folder containing images, and your data will be displayed.

Geeqie is an image viewer, not an image editor. However, Geeqie can be configured to call external image editors such as the Gimp, resulting in a very powerful combination.

Features

Basic Features

- Icon view
- Full Screen
- Slide Show
- File Management
- Collections

Advanced Features

- Search and Select

  Image Search based on many parameters
- Finding Duplicates

  Find image duplicates based on image content
- List panes

  View IPTC, and view and edit Exif and XMP data stored in images
- Sort Manager Sidebar

  Simple Sort Manager to quickly sort a folder of images into other folders
- Rename

  Rename images in bulk
- Image Overlay

  Overlay the image with a panel showing user-definable data
- Keywords

  Add XMP keywords to images, and recall those images with the search function

- Preferences

  Geeqie is a very configurable application

“An application that displays images.” - This was, and still is, the primary goal of Geeqie. Over time image management features have been added, but the core purpose of Geeqie remains the same.

Chapter 2. Main Window

Table 2.1.

Layout window Title bar Tools Main menu Main toolbar	Location bar Shortcuts pane Folders pane Files pane	Image pane Info sidebar (optional) Sort manager (optional) Status bar Status toolbar

The Geeqie main window is designed for quick and easy navigation of images. The window consists of three regions: the File Pane , the Folder pane , and the Image Pane (optionally bordered by Sidebars) , all bordered by Menus and Status Bar .

The Focus directories pane, Focus files pane, and Focus address bar actions move keyboard focus directly to those areas. Assign shortcuts in Keyboard Options; these actions have no default shortcuts. They reveal hidden panes and also work when the file list is floating. Focusing the address bar selects its text.

Multiple instances of the main window may be started via the Windows → New Window (**Ctrl+N**) menu item, and the Layout of each may be individually configured in the Window Layout dialogue.

Multiple windows normally belong to one Geeqie application instance, including windows opened by separate command line launches. They may be closed individually or simultaneously. If File → Quit (**Ctrl+Q**) is pressed, all windows belonging to that application instance will be closed. Window restoration at the next startup depends on the Remember session setting in Window Options.

If Windows → Close Window (**Ctrl+W**) is pressed, only that window will be closed. Closing the last main window quits the application instance. To start an independent instance, see Independent Instances.

Several window layouts may be defined and displayed at any time, as described in the Windows menu .

File Pane

The file pane displays the images contained in the active folder. It has two modes of display, allows multiple selection, and can initiate drag and drop operations.

A **context menu** can be displayed by clicking the right mouse button on the file pane or pressing the menu key while the file pane has the focus.

List view

This mode can be enabled by selecting **List** from the **View** menu, pressing `Ctrl + L`, or unselecting **View as icons** from the context menu.

The list view mode displays images in a standard vertical list. Three columns provide the file name, size in bytes, and the time the contents were last modified.

Thumbnails can be displayed next to each file in the list, to toggle thumbnails on or off by selecting **Thumbnails** from the **View** menu, pressing `T`, clicking on the tool bar, or selecting **Show thumbnails** from the context menu.

Grouped sidecar files are represented by the main file with an expander. Clicking on the expander shows sidecar files.

Changing the file list font

The file list normally uses your desktop's interface font. To change the font for the file names and details in list view, create or edit `~/.config/gtk-4.0/gtk.css`. The file must be named `gtk.css`, in the `gtk-4.0` directory (with a hyphen). If that directory already contains `settings.ini`, leave it in place and create `gtk.css` alongside it. Put the following rule in `gtk.css`, not in `settings.ini`; no `geeqie.ini` file is needed:

```
.geeqie-file-list {
    font-family: monospace;
}
```

If `XDG_CONFIG_HOME` is set, use `gtk-4.0/gtk.css` inside that directory instead. Create the directory and file if they do not exist, and preserve any existing rules.

You can replace `monospace` with an installed font name, such as `"DejaVu Sans Mono"`, and optionally add `font-size: 11pt;` inside the braces.

Restart Geeqie to apply the change. This rule applies only to the file list; the folder pane and icon view keep their usual fonts. To restore the default font, remove the rule and restart Geeqie.

Icon view

This mode can be enabled by selecting **Icons** from the View menu, pressing `Ctrl + I`, or selecting **View as icons** from the context menu.

The icon view mode displays images as thumbnails in a grid, and by default a file name below each thumbnail.

The file name displayed below each thumbnail can be turned on or off by selecting **Show filename text** from the context menu.

Grouped sidecar files are represented by the main file. For accessing sidecars switch to List mode.

File Filter

A filter box can be opened by selecting **Show File Filter** from the View menu.

Characters you type into this box will be used for pattern matching to select the files displayed. Regular expressions are used in this field.

For basic pattern matching "JPG" will match any filename containing those characters.

If you type a newline, the text will be saved in the dropdown box list. The last 10 items are saved.

To delete an item, select it from the drop-down list, click on the text box to select the text, and then press **Delete** followed by **Enter** .

By default filtering is case-insensitive. A checkbox labelled "Case" permits the filter to be made case-sensitive.

Additionally a drop-down box permits any of the file Classes to be selected or deselected from the viewed list.

Also additionally a drop-down box permits any of the file Rating to be selected or deselected from the viewed list.

Multiple selection

It is possible to select more than one image from the file pane. Multiple files can be selected with several methods:

Mouse

- `Ctrl` + Primary mouse button will add or remove the file from the selection.
- Middle mouse button will toggle selection of the file.
- `Shift` + Primary mouse button will select all images between the clicked and focused image, or it will unselect all those images if the clicked image is already selected.
- `Shift + Ctrl` + Primary mouse button works as Shift + Primary mouse button, but will not change the selection of files outside the region between the focus and clicked image.

Keyboard

- `Ctrl + Arrows` will move the focus without changing the selection.
- `Ctrl + Space` will toggle selection of the focused image.
- `Shift + Arrows` will select a region of images.
- `Shift + Ctrl + Arrows` works as `Shift + Arrows`, but will not change the selection of files outside the current focus region.

To select all files press `Ctrl + A`, or to unselect all image press `Ctrl + Shift + A`. These operations are also available from the **Edit** menu as **Select all** and **Select none** .

The current number of selected images and their total size will be indicated in the File list section of the status bar .

Sorting

The order of the images can be changed by clicking the sort area of the status bar or from the context menu. The sort methods are by file name, number, file date, or file size. The number method will sort file names by their natural order, for example files with names of file_10, file_12, and file_9 will appear in order file_9, file_10, and file_12.

Selecting the ascending menu item will toggle between increasing and decreasing sort order.

Context menu

Right clicking the mouse or pressing the menu key while the file pane has focus will display a menu. The menu functions will perform the same as those that match the window's menu bar , however a few functions are unique to the context menu:

Rebuild selected thumbnails	Removes the cached thumbnails for the selected files and generates new thumbnails from the source images. Use this when a thumbnail is incomplete or outdated. If the right-clicked file is not selected, only that file is processed. Source images, similarity data, and metadata are unchanged. In list view, new thumbnails are loaded when thumbnail display is enabled.
View in new window	Opens a new window containing a list of the currently selected images.
Copy Image	Loads the file that was right-clicked and copies its decoded image pixels to the clipboard. The thumbnail, original encoded file, metadata, and sidecar files are not copied.
Sort	Changes the sort order of file pane.
View as icons	Toggles between the list and icon view modes.
Show thumbnails	Toggles display of thumbnails in the list view display mode.
Show filename text	Toggles display of file names in the icon view display mode.
Show star rating	Toggles display of the "star rating" (Xmp.xmp.Rating).

Drag and Drop

Drag and drop can be initialized with the primary or middle mouse buttons in the file pane. Dragging a file that is selected will include all selected files in the drag. Dragging a file that is not selected will first change the selection to the dragged file, and clear the previous selection.

Collections

Geeqie provides the ability to build lists of images, regardless of the location of the files on disk.

When saving a new collection, select Save paths relative to the collection file to store image paths relative to the folder containing the `.gqv` file. Paths begin with `./` or `../`. You can move the collection and images together while preserving their folder layout. The setting is retained when the collection is reopened or saved to another location. Absolute paths remain the default.

The files pane displays and edits collections in the main window.

Collection entries are retained when their files are unavailable, including files on disconnected drives and files deleted outside Geeqie. Cached thumbnails are displayed when available; opening an unavailable image shows a broken image indicator. Saving the collection preserves these entries. Use Remove from collection to remove entries you no longer want.

To create a new empty collection in a new default window, select File → New collection (**C**) .

To open an existing collection, select File → Open collection (**O**) . The dialog combines collection files in the default collections directory with recently opened or saved collections, including those stored elsewhere. Each entry shows its filename followed by its full path. Duplicate paths appear only once, and entries are sorted by filename, then path. Click an entry or select it and press **Return** to display the collection in the files pane. Press **Escape** to close the list.

Unavailable or unreadable collection files remain in the list but are grayed out and cannot be selected. Their history entries are retained between sessions, so collections on disconnected drives can be opened again when the drives are available. The Recent collections list maximum size setting in

Behavior Options controls how many collections are retained in history; it does not limit the files listed from the default collections directory.

When images are moved or renamed within Geeqie, existing collection files in the default collections directory and collection history are updated to use the new paths.

File → Open recent uses GTK's shared recent-files list instead of collection history. Its default Geeqie image files filter matches Geeqie's configured file extensions; select All files to show other recent files.

To delete or rename a collection file, use your file manager. The Open collection list does not provide file management actions.

Viewing a collection in the files pane

When browsing a directory containing collection files, the folder pane displays them as folders with a pictures-folder icon. Click one to display its members in the files pane. This works in both folder-list and folder-tree views; these entries are not actual disk directories.

The files pane supports list and icon views, selection, filtering, sorting, and the usual file operations. The main path bar shows the collection path, or its name until it is saved.

Copy, move, rename, and delete operate on the actual files. Use Remove from collection in the files pane context menu to remove membership while leaving the files on disk. Dropping files into the pane adds them to the collection. When a drop includes folders, choose whether to add their contents, add contents recursively, or skip the folders. The same menu can append another collection, save or save as, and change or randomize the collection order.

Use Go to original on a collection member to open its containing folder in the files pane.

Use **Ctrl**+**Alt**+**S** to save the current collection, **Ctrl**+**Shift**+**S** to save it as another file, or **Ctrl**+**Alt**+**Backspace** to remove selected files from the collection.

When the files pane shows a collection, its context menu includes Info text. Select Show info text to display each entry's text in the list or icon view. Select Edit info text to create or change the text for the clicked file or selected files. Enter the text in the dialog and select Save. When multiple files are selected, the same text is applied to each one.

In list or icon view, drag images within the files pane to change their collection order. Drop before or after a row or icon, or in empty space to append. A marker indicates the insertion side. Selected images move together. Reordering switches the pane to collection order; save the collection to retain the new order.

You can drag images between the files panes of two main windows, whether each pane shows a folder or a collection. Dropping onto a collection adds references to the images without moving the files or removing them from the source collection. The drop position determines where new entries are inserted in both list and icon views. Dropping onto a folder uses the usual file copy or move operation.

Use Windows → New window → New window from current to show the same collection in another main window. Each window can then browse independently.

Select Collection order from the sort menu to use the collection's order. Other pane sorts do not change the saved collection order. Select a directory to return to directory browsing. If this is the last window showing a modified collection, Geeqie asks whether to save or discard the changes, or cancel navigation. The same choices are offered when closing or quitting. Collection mode is not restored after restarting Geeqie.

Folder pane

The folder pane contains the menu bar, tool bar, path entry, and folder list.

Menu bar

The main menu of the window. The functions available from the main menu are described in the Menus chapter.

Tool bar

The tool bar includes buttons for often used operations. The tool bar can be hidden or shown by selecting **Hide toolbar** from the **View** menu.

Table 2.2. Tool bar

Icon	Tooltip	Description
	Show thumbnails	This button is active when the file list is in list view mode, and activating it will toggle the display of thumbnails.
	Back	Move back in folder history
	Up	Move up one folder
	Home	Change the main window to display the contents of the user's home folder
	Refresh folder display	Refresh the contents of the file pane
	Zoom in	Increase the image display size.
	Zoom out	Decrease the image display size.
	Zoom to fit	This will size the image to fit within the window
	Zoom 1:1	Restore zoom to the image's original size.
	Preferences	Displays the Preferences window .
	Float file list	Toggle the folder pane, file pane, and status bar between the main window and a secondary window.

Path entry

This displays the active folder of the window and allows typing a location.

A folder or file can be typed into the entry to quickly change to a desired location or image, the main window will update after pressing the `Enter` key. Tab completion is also available in the path entry.

Note

Tab completion uses the Tab key, press the Down key to move the focus from the entry.

Next to the path entry is a button with a down arrow, clicking this will display a drop down list containing the most recent locations viewed. The list is ordered such that the most recent location is at the top. Selecting a location from the drop down list will change the window to display the location.

An Internet URL may be typed or drag-and-dropped into this field. The file will be downloaded to a temporary file and displayed.

Folder list

This pane consists of two sections: a folder pane and a shortcuts pane. The shortcuts pane may be hidden or revealed simply by moving the vertical divider to or from the extreme left.

Folders

This displays a list of sub folders contained in the active folder, or a tree that includes all folders in the file system.

A **context menu** can be displayed by clicking the right mouse button on the folder pane or pressing the menu key while the folder pane has the focus

Shortcuts

This is a list of shortcuts (bookmarks) to commonly used folders, images and collections. A context menu will appear by right clicking on an item, or pressing the menu key when the item has the focus.

Folders or images can be added to the list by activating the Add button at the bottom of the pane, or dragging a folder or image onto the list.

Collections may be added by opening the File → Open collection... (**O**) dialog and dragging a collection onto the list.

To edit an item select **Properties** from its context menu. A dialog will appear allowing the Name, Path, and Icon for the item to be changed. Select OK to close the dialog and apply the changes, or Cancel to close the dialog and discard the changes.

To change the order of the list, select **Move up** or **Move down** from the item's context menu. To move an item with the keyboard move the focus to that item, then press **Shift**+**Up** or **Shift**+**Down** to move the item.

To remove an item from the list select **Remove** from it's context menu.

List view

This mode can be enabled by from the **View** menu, from the context menu, or pressing `Ctrl + T`.

The list view mode displays sub folders contained in the active folder in a standard vertical list. The parent folder can be selected with the folder named "..".

Tree view

This mode can be enabled by selecting **Tree** from the View menu, pressing `Ctrl + T`, or selecting **View as tree** from the context menu.

The tree view mode displays all folders in the file system, starting with the root folder and branching out to include sub folders. A folder branch can be expanded or collapsed by clicking the expander next to each folder icon, or by pressing the respective `+` or `-` keys when the folder branch has the focus.

Sorting

The order of the folder names will match the order selected for the file pane. Sort order can be changed by clicking the sort area of the status bar or from the context menu of the file pane.

Context menu

Right clicking the mouse or pressing the menu key while the folder pane has focus will display a menu. The menu functions will perform the same as those that match the window's menu bar , however a few functions are unique to the context menu:

Table 2.3. Context menu

Icon	Text	Description
	Up to parent	This will change the window to display the contents of the parent folder.
	Slide show	Start a slide show that contains all images in the selected folder.
	Slide show recursive	Start a slide show that contains all images in the selected folder, and all images in the sub folders of that folder.
	Find duplicates	This will open a new find duplicates window containing all images in the selected folder.
	Find duplicates recursive	This will open a new find duplicates window containing all image in the selected folder, and all images in the sub folders of that folder.
	New folder	Creates a new folder that is a sub folder of the selected folder.
	Rename	Renames the selected folder. **Warning** **Warning:** Renaming a folder will not update metadata, such as keywords, or cached thumbnails for images contained in that folder, the data will be lost.
	Delete	Delete file or folder
	Copy path	Copy selected item-path(s) to clipboard [a] . Each data item copied will be enclosed in quotes.
	Copy path unquoted	Copy selected item-path(s) to clipboard. The data will not be enclosed in quotes.
	View as list	Switches to list tree mode of the folder pane.
	View as tree	Switches to tree mode of the folder pane.
	Show hidden files	Switches to list tree mode of the folder pane.

Icon	Text	Description
	Refresh	Switches to tree mode of the folder pane.

[a] The clipboard used is selected here.

Drag and drop

Image files can be dropped onto folders displayed in the folder pane. A menu will appear allowing you to select to move or copy the dropped images into the target folder. The list will automatically scroll up or down when the drag pointer is near the respective top or bottom of the list. When in tree mode, hovering over a folder for a brief period of time will expand that folder.

A folder can be dragged onto another folder in the same folder pane or in another Geeqie window to copy or move it there. The entire folder is transferred, including hidden files, sidecars, and subfolders, regardless of the file display filters. The action follows the drag and drop preference, with a Copy/Move/Cancel menu when the preference is Ask.

Transfer one folder at a time. If a folder with the same name already exists at the destination, choose Merge to combine their contents, or Duplicate to keep the incoming folder separate as `Folder (Copy)`. If that name already exists, Duplicate uses `Folder (Copy 2)` or the next available number. The transfer still follows your original Copy or Move action. Existing destination folders keep their attributes. A folder cannot be transferred into itself or one of its subfolders. Progress is displayed and the operation can be canceled.

Within a folder merge, matching subfolders also offer Merge or Duplicate. Other name conflicts offer Skip, Rename, Duplicate, and Cancel. Skip leaves the source item in place. Rename transfers it using the name you enter. Duplicate automatically chooses an unused name, such as `image (Copy).jpg`, preserving the file extension. Existing files are never replaced, and symbolic links at the destination are not followed as folders.

Select Apply this action to all files to repeat Skip or Duplicate for all remaining conflicts in this transfer. Selecting it with Merge repeats Merge for subsequent folder conflicts. Rename remains a manual action for one item. These choices are reset for the next transfer. Cancel always stops the transfer.

During a merge move, selected items are copied before source removal begins. Skipped items and their containing source folders remain in place. Canceling or failing during the copy stage leaves all source contents intact; completed copies remain at the destination. Canceling during source removal leaves the completed copies and any remaining source contents in place.

Folders can also be dragged to add their images to a collection in the file pane.

Using `Shift` key while dragging is a shortcut to move action. Menu will not appear but move will occur.

Using `Ctrl` key while dragging is a shortcut to copy action. Menu will not appear but copy will occur.

Image Pane

The image pane is the region of the main window that displays the selected image. The image pane allows adjustment of display size and orientation; it also provides several methods to manipulate the image with the mouse and keyboard.

At 100% zoom (1:1), one image pixel corresponds to one physical display pixel, independently of desktop scaling. Interface text and controls retain the desktop scale. Moving the window between displays preserves the selected zoom and adjusts the image size accordingly.

Image display

The image is drawn to the screen as it is decoded from disk. The image can be panned and zoomed while it is still loading. Image adjustment operations for rotation, mirror, and flip will delay until the image is completed. The adjustment queue will remember one operation and will retain the most recently requested function, which will be performed once the image has finished loading.

The quality of the image, especially when using the zoom feature, is dependent on the settings that are available on the Image tab of the preferences dialog.

To increase the speed of loading the next image when navigating the image list in sequence, enable the Preload next image option.

Image Overlay

An image overlay can be turned on and off by repeatedly pressing **I** . This overlay can either display image data such as f-stop, shutter speed and descriptive text, or a histogram.

Data the overlay will always show is:

- Marks , if they are selected

If the histogram is being displayed, **K** will switch the histogram between colors, and **J** will toggle between a linear or logarithmic histogram.

Data the overlay displays is customizable through the Image Overlay Template option in the Windows tab in the Preferences dialog.

Image panning and manipulation

When the image size is larger than the image pane, the visible region can be changed with the keyboard or mouse. The keyboard can also be used to adjust the orientation of the image.

Keyboard and mouse

Keyboard Shortcuts describes how you may manipulate the image displayed.

Note

The rotation, mirror, and flip functions do not change the image data, they merely change the image as you view it. You may store the modified orientation into metadata manually with Edit → Orientation → Apply the orientation to image content , or this may happen automatically if *Write altered image orientation to the metadata* is set in Metadata Options

Mouse

Press the left mouse button on the image and hold it, then move the mouse to pan the image in the desired direction. Releasing the mouse button will stop the panning operation. The mouse wheel can also be used to scroll the image when holding down the Shift key [1] .

To scale the image size hold down the Ctrl key and scroll the mouse wheel up to zoom in, or down to zoom out.

[1] The mouse wheel behavior for scrolling and image step can be swapped with the "Mouse wheel scrolls image" option located on the Behaviour tab of the preferences dialog.

Bird's-eye view

When *Show bird's-eye view when zoomed in* is enabled in the Behavior Options, zooming or panning an image that is larger than the image pane temporarily displays a small overview of the complete image in the upper-right corner. A white rectangle in the overview indicates the region currently visible in the image pane. This feature is disabled by default.

Click a position in the overview to center that part of the image in the image pane. Press and hold the primary mouse button in the overview, then move the pointer to pan directly to another part of the image.

The overview remains visible while it is being used and hides automatically after a short period without zooming or scrolling. It is not displayed when the complete image fits inside the image pane.

Smooth Scroll

To start the smooth scroll function hold down the Control key and click the left mouse button. An icon will appear at the location of the mouse pointer, the mouse can then be positioned to pan the image. The image will pan in the direction of the mouse pointer relative to the icon. The pan speed will increase in proportion to the distance that the pointer is from the center of the icon. To end the smooth scroll function click the left mouse button within the image pane.

Mouse Wheel Mode

A single click on the mouse middle button will toggle the function of the mouse wheel between "change image" and "change zoom".

Note

This option cannot be used if either the Preference *Mouse wheel scrolls image* or *Navigation by left or middle click on image* are selected.

In Split View mode, each window can be clicked individually so that, in Quad View for example, Connected Zoom can be on 2, 3 or all 4 windows.

Navigation

The displayed image can be changed to the next or previous image using the keyboard or mouse.

Keyboard

The PageDown and PageUp keys will change the image to the next or previous image, respectively, in the list. The Home and End keys will jump to the first and last image, respectively. When the image has the focus, the SpaceBar and BackSpace keys can also be used to change the image.

Mouse

To change to the next image in the list click the left mouse button, click the middle mouse button to change to the previous image. The mouse wheel can also be used to change the current image .

Context menu

Right clicking the mouse or pressing the menu key while the image has the focus will display a menu. The menu functions will perform the same as those that match the window's menu bar , however a few functions are unique to the context menu:

View in new window	Opens a new window containing a second copy of the current image.
Pause slide show, Continue slide show	Controls the pause state of an active slide show, the text of the menu will change to reflect the action that will be performed.
Copy Image	Copy the decoded image pixels to the clipboard for use by applications such as GIMP. This does not copy the original image file or its metadata.

Drag and Drop

Files and folders can be dropped onto the image pane, dropping a file will display the file and dropping a folder will change the main window to display the contents of the folder. When dropping multiple items on the image pane, only the first item in the list will be used, all additional items will be ignored.

A drag and drop operation can be initiated from the image pane with the middle mouse button, the dragged item will contain the file path of the displayed image.

Displaying images side by side

Using **View** → **Split** item from the main menu, image pane can be split into either two, three or four panes, allowing one to display images side by side. You may use drag'n drop to change images in each pane.

Connected zoom is possible through the **View** → **Zoom** → **Connected Zoom** menu item. Zoom is then applied to all displayed images.

Connected scrolling is also possible using Shift and primary mouse button.

To select images in split panes for file operations, click the checkbox in the upper-left corner of each desired pane. The checkbox appears when the pointer is over a pane and remains visible while selected. Its tooltip explains its purpose. You can also hold Ctrl and Shift and click a pane with the primary mouse button. Click a selected checkbox again, or repeat the gesture, to deselect a pane. Selected panes have a colored border, independent of the active pane.

Press Delete, or choose File → Move selection to Trash…, to review and delete the selected images together. The confirmation lists full paths and includes grouped sidecars according to the normal deletion rules. Existing Trash settings apply.

While panes are selected, selection actions in the image-pane and file-list right-click menus use those images. This includes copy, move, rename, deletion, clipboard paths, plugins, and collections. Right-clicking another image preserves the pane selection. Copy Image is available for a single selected image; clipboard path actions support multiple selected images.

Escape clears the pane selection. Changing a displayed image or the split layout also clears the selection. Delete and Shift+Delete use the pane selection when present, otherwise they use the file-list selection. Shift+Delete requests permanent deletion.

As a general rule, hotkeys and mouse without shift affects just one image and with shift all images.

Navigation

Moving around in the file system

To select the folder containing the desired images for display, change to the image's folder using the folder view. Single clicking a folder with the primary mouse button will change the main window to

display that folder's contents. When the folder view is displayed as a list, use the folder named ".." to change to the parent folder (the parent folder is the folder that contains the current folder).

To select a folder with the keyboard, move the focus to the desired folder and press the Space or Enter key.

Moving around within a folder

After selecting the desired folder as above, the desired image can be viewed by selecting it from the file list. To select a file for display single click it with the primary mouse button.

To select a file with the keyboard, move the focus to the desired file. As you move the focus within the file list the image display will update to match the highlighted file.

Going home

To change to your home folder, click the Home button in the tool bar. Home folder can be set in window preferences.

Direct path entry

The current path is displayed in an entry field above the folder list. A folder or file can be typed into the entry to quickly change to a desired location or image, the main window will update after pressing the Enter key. Tab completion is also available in the path entry.

Next to the path entry is a button with a down arrow, clicking this will display a drop down list containing the most recent locations viewed. The list is ordered such that the most recent location is at the top. Selecting a location from the drop down list will change the window to display the location.

Stepping through images sequentially

To change the displayed image to next or previous one within the same folder, use the respective PageDown and PageUp keys. The mouse can also be used to change the image by clicking the image display with the respective primary and middle buttons.

To change to beginning or end of the image list, use the respective Home or End keys.

To navigate between sibling folders, use Ctrl+Shift+Right to go to the first image in the next folder, or Ctrl+Shift+Left to go to the last image in the previous folder. When the Behavior Options preference Auto next folder is enabled, reaching the last image in a folder with Next Image will automatically advance to the first image in the next sibling folder, and reaching the first image with Previous Image will automatically go to the last image in the previous sibling folder.

Menus

This section describes the functions of each item on the menu bar in the Geeqie main window.

File Menu

Open File (**Ctrl+N**)	Opens a file using the system default dialog. A limited range of file types may be displayed in the preview pane.

New collection (**C**)	Opens a new default window with an empty collection in its files pane.
Open collection (**O**)	Lists collection files from the default collections directory and collection history, showing each filename followed by its full path. Duplicate paths appear only once. Click an entry or press **Return** to open it in the files pane; press **Escape** to close the list. Unavailable entries are grayed out and cannot be selected. See Collections.
Open recent	Lists existing local files from GTK's shared recent-files database, with the most recent first. The default Geeqie image files filter matches Geeqie's configured file extensions; select All files to show other recent files. Select an entry and click Open or press **Return** to open it. This list is independent of collection history and its size limit.
Save collection	Saves the collection displayed in the files pane. For a new collection, opens the save dialog.
Save collection as…	Saves the displayed collection to a selected file.
Search… (**F3**)	Opens a new Search window.
Find duplicates... (**D**)	Opens a new Find Duplicates window.
Print.. (**Shift+P**)	Opens a print dialog.
New folder… (**Ctrl+F**)	Creates a new folder in the current path.
Copy… (**Ctrl+C**)	Opens a dialog to select a destination to copy the currently selected files.
Move… (**Ctrl+M**)	Opens a dialog to select a destination to move currently selected files.
Rename… (**Ctrl+R**)	Renames the current file selection.
Open With...	Calls the standard GTK application chooser for a single file.
Delete… (**Ctrl+D**)	Deletes the current file selection.
Symlink	Creates a symbolic link. This is implemented via a plugin .
Close window (**Ctrl+W**)	Closes the window containing the menu, if the window is the last main window Geeqie will quit.
Quit (**Ctrl+Q**)	Quits the Geeqie application instance and closes all its windows. If collections or image metadata have been changed but not saved, the user will be prompted for confirmation. Other independent Geeqie instances are not closed.

Go menu

The Page commands are for files which contain multiple images e.g. .tiff files

First Image (**Home**)	Goes to the first image.
Previous Image (**Backspace**)	Goes to the previous image.
Next Image (**Space**)	Goes to the next image

Last Image (**End**)	Goes to the last image
Show current image	Scrolls the files pane to bring the currently displayed image into view. The selection and keyboard focus stay unchanged. You can assign a keyboard shortcut in Preferences.
Image Back	Each image you view is appended to a chain. You may traverse forwards and backwards through this chain using this and the following key. Move to the previous image in the chain.
Image Forward	Move to the next image in the chain.
Back	Back Goes to the previous folder
Next folder (**Ctrl+Shift+Right**)	Goes to the first image in the next sibling folder.
Previous folder (**Ctrl+Shift+Left**)	Goes to the last image in the previous sibling folder.
Home (**Home**)	Goes to the the Home folder as defined in Window Layout
First Page (**Ctrl+Home**)	Goes to the first page.
Last Page (**Ctrl+End**)	Goes to the last page.
Previous Page (**Ctrl+PageUp**)	Goes to the previous page.
Next Page (**Ctrl+PageDown**)	Goes to the next page.

Select menu

Select all (**Ctrl+A**)	Selects all files in the current list
Select none (**Ctrl+Shift+A**)	Deselects all files
Invert selection (**Ctrl+Shift+I**)	Inverts current selection
Rectangular selection (**Alt+R**)	Toggles rectangular selection mode for icons
Copy to clipboard	Copy selected item-path(s) to clipboard [2] . Each data item copied will be enclosed in quotes.
Copy to clipboard (unquoted)	Copy selected item-path(s) to clipboard. The data will not be enclosed in quotes.
Cut File to Clipboard (**Ctrl+X**)	Place the selected files on the clipboard for moving when pasted into a compatible file manager. Selecting this command does not immediately move the files.
Show marks (**M**)	Displays marks in the file list
Clear marks	Clear all marks for all images **Warning** Marks that are linked to keywords will also be cleared. This may result in a metadata write operation being triggered.
Mark n	Opens submenu for manipulating mark n

[2] The clipboard used is selected here.

Mark n → Toggle Mark n (**n**)	Set or unset mark n on selected files
Mark n → Set mark n	Set mark n on selected files
Mark n → Clear mark n	Clear mark n on selected files
Mark n → Select mark n (**Ctrl+n**)	Selects all files having mark n set
Mark n → Add mark n	Adds files having mark n set to selection
Mark n → Intersection with mark n	Intersects current selection and mark n
Mark n → Unselect mark n	Removes files having mark n set from current selection
Mark n → Filter mark n	Shows only the images that have mark n set.

Edit menu

Orientation	Adjust image orientation
Orientation → Rotate clockwise (**]**)	Rotates the current image clockwise 90 degrees, does not modify the file on disk.
Orientation → Rotate counterclockwise (**[**)	Rotates the current image counterclockwise 90 degrees, does not modify the file on disk.
Orientation → Rotate 180 (**Shift+R**)	Rotates the current image 180 degrees, does not modify the file on disk.
Orientation → Mirror (**Shift+M**)	Swaps the left and right sides of the image so that the result is similar to that of looking at the image in a mirror, does not modify the file on disk.
Orientation → Flip (**Shift+F**)	Swaps the top and bottom sides of the image, does not modify the file on disk.

Warning

The above orientation commands - Rotate, Mirror and Flip - do not change the image on disk. The changes will be lost when Geeqie closes. Refer to the Metadata tab of the Preferences Dialog if you wish these changes to be written to metadata.

Refer also to the FAQ .

Orientation → Original state (**Shift+O**)	Removes all temporary effects applied through the above commands, reverting the image according to its Exif Orientation tag.
Auto rotate image using EXIF information	Auto rotates images based on metadata stored in images that have Exif data. Refer also to the FAQ .
Orientation → Apply the orientation to image content	This command affects only jpeg, tiff and png files. The image file on disk is rotated to the orientation as specified by its Exif Orientation tag. JPEG transformations preserve the compressed image coefficients without another lossy compression step. File

size and checksums can change because the compressed representation and metadata are rewritten. Transformations that would discard edge pixels at JPEG block boundaries are refused, leaving the file unchanged.

Warning

The plugin org.geeqie.rotate.desktop, using external commands exiftran, exiv2, jpegtran and mogrify, is used to implement this command. Refer to Plugins section for details.

Orientation → Rotate counterclockwise (**[**)	Rotates the current image counterclockwise 90 degrees, does not modify the file on disk.
Orientation → Rotate 180 (**Shift+R**)	Rotates the current image 180 degrees, does not modify the file on disk.
Orientation → Write orientation to file	Saves the current image orientation to the disk file. This operation can only be carried out on jpeg, tiff and png files. JPEG transformations preserve the compressed image coefficients without another lossy compression step. File size and checksums can change. Transformations that would discard edge pixels at JPEG block boundaries are refused, leaving the file unchanged.

Note

The Preferences/Metadata option "Write altered image orientation to the metadata" must be off.

Exiftran, exiv2 and jpegtran are required for JPEG files; ImageMagick's mogrify is required for TIFF and PNG files.

Orientation → Write orientation to file (preserve timestamp)	As above, but the file date and time are preserved.
Rating	Set a Rating value for each image.
Rating → n (**Alt+Keypad+n**)	"n" is in the range 0 to 5. Sets the Rating value for the image.
Rating → -1 (**Alt+Keypad +Minus**)	Sets the Rating value to -1 for the image.
Save metadata (**Ctrl+S**)	Saves queued metadata changes. Autosave can be configured too.
Keyword Autocomplete (**Alt+K**)	Refer to Info Sidebar.
Draw rectangle	Entering rectangle mode selects the whole image, or the largest rectangle matching the selected aspect ratio. Drag inside the rectangle to move it, or drag an edge or corner to resize it. A single click leaves the rectangle unchanged. Drag outside the rectangle to replace it. Press Escape to leave rectangle mode without changing the image.
Aspect ratio	Select an aspect ratio for the above action. Enable Portrait orientation to swap width and height, for example from 16:9 to 9:16. Changing the ratio or orientation updates the current rectangle around its center, shrinking it if necessary to stay

	within the image. The orientation is remembered for new rectangles.
Save Cropped Copy…	Save the selected rectangle as a new JPEG or PNG image. The original file is preserved. The copy uses the loaded image pixels at their available resolution and applies the displayed orientation. For RAW files, these pixels may be an embedded preview rather than full sensor data; the output dialog shows the available crop dimensions. JPEG quality defaults to 90. PNG preserves transparency; JPEG replaces transparency with white. Embedded color profiles are preserved when available. With Exiv2 support, Preserve metadata copies descriptive, camera, exposure, and GPS metadata and updates orientation and dimensions. Embedded thumbnails and source editing instructions are omitted.
Preferences (**Ctrl+O**)	Opens the Preferences Dialog for changing of global preferences.
Configure Plugins…	Opens the Plugins Dialog for changing of desktop files.
Configure This window…	Opens the Window Preferences Dialog for changing of options that can differ for each window.
Cache maintenance	Opens a dialog with utilities for thumbnail and keyword Cache and Data Maintenance .

Plugins menu

Plugins	This menu contains the plugins (desktop files) that were found on the system. See the Plugins section for details.

View menu

View in new window (**Ctrl+V**)	Opens selected image in its own window
Pan view (**Ctrl+J**)	Opens current folder in Pan view window
Exif data (**Ctrl+E**)	Toggles display of the Exif sidebar.
Files and Folders+Folder List (**Meta+L**)	Set the folder view mode to list.
Files and Folders → Folder Tree (**Ctrl+T**)	Set the folder view mode to tree.
Files and Folders → Image List (**Ctrl+L**)	Changes the file list to display files in list format.
Files and Folders → Icons (**Ctrl+I**)	Changes the file list to display files as icons.
Files and Folders → Show Thumbnails	Toggles the display of image thumbnails in the file list, this item is not available when the file list is in icon view mode. A check appears next to this item when thumbnails are enabled.
Files and Folders → Float file list (**L**)	Splits the 3-pane main window into two parts:

	• File pane and Folder pane, along with menu and status bar • Image pane and its associated sidebar and sort manager (if displayed)
Files and Folders → Hide file list (**Ctrl+H**)	Hides the File and Folder panes, along with menu and status bar. `Right-click` on the Image pane provides a menu item to redisplay these items.
Zoom	
Zoom → Connected zoom	Actions in this menu are identical to those in **Zoom** menu, but apply to all images displayed in split view.
Zoom → Zoom in	Increases the zoom by one increment, resulting in a larger image.
Zoom → Zoom out	Decreases the zoom by one increment, resulting in a smaller image.
Zoom → Zoom to fit	Automatically adjusts the zoom so that the image will fit within the display pane. The effect on images with an original size smaller than the display pane can be changed with the *Allow enlargement of image for zoom to fit* option located on the Image tab of the preferences dialog.
Zoom → Fit Horizontally	Adjusts the zoom so image height fit within the display pane.
Zoom → Fit Vertically	Adjusts the zoom so image width fit within the display pane.
Zoom → Zoom n:m (**Z**)	Set zoom to the n:m value. 1:1 is the original size of the image. When n is greater than m, zoom in occurs else zoom out occurs.
Zoom → Zoom n:m (**Z**) Split	
Split → Horizontal	Splits main image view in two horizontal panes.
Split → Vertical (**U**)	Splits main image view in two vertical panes.
Split → Triple	Splits main image view in three horizontal panes.
Split → Quad	Splits main image view in four panes.
Split → Single (**Y**)	Reverts to normal view mode (one image).
Image Overlay+Image Overlay	Displays Image Overlay information box (OSD), use twice to add histogram to it.
Image Overlay+Histogram channels	Switches channels displayed in the histogram (red, green, blue, RGB, value, maximum value).
Image Overlay+Histogram log mode	Toggles between linear and logarithmical histogram.
Full screen	Toggles the full screen window display.
Over/Under Exposure (**Shift+E**)	Toggles highlighting of pixels that are either over or under exposed. This function operates on the displayed pixels. If a raw file is being displayed, the embedded jpeg data is being used. If the image is zoomed, the zoom type affects the results. Two-pass rendering also has some influence.

	If a pixel has any channel at zero or full-scale, the pixel is displayed in red.
Keywords	Toggles display of the Keywords sidebar.
Sort manager	Toggles display of the Sort Manager sidebar.
Hide Bars and Files (`)	This item is intended to be used by its shortcut key. When pressed, the info sidebar, sort manager, file list, menu and status bars will be hidden (if displayed). Pressing the shortcut key again will revert the display to its previous layout.
Hide Selectable Bars (**Control+`**)	This item is intended to be used by its shortcut key. When pressed, bars selected in Preferences/General will be hidden. Pressing shortcut key again will revert the display to its previous layout.
Toggle slideshow	Begins or ends a slide show, the slide show status is displayed in the status bar.
Pause slideshow	Pauses a slide show, the slide show status is displayed in the status bar.
Faster	Decreases slide show interval in units of 0.5 seconds.
Slower	Increases slide show interval in units of 0.5 seconds.
Refresh	Updates the contents of the folder and file lists to match what is currently on disk, the current image will also be reread from disk.

Windows menu

This menu permits you to define several different window layouts that can be recalled at any time.

New window → default (**Ctrl+N**)	Opens a new window with the default layout. The window is given a name in the form "lw<n>".
New window → from current	Opens a new window with the layout of the window that has focus. The window is given a name in the form "lw<n>".
Rename window	If a window is given a name that does not begin with the characters "lw", the layout will be saved and be available for loading at a later time..
Delete window	Delete a saved window layout from the list.
Close window	Closes the current window. Closing the last main window quits the application instance.

Help menu

Contents (**F1**)	Opens the Geeqie user manual in a new browser window.
Search and Run Commands (/)	Opens a text box in which you may type characters to find a command. The characters you type will be matched to the first characters of any word in all the commands in Geeqie - both built-in commands and plugins - and matching actions will be

	displayed in a pop-up. You may scroll to any of the matching commands and press `Enter` to execute the command. Pressing `Enter` without selecting from the list will execute the first action in the list. Press `Esc` to close the box.
On-line help search	Use a web browser to search Geeqie's on-line help files. The search engine used is defined in Preferences General .
Keyboard shortcuts	Displays a window listing the default keyboard shortcuts.
Keyboard Map	Displays an image showing the currently allocated keyboard shortcuts.
Release notes	Displays a window containing the release notes for the version of Geeqie in use.
About	Opens a window containing the Geeqie version, credits, website, and email contact information.
Log window	Opens the log window, mostly used for debugging purposes.

Tear off menus

Menus can be permanently displayed by selecting the dotted line item at the top of each menu category. A tear off menu will behave the same as an ordinary window. To close a tear off menu, select the dotted line item at the top of the menu.

Changing the main window menu bar keyboard shortcuts

The section Keyboard Options describes how to modify the keyboard shortcuts used in the main menu.

Status Bar

The status bar contains information and details about the contents of the main window. Information displayed includes the progress of generating thumbnails, the number of files in the file list with byte size, the current file selection count including byte size, details about the currently displayed image, and the zoom and image scroll settings. The current sort preference is also displayed, and can be adjusted from the status bar.

The status bar is comprised of five sections, from left to right these are a progress bar, sort method, file list counts, image details, and zoom and scroll setting. Each section is described below.

Progress Bar

The Progress bar updates to display the current state of thumbnail generation, or the reading of metadata in the current folder.

When “Loading thumbs...” is displayed, thumbnails are currently being generated when Geeqie is idle; the progress bar will update to display the percentage of thumbnails that are completed.

When “Loading meta...” is displayed, certain metadata is being loaded when Geeqie is idle; the progress bar will update to display the percentage of files that have been read. Refer to Preferences metadata.

Sort method

This is the method used to sort image names in the file list. Clicking or activating this field will display a menu to allow selection of the sort method.

Name	Images are sorted by file name.
Number	Images are sorted by natural numerical order, such that, for example, 9_name.jpg will be listed before 10_name.jpg.
Date	Images are sorted by file modification date.
File Date	Images are sorted by one of four types of file date. They are described in the Reference section .
Size	Images are sorted by file size on disk.
Rating	Image are sorted by Xmp.xmp.Rating.
Class	Image are sorted by class. The class types, and sort order, is: • Unknown • Image • Raw Image • Video • Metadata
Ascending	Toggles between increasing and decreasing sort order. A check will appear next to this entry to indicate ascending sort order.

Note

When images have equal rank, for example in rating or class sorts, within each section images will be sorted by filename.

File list

This section displays the count of files that appear in the file list, including the total byte size. The number of files selected is displayed in parenthesis, including the total byte size of the selected files. The slide show status is also displayed here; when a slide show is running, “Slideshow” will appear, the text will change to “Paused” when a slide show is in the paused state.

Image Details

This section displays the dimensions (width x height) and file byte size of the image that is active in the image pane. When the format of the file in the image pane can not be determined the dimensions will show as “(0 x 0)”, in addition “(no read permission)” may appear if the file permissions do not allow reading the contents of the file.

Zoom and Scroll

The button label displays the current zoom ratio. A ratio of 1:1 is the image's original size. When the left number is larger the image is displayed larger than original size, when the right number is larger the image is displayed smaller.

A tilde (~) appears within the ratio display when the zoom is set to fit the image within the display area. In this zoom mode the ratio is automatically adjusted, and the displayed ratio may not be the actual ratio because the status bar display rounds the actual value to the nearest tenth (0.1).

Clicking this button permits control of the behavior of the zoom and scroll settings used when changing the displayed image.

Zoom to original size	The new image is set to its original size.
Fit image to window	The new image's zoom is changed so that the image will fit within the current view area.
Leave zoom at previous setting	The zoom setting is unchanged, the new image will be scaled the same as the previous image.
Scroll to top left corner	The new image is displayed from top left corner.
Scroll to image center	The new image is centered
Keep the region from previous image	The new image is positioned as the previous one, whenever possible.

Buttons

The Status bar toolbar buttons, the defaults are shown below, correspond to selected menu action. The buttons displayed may be changed in the Toolbar tab of Preferences.

Exif rotate

Toggles the auto-rotation of images if they have Exif orientation data. Refer also to the FAQ .

Show Pixel Info

Shows coordinates and RGB[A] value under cursor.

Enable Color Management

Enables color profiles.

Save Metadata

Opens dialog for review and saving of metadata changes.

Layout

The main window consists of four areas, the folder pane , file pane , image pane , and the status bar . The status bar is always at the bottom of the window. The three remaining areas, or panes, can be arranged and sized by the user.

Selecting the location of each pane

The arrangement of the folder, file, and image pane is selected on the Window Layout dialogue.

Changing the size of a pane

Between the panes of the main window are two dividers that can be moved with the mouse to adjust the size of each pane. Press and hold the primary button, then move the divider to the desired location and release the button.

Folder pane

The folder pane contains the menu bar, tool bar, path entry, and folder list. The list can be set to display the sub folders of the active folder or a tree that starts with the root folder and contains all folders in the file system.

File pane

The file pane lists the images contained in the active folder. The file pane can be set to display as a vertical list or an icon view.

Image pane

The image pane is where the active image is displayed.

Chapter 3. Sidebars

A sidebar is a panel that appears next to an Image Pane , and displays additional information about the image.

Info Sidebar

The Info Sidebar shows various information about the image and can also be used for metadata editing.

To toggle display of the Sort Manager select View → Info Sidebar (**Ctrl+K**) .

It consists of several panes. Left-clicking on the pane title expands or collapses the pane. Right-clicking on the pane title or on empty space opens a context menu allowing you to add and remove panes, or move them up and down.

Adding a metadata key pane

Select Metadata key… from the Add menu to create a pane for an individual metadata field. Enter the fully qualified XMP, Exif, or IPTC key and, optionally, a pane title. If the title is left empty, Geeqie uses the description of the metadata key when one is available.

Example keys include `Xmp.dc.title`, `Exif.Photo.DateTimeOriginal`, and `Iptc.Application2.Headline`. The OK button is enabled only when the key begins with `Xmp.`, `Exif.`, or `Iptc.`.

Note

XMP fields in these panes are editable. Exif and IPTC fields are displayed read-only.

The new pane is saved as part of the current layout configuration.

Histogram

This histogram is identical to that shown in the Image Overlay . It can be configured by right-click to modify the display of color or linear/logarithmic scale.

Title

>

Note

Title text is stored in the *Xmp.dc.title* tag.

Description

Note

Description text is stored in the *Xmp.dc.description* tag.

Keywords

Keywords, together with a flexible search tool, are essential for handling large image collections.

Geeqie has free-form keywords and a hierarchical method of adding new keywords to an image.

Note

Keywords are stored in the *Xmp.dc.subject* tag.

Note

Keywords are not case-sensitive unless selected as so in Preferences

The Keyword pane in the sidebar consists of two sections:

- The left hand section which is for free-form keywords
- The right hand section which is a hierarchical structure, allowing sets of keywords to be added.

The contents of the left-hand box is the data written to file.

The right-hand section has two types of entries :

- Keywords
- Helpers

An example of keyword hierarchy is:

```
[ ] animal
[ ]   mammal
[ ]     dog
[ ]     cat
[ ]   insect
[ ]     fly
[ ]     dragonfly
daytime
[ ]   morning
[ ]   afternoon
```

In this example *daytime* is a Helper, and all the others are keywords. Helpers are not stored in metadata - their purpose is only as an organisational aide.

In the above example if *cat* is clicked, both *mammal* and *animal* will also automatically be checked, and all three transferred to the left hand box, ready for writing as metadata.

If a hierarchy of keywords are checked, and the top-level item then un-checked, all lower level keywords will also be unchecked and will be removed from the left-hand pane.

Warning

This action cannot be undone.

Irrespective of their position in a hierarchy in the right-hand box, all keywords are written as individual entries into metadata. When metadata for an image is read in, Geeqie will attempt to recreate any hierarchies within the data, and display it accordingly in the right-hand box.

Right-click on the left-hand box allows the selected keywords to be written to the currently selected files - the keywords may be either added to the existing contents, or entirely replace the existing content.

Warning

Keyword completion is used when selecting keywords. If the selected area includes part of a keyword, the selection will automatically be expanded to cover the whole keyword. If no text is selected, all keywords will be used.

Right-click on the right-hand box allows a connection to be made between a keyword and a mark. The Marking Images section shows how marks and keywords inter-operate.

The ordering of keywords in the right-hand box may be changed by drag-and-drop.

Changes to the keyword hierarchy are automatically saved in `geeqierc.xml` about two seconds after the first edit in a group of changes. This also saves the current application settings. The hierarchy is saved again when Geeqie exits normally. A save failure is reported through a notification and the log window.

The meta-data write operation for a file is triggered either manually or as defined in Metadata .

If the right-click menu item Revert is executed at any time after a Show all or Collapse all , the keyword layout will be restored to the state prior to the Show all or Collapse all .

If the right-click menu item Revert is executed at any time before a Show all or Collapse all , the keyword layout will be restored to the state existing at start-up.

Note

The selections in the sub-menu On any change will affect the operation of the Revert option.

Keyword Autocompletion

The text box at the bottom of the keywords pane is used for autocompletion. Any keywords typed into the standard keyword box or the autocompletion box will be remembered as candidates for future autocompletion.

Frequently used sets of keywords can be entered as comma-delimited lists.

The list of keywords used for autocompletion can be edited on the Keywords tab of the Preferences dialog.

The menu action "Keyword autocomplete", set to `Alt + K` by default, will shift the keyboard focus to the autocomplete box. Pressing `Alt + K` a second time will shift the keyboard focus back to the previous object.

List panes

There are five list panes, whose contents can be configured to display Exif, XMP or IPTC metadata tags. The five panes are:

- Title
- Exif
- File Info
- Location and GPS
- Copyright

Any metadata tag can be displayed in any pane - the pane titles merely exist as an organisational aide.

It is possible to reorganise the data displayed by drag-and-dropping any item within a pane, or from one pane to another (although in some cases it makes little sense to do so).

Note

It is not possible to undo these actions, nor is there a way to reset the display to its original settings.

It is also possible to drag new items from the Exif Window onto any pane.

As an aide, in addition to standard metadata tags, Geeqie can call lua scripts and also provides certain pre-formatted tags. These tags are:

Table 3.1. Pre-formatted tags

Tag	Replaced by Meta data
formatted.Camera	Camera and software name
formatted.DateTime	Image Date
formatted.MediaDate	Media date, with Exif, QuickTime and file modification date fallbacks
formatted.ShutterSpeed	Shutter speed in seconds (e.g. 1/60s)
formatted.Aperture	Aperture (e.g. f5.6)
formatted.ExposureBias	Exposure bias (e.g. +0.3)
formatted.ISOSpeedRating	ISO sensitivity (e.g. 100)
formatted.FocalLength	Focal length
formatted.FocalLength35mmFilm	Focal length 35mm
formatted.SubjectDistance	Subject distance
formatted.Flash	Flash
formatted.Resolution	Resolution
formatted.ColorProfile	Color profile
formatted.GPSAltitude	Altitude above sea level
formatted.GPSPosition	Latitude, Longitude
formatted.localtime	Local time corrected for UTC offset, DST correction [a]
formatted.timezone	Timezone indicated by GPS lat/long values [b]
formatted.countryname	ISO 3166 country name indicated by GPS lat/long values
formatted.countrycode	ISO 3166 two-letter abbreviated country name indicated by GPS lat/long values
file.size	file size in bytes
file.date	file date and time in human readable form
file.mode	file mode flags
file.ctime	refer to operating system documentation for the meaning of ctime
file.owner	the file's owner. Refer to operating system documentation for the meaning of file permissions
file.group	the file's group. Refer to operating system documentation for the meaning of file permissions
file.class	the file's class e.g. Image, RAW image etc.
file.link	if the file is a symbolic link, shows the path of the source file
file.page_no	if the file is a multi-page file, e.g. .tiff, the current page number and page total is displayed (e.g. [2/10])

lua.lensID	lens identification data [c]
lua.<lua script>	Call a lua script [d]

[a] Refer to UTC and DST section.

[b] Refer to UTC and DST section.

[c] Requires lua and exiftool.

Note

This function may be slow.

[d] The output is expected to be text.

Right-clicking on an item opens the context menu:

Add entry	Key	Exif, IPTC, XMP or Geeqie formatted exif tags.
	Title	Displayed title text.
	Show only if set	Inhibit display if the tag is empty.
	Editable	Displays the value as editable.

Note

Geeqie internally converts Exif and IPTC metadata to XMP on read and back on write, thus any changes made to Exif and IPTC here will be lost on write. You must edit the corresponding XMP instead.

An exception is Makernote tags which don't have an equivalent in XMP. However Makernotes cannot be written to an external file (a sidecar or Geeqie metadata file).

Show hidden	Shows all entries regardless of *Show only if set* .

Map Display

Geeqie can display maps created by the openstreetmap.org [https://www.openstreetmap.org] . To use this facility, Geeqie must have been compiled with the --enable-map option.

- Image location display

  Selected geocoded images will show as small icons on the map. Clicking an icon expands or collapses its image thumbnail and details. Clicking the Show image button in the details or Ctrl-clicking an icon displays that image in the main window while preserving the selection.

  Right-click on the map will show other map options.

  Middle-click controls the map-centering function.
- Geo-coded search

  Left-click stores the latitude/longitude under the cursor into the clipboard. This may be used to define the origin of a geocode search .
- Geo-coding Images

  If an image is not geocoded, the filename or icon on the file pane can be dragged-and-dropped onto the map. The image latitude and longitude xmp meta-data will be updated to the drop position on the map.
- Map Centering

  If an internet URL containing a valid latitude and longitude is dropped on the map, the map will be re-centered on that location. The zoom level will not change. Refer to Decoding Latitude and Longitude .

Sort Manager Sidebar

The Sort Manager sidebar provides a convenient method to quickly sort many files into a set of folders or collections.

Drag the divider beside the Sort Manager to adjust its width. The chosen width is retained when the sidebar is hidden or the window is resized, and is saved with the window layout. Enable Remember session in Window Options to restore it at startup. The Info sidebar has its own divider.

To toggle display of the Sort Manager select View → Sort Manager (**Shift**+**S**) .

The mode of the Sort Manager is selected by the drop down list under the sidebar title. The Folder mode allows moving or copying the visible image to a folder, and the Collection mode allows adding the visible image or file selection to a collection.

Using destination shortcuts

The first nine destinations are numbered in their displayed order. Assign shortcuts to Sort Manager: Process using destination 1 through Sort Manager: Process using destination 9 in Keyboard preferences to process images directly using those entries. No shortcuts are assigned by default. These actions use the current mode and operation, highlight the chosen destination, and work only while the Sort Manager is visible. They do nothing when that numbered entry is absent or its destination no longer exists.

Numbers follow the currently displayed list. Reordering destinations, switching between Folders and Collections, or navigating to a source folder with its own destinations can change the target of a shortcut. Leave Use destinations for this folder disabled if you want to use the shared folder list

everywhere. The highlighted-destination action remains available for repeating an operation and for destinations beyond the first nine.

Hold **Ctrl** and click a folder or collection to select it as the destination without processing any images. The destination remains highlighted when focus returns to the file list. Folder and collection destinations are remembered separately when switching modes and across restarts.

To process images without clicking the destination again, assign a keyboard shortcut to Sort Manager: Process using highlighted destination in Edit → Preferences → Keyboard. No shortcut is assigned by default. This action is available while the Sort Manager is visible.

In Folders mode, the shortcut applies the selected Copy, Move, or Desktop action to the selected images using the highlighted folder. In Collections mode, it adds the current image or selected images to the highlighted collection, according to the selection style. Clicking a destination normally both highlights it and performs the operation.

The Help button opens this Sort Manager help page.

Folder view

The Sort Manager folder view consists of a vertical list for bookmarking folders, a section to select the operation to perform, and a set of buttons along the bottom.

Clicking or activating an item in the list will copy, move, or link the active image's file to that location. The active image will then change to the next image.

Bookmark list

This is a list of possible folders to use as a destination for the selected operation. A context menu will appear by right clicking on an item, or pressing the menu key when the item has the focus.

Folders can be added to the list by activating the Add button at the bottom of the sidebar, or dragging a folder onto the list.

By default, folder destinations are shared across source folders and Geeqie windows. Enable Use destinations for this folder to use a separate destination list for the currently displayed source folder. The list starts empty; use Add or drag destination folders onto it.

For example, configure Nature, Sports, and People as destinations while viewing Downloads, then configure Cycling, Skiing, and Swimming while viewing Sports. Geeqie switches lists when you navigate. The setting applies to that exact source folder, without inheritance by subfolders. Windows displaying the same source folder share its list.

Turning the option off restores the shared list and retains the folder's separate list for later use. Switching lists clears the highlighted destination, so select a destination before using the keyboard action to process images. Folder associations and destination lists are saved in Geeqie's configuration; no configuration files are created in image folders. Renaming or moving a source folder does not transfer its association to the new path. Collections mode is unaffected.

To edit an item select **Properties** from its context menu. A dialog will appear allowing the Name, Path, and Icon for the item to be changed. Select OK to close the dialog and apply the changes, or Cancel to close the dialog and discard the changes.

To change the order of the list, select **Move up** or **Move down** from the item's context menu. To move an item with the keyboard move the focus to that item, then press **Shift**+**Up** or **Shift**+**Down** to move the item.

To remove an item from the list select **Remove** from it's context menu.

Copy, Move, Link and Desktop actions

Clicking a folder sets and highlights the destination and processes the selected images. Hold **Ctrl** while clicking to set the destination only. The Sort Manager: Process using highlighted destination action repeats the selected operation using that folder. A shortcut can be assigned in Keyboard preferences.

Plugins whose desktop files contain `X-Geeqie-Filter=true` also appear in the folder action list. Selecting a plugin and clicking a destination passes that folder to the plugin. See Geeqie desktop file keys for the filter key and how a plugin retrieves its destination.

Add button

Activate this button to add a folder to the bookmark list. A dialog will appear to allow selection of the folder to use for the new item.

Undo button

Activate this button to undo the last copy, move, or link operation. The undo function can only reverse the most recent operation.

Collection view

The Sort Manager collection view consists of a vertical list of collection, a section to determine the current selection, and a set of buttons along the bottom.

Clicking or activating an item in the list sets it as the destination and adds images according to the selection style below. Hold **Ctrl** while clicking to set the destination without adding images. The destination remains highlighted when focus returns to the files pane. The highlighted folder and collection destinations are remembered separately across restarts.

The Sort Manager: Process using highlighted destination action adds images to the highlighted collection according to the selection style below. It works while the Sort Manager is visible. In Folders mode, it processes the selected images using the highlighted folder and the selected Copy, Move, or Desktop action. No shortcut is assigned by default; one can be assigned in Keyboard preferences.

Collection list

This is the list of collections located in the folder listed in Configuration Files , together with collections added from other folders. Activating a member of this list will result in the current image selection to be added to the collection. If the image already exists within the collection, it will not be added a second time.

An existing collection or a new empty collection can be added to the list by activating the Add button at the bottom of the sidebar.

Choosing the selection style

The images to add when activating a collection is determined by the selection style:

Add current image	The visible image is added to the collection.
Add selected images	All images selected in the file list will be added to the collection.

Add button

Activate this button to choose an existing .gqv collection or enter a new .gqv filename. Existing collections are added without changing their contents; a new filename creates an empty collection. Collections outside the default collections folder are remembered and remain available in the list. The added collection becomes the highlighted destination.

Undo button

Activate this button to undo the last add operation. The undo function can only reverse the most recent operation.

Chapter 4. Stand-alone Windows

This section describes the windows that may be displayed simultaneously with the Geeqie main window. They are initiated via the **View** and **File** menus.

Image Window

An image window is a window that displays only the image, without menus or sidebars.

An image window can always be opened from the context menu of all images and files. Keyboard shortcuts exist for most windows that display images or files, however the shortcut can vary dependent on the type of window containing them.

The image pane contained in an image window behaves and operates similar to the Image Pane of a Main Window, including the Image Overlay . The keyboard shortcuts are the same, and the context menu is the same except *Hide file list* is replaced by *Close window* . The slide show and full screen functions are also available.

Navigation

When the image window is initiated from an image selection or from a collection in the files pane, the displayed image can be changed forward or backward in the list with the respective **PageDown** and **PageUp** keys. In addition, all navigation methods for the mouse and keyboard are similar to that of the Image Pane of a Main Window.

Drag and Drop

Images can be dragged and dropped onto an image window, but the process is somewhat different to that of an Image Pane . If multiple images, a folder or an image from a collection in the files pane is dropped onto the image window, you can cycle through all these images using the navigation keys.

The dropped item can be one or more files, folders, or an image from a collection in the files pane:

One or more files	The image window will display the first image; the list of dropped images can be navigated similar to a main window's image pane.
One or more folders	When the dropped list of items includes a folder, a menu will appear allowing to skip the folders, add their contents, or add their contents recursively. After choosing from the menu the requested files will be added to the image window.
Image from a collection	The image window will change to display the image and become linked to the collection containing the image. Images in the collection can then be navigated and viewed from the image window.

Changing the image window by dropping a new item onto it will cause the previous list of images or linked collection to be discarded.

To update the image display in the event that the source file has changed, press **R** .

Exif Window

The Exif window shows the Exif, IPTC and XMP metadata that are present in the image file.

Unlike the other parts of the program, this window shows the "raw" metadata read from the file, without any post-processing. Thus, this window is ideal for metadata debugging.

The Exif pane in the Info Sidebar is more suitable for general use, as it displays merged data from sidecar and private metadata files.

To facilitate including extra items in the Exif pane in the Info Sidebar , it is possible to drag-and-drop an item from the Exif window onto the Sidebar pane.

A search of the Exif data can be initiated by clicking anywhere on the column you wish to search, and then type `Ctrl + F` or press any alpha-numeric key. The search is case-insensitive

Pan View

This window offers alternative views of the *current directory and its children* .

It may be opened by View $\rightarrow$ Pan view (**Ctrl+J**) . You can pan the view as you pan an image in normal view mode, using left mouse button and drag.

A primary mouse button click on any image will display information about the image. Secondary mouse button will show a context menu.

The Keyboard Shortcuts available are listed in the Reference section.

Note

Pan view recursively visits each folder under the specified folder. This can consume considerable computer resources.

Enable thumbnail caching to reuse thumbnails in Pan View. Both Geeqie's own cache and the shared thumbnail cache are supported. With Geeqie's own cache, new thumbnails are saved at the size configured in Preferences and scaled to the selected Pan View size. Changing the Pan View size does not regenerate valid cached thumbnails.

Select a folder

Location	The initial location is the current Geeqie folder when the Pan view window is opened.

Choose a view mode

A select box let you choose among different view modes:

Timeline	Shows images grouped by date.
Calendar	Shows a calendar with the number of images indicated for each day (as dots and number). Left mouse button click will show images in a pop-up.
Folders	Show images grouped by folder. Each box represents a folder. Parent and children folders are shown as boxes enclosing other boxes.
Folders (flower)	Show images grouped by folder. Each box represents a folder. Parent folders are linked to children by a line. The top folder is the center of the flower.

Grid	All images are shown on a grid.

Timeline and Calendar normally use the file modification date. To group images by the time they were photographed, right-click in Pan View and enable Sort by original capture date. This uses `Exif.Photo.DateTimeOriginal`; images without a usable original capture date retain their file modification date. The setting is remembered when Pan View is closed.

How each image should be represented

Another select box let you choose the representation of each image:

Dots	Each image is represented by a dot.
No images	Each image is represented by a square box.
Small thumbnails	Each image is represented by a small thumbnail.
Normal thumbnails	Each image is represented by a normal thumbnail.
Large thumbnails	Each image is represented by a large thumbnail.
1:10 (10%)	Each image is represented by reduced image (1/10 of the original's dimensions).
1:4 (25%)	Each image is represented by reduced image (1/4 of the original's dimensions).
1:3 (33%)	Each image is represented by reduced image (1/3 of the original's dimensions).
1:2 (50%)	Each image is represented by reduced image (1/2 of the original's dimensions).
1:1 (100%)	The original image is shown at its normal dimensions.

Find

At the bottom of the window:

Find button	Shows a text field for image search. One can enter either: • filename or part filename • file date in any of the following forms, where the separator may one of `/-,.` or a space: • yyyy-mm-dd • yyyy-mm • yyyy **Enter** will start the search, pressing it again will move to the next match. The matched file is made visible and details are shown for it.

Chapter 5. Search and Select

Geeqie provides comprehensive set of search methods, including not only the standard functions such as file date or size, but also image similarity.

There is a duplicate file finder, which uses search parameters such as image dimensions or checksum. Two sets of image files may be compared.

Geeqie also has a set of temporary "marks" or "tags" which may be allocated to any image as an aide to file organising.

Caching Data For Searches

Searching large numbers of files for similarity can take significant cpu time. Part of this is the time to compute the similarity matrix for each file.

If the Cache Thumbnails option is selected in Preferences/General, the similarity matrix, checksum, image size and Exif original date may also be cached. If Perceptual hash algorithm is selected in the Duplicates window, all eight orientation hashes are also cached. This will reduce the time needed for future searches.

> **Note**
> If you frequently search on similarity and your images are in a tree arrangement under a single point, the Create file similarity cache from the top of the tree will generate the similarity data for all images.

Similarity data are stored in a folder hierarchy that mirrors the location of the source images. The data have the same name as the original appended by the file extension .sim.

The root of the hierarchy is:

```
$XDG_CACHE_HOME/geeqie/thumbnails/
```

or, if $XDG_CACHE_HOME is not defined:

```
$HOME/.cache/geeqie/thumbnails/
```

Search Window

Geeqie provides a utility to find one or more images based on various parameters.

To display a new Search Window press **F3** or select Search from the File menu.

To start a search activate the `Find` button or press `Ctrl + Enter`. A search in progress can be stopped by activating the `Stop` button or press `Ctrl + Enter`.

The progress of an active search is displayed as a progress bar at the bottom of the window. The progress bar will also display the total files that match the search parameters, and the total number of files searched.

When a search is completed, the total number of files found and their total size will be displayed in the status bar.

Search location

One of several locations can be chosen as the source to use in the search.

Path
: The search will include files located in the selected folder, enable the Recurse check box to include the contents of all sub folders.

Descriptions
: This option is only for use with GqView legacy metadata.

  The search will include all files that have a keyword or description associated to it.

  Note
  Only keyword and description associations stored in the user's home folder are included in this search type.

Results
: The search will include all files currently listed in the results list. Use this to refine a previous search.

Collection
: The search will include all files in the selected collection.

Search parameters

Each search parameter can be enabled or disabled with the check box to its left. For a file to be a match, all enabled parameters must be true.

File
: If "name is" is selected, the search will match if the text exactly matches the file name. If "name contains" or "path contains" is selected, the search will match if the entered text appears within the file name or path.

  The text comparison can be made to be case sensitive by enabling the Match case checkbox.

  The search can be restricted to symbolic links by enabling the Symbolic Link checkbox.

  If "contains" or "path contains" is selected, Perl Compatible Regular Expressions are used.

File size
: The search will match if the file size on disk is equal to, less than, greater than, or between the entered value, depending on the method selected from the drop down menu. The *between* test is inclusive - for example a file of size 10 will match if the size parameters are between 10 and 15.

File date
: The search will match if the file date is equal to, before, after, or between the entered date, depending on the method selected from the drop down menu. The *between* test is inclusive, for example a file with date of 10/04/2003 will match if the date parameters are between 10/04/2003 and 12/31/2003.

  For convenience, the button displays a pop up calendar to enter the date.

  One of four date types may be selected. They are described in the Reference section .

  Note
  If an image does not have an exif date, it will default to 01 January 1970.

Image dimensions

The search will match if the image dimensions are equal to, less than, greater than, or between the entered values, depending on the method selected from the drop down menu. The *between* test is inclusive.

The image dimensions test is simple, both width and height must be within the allowed values for a match.

Image content

The search will match if the image contents are similar within the percentage value, inclusive. This uses the same test and data that is used to determine image similarity when Finding Duplicates . The entry is for entering the path for the image to use in this test. Choose Standard similarity algorithm or Perceptual hash algorithm from the drop down menu. The selection applies to this search window. Similarity percentages differ between algorithms; see Similarity Algorithms for threshold details.

Keywords

The search will match if the file's associated keywords match all, match any, or exclude the entered keywords, depending on the method selected from the drop down menu. Keywords can be separated with a space, comma, or tab character.

Description

The search will match if the file's Description field contains the entered pattern. Perl Compatible Regular Expressions are used.

Exif

The search will match if the file's Exif Tag contains the pattern entered in the Value box. Perl Compatible Regular Expressions are used.

> **Note**
>
> The value checked will be against the formatted value e.g. for focal length search for 67.5 mm and not 675/10. See also pre-formatted tags

Geocoded position

The search will match if the file's GPS position is less than or greater than the selected distance from the specified position, or is not geocoded, depending on the method selected from the drop down menu. The search location can be specified by

- Type in a latitude/longitude in the format `89.123 179.123`
- Drag-and-drop a geocoded image onto the search box
- If Geeqie's map is displayed, a left-click on the map will store the latitude/longitude under the mouse cursor into the clipboard. It can then be pasted into the search box.
- Copy-and-paste (in some circumstances drag-and-drop) the result of an Internet search.

> **Note**
>
> In this last case, the result of a search may contain the latitude/longitude embedded in the URL. This may be automatically decoded with the help of an external file:- `$HOME/.config/geeqie/geocode-parameters.awk` See Decoding Latitude and Longitude for details on how to create this file.

Class — The search will match if the file's class is, or is not, one of the following types.

- Unknown
- Image
- Raw Image
- Video
- Document
- Metadata
- Unknown
- Broken (Of class Image, Raw Image, Video, or Document, but Geeqie cannot display an image)

Marks — The search will match if the file does or does not have a mark attached to it. Refer to Marking Images .

Results list

Files that match the parameters of the search are shown in the list. By default they are displayed in the order that they are found. To sort the results list by a column, activate or click the column title. Activating a column that is already used to sort the list will toggle between increasing and decreasing sort order.

A **context menu** is available for the result list by right clicking the mouse or pressing the Menu key when a row has the focus.

The Keyboard Shortcuts available are listed here.

The image Dimensions column of the result list will only contain dimension information when dimensions are included in the search parameters.

Status bar

At the bottom of the search window is an area that includes the following items from left to right:

Thumbnails — Enable this check box to display a thumbnail next to each image in the results list.

File count display — Displays the total count of files in the results list, and their size on disk. The count of selected files will appear in parenthesis.

Search progress — When a search is active, the text “Searching...” will appear here. Two numbers in parenthesis will also be displayed, the first number is the number of files that match the search parameters and the second number is the total number of files that were searched.

`Find` — Activate the find button to start a search with the selected parameters.

`Stop`	Activate the stop button to end a search that is in progress.

Drag and Drop

Drag and drop can be initiated with the primary or middle mouse buttons. Dragging a file that is selected will include all selected files in the drag. Dragging a file that is not selected will first change the selection to the dragged file, and clear the previous selection.

Finding Duplicates

Geeqie provides a utility to find images that have similar attributes or content.

To display a new Find Duplicates Window select File → Find duplicates .

Adding files to be compared

Add files to be compared using drag and drop. Drop files or folders onto the Find Duplicates window to add them to the list of files to compare. When one or more folders are dropped onto the window a menu will appear allowing you to choose the desired action:

Add contents	The contents of dropped folders will added to the window.
Add contents recursive	The contents of dropped folders and all sub folders will be added to the window.
Skip folders	Ignore folders contained in the drop list.
	When files are added to the window, the comparison is restarted to include the new files.

Note

Symbolic links to folders will be followed, but symbolic links to files will be ignored. Only actual files will be tested.

Comparison methods

The attribute to use for two images to match can be selected with the **Compare by:** drop down menu. Each method is explained below:

Name	The file name.
Name case-insensitive	The file name but ignoring case.
Size	The file size.
Date	The file date.
Dimensions	The image dimensions.
Checksum	The MD5 file checksum.
Path	The complete path to file.
Similarity (high)	Very similar image content. Preset to 95%.
Similarity	Similar image content. Preset to 90%.

Similarity (low)	Slightly similar image content. Preset to 85%.
Similarity (custom)	The percentage value used to consider two images a match is configured in the spin box at the bottom of the window. With the standard algorithm, setting a value below 85 is not likely to be useful. When pHash is selected in the Duplicates window, percentages measure matching hash bits instead of RGB similarity; see Similarity Algorithms for threshold details.
Name $\neq$ Content	Show images with the same name but different content. **Note** Because pairs of files are being compared, when there are multiple hits care should be taken when analyzing the results.
Name case-insensitive $\neq$ Content	Show images with the same name, ignoring case, but different content. **Note** Because pairs of files are being compared, when there are multiple hits care should be taken when analyzing the results.
Show all	Do not compare. Show all images.

Results list

Files that match with the selected comparison method will appear in the list. Matching files are grouped in alternating color.

The order of the result list can be changed by clicking on the column header. This will re-order the images within each set. When comparing by image content similarity, the matching sets will be sorted by order of rank starting with the files that are most similar.

A **context menu** is available for the result list by right clicking the mouse or pressing the Menu key when a row has the focus.

Groups in the results list may be selected by using the keyboard. Refer to Keyboard Shortcuts.

The **selection** can be changed using the keyboard and mouse the same as in a file pane of the main window.

The image Dimensions column of the result list will only contain dimension information when comparing by dimensions, or when the data is easily available from memory or has been read from the cache.

Selecting duplicates by file size

The results context menu's Select group 1 duplicates command selects the first displayed file in each match group. Select group 2 duplicates selects all remaining files in each match group, including groups containing more than two files. These commands replace the current selection. Here, group 1 and group 2 refer to row positions within each displayed match group, not the two input file sets used by Compare two file sets.

To retain the largest file in each match group, click the Size column header until the largest file appears first in each group, then choose Select group 2 duplicates. All other files in every match group will be selected. Review the selection before deleting or moving files.

To retain the smallest file instead, sort with the smallest file first and use the same selection command. Sorting changes clear the selection, so sort before selecting. Size refers to bytes, not image dimensions. When sizes are equal, one file remains unselected without any quality-based preference; a larger file is not necessarily a higher-quality image.

Exporting results for scripts

After comparison finishes, use the results context menu's Export submenu to choose Export to JSON or Run script with JSON…. JSON exports all result groups, including unselected members, and records which files were selected. CSV and tab-delimited exports continue to export selected rows.

To run a script, choose an executable script with an interpreter line, such as `#!/usr/bin/env python3`. Geeqie passes one argument: the path of a temporary JSON snapshot. The script runs asynchronously. Geeqie removes the snapshot when the script exits, and reports launch failures or unsuccessful exit status. The snapshot is captured when the menu command is invoked; later selection changes do not affect it.

The command `geeqie --dupes-export-json` writes JSON from the most recently opened Find Duplicates window to standard output without changing its selection:

```
geeqie --dupes-export-json > duplicates.json
```

Check `schema` (`org.geeqie.duplicates`), `version` (`1`), and `status` before processing. A status of `ready` means comparison is complete; `incomplete` and `no_window` return an empty groups array. An empty completed search also has status `ready`.

Completed exports include stable, untranslated criterion names in `criteria`, the numeric `match_mask` from `DupeMatchType`, `two_sets`, `algorithm` (`standard` or `phash`), `threshold` in percent, `ignore_orientation`, and `ignore_same_folder`. Algorithm and threshold are null for comparisons that do not use similarity.

Each group has an export-local numeric `id`, a `representative_id`, and a `members` array. Each member includes its file URI as `id`, its `path`, source `set` (1 or 2), `selected` flag, byte `size`, modification time `mtime` in seconds since the Unix epoch, and `width` and `height` when known. Unknown dimensions are null. A file may occur in multiple groups and retains the same URI identifier.

The `similarity_to_representative` value is the numeric percentage stored for that member's match to its group representative. It is null for the representative itself and for comparisons without similarity. It does not describe every possible pair of group members. The root `score_semantics` field is `member_to_representative_percent`. JSON strings escape quotes, backslashes and control characters; numbers do not depend on locale.

For example, an executable Python script can read the snapshot and print each group's paths:

```
#!/usr/bin/env python3
import json
import sys

with open(sys.argv[1], encoding="utf-8") as source:
    results = json.load(source)
if (results["schema"], results["version"], results["status"]) != ("org.geeqie.du
    raise SystemExit("No completed duplicate results")
for group in results["groups"]:
    print(group["id"], [member["path"] for member in group["members"]])
```

Status bar

Along the bottom of the Find Duplicates window is an area that displays the count of files contained in the window, and the number of files in the result list.

The status bar will also display the status of an active compare operation using the progress bar. A compare operation involves 2 or 3 stages, depending on the type of comparison. These are the stages in order:

1. If necessary, extra data is read into memory for the comparison stage and the progress bar will indicate this stage with text such as “Reading dimensions...”, “Reading checksums...”, or “Reading similarity data...”.

2. The images are compared using the selected method, the progress bar will indicate this stage with the text “Comparing...”.

3. The results are sorted for display, the progress bar will indicate this stage with the text “Sorting...”.

   Stage 1 is only used for the Dimensions, Checksum, and Similarity compare methods.

   If the time to complete a stage will be significant, an estimated time to completion will also be displayed in the progress bar. The estimated time only refers to the current stage, other stages are not included in the estimate. The time estimate is displayed using the format MINUTES:SECONDS.

Thumbnails

Thumbnails can be displayed beside each image in the result list by enabling the Thumbnails check box.

Ignore Orientation

When selected and a similarity compare is being used, the images are checked against 4 rotations: 0°, 90°, 180°, 270°, plus flip and mirror.

Sort

The normal sort order is for groups (in the case of Similarity checks) with the highest number of near-100% matches to be at the top of the list.

If this box is checked, groups with the lowest number of matches are placed at the top of the list.

Ignore matches within the same folder

Enable this check box to compare files only when their immediate parent folders differ. This is useful when each folder contains a burst or sequence of similar images and you want to find matches across folders. Subfolders are treated as separate folders. The option applies to all comparison methods and to both single-set and two-file-set searches.

The option is off by default in each new Find Duplicates window. Changing it restarts the comparison. Each result group contains files that match its representative in another folder; two members of the same group can still belong to the same folder. A file can appear in more than one group when necessary to show cross-folder matches.

Compare two file sets

Sometimes it is useful to compare one group of files to another, different group of files. Enable this check box to compare two groups of files. When enabled, a second list will appear and files can be added to this list using the same methods for the main list.

When comparing two file sets the results list will display matches between the two lists. For each match group, the first file is always from the main group, and the remaining files are always from the second group.

Drag and Drop

Drag and drop can be initiated with the primary or middle mouse buttons. Dragging a file that is selected will include all selected files in the drag. Dragging a file that is not selected will first change the selection to the dragged file, and clear the previous selection.

Image Data Window

`Ctrl+Shift+Right Mouse click` : Use this to display a dialog containing the data stored for the clicked image file. This is usually only useful for debugging purposes.

Marking Images

Marks are intended as a temporary method of grouping and filtering sets of images.

Keywords are intended as a permanent method of identifying sets of images, and to this end it is possible to link any mark with any keyword.

The Select menu gives access to the marks operations of setting, filtering and intersection.

There are 10 individual marks, any of which can be associated with an image simply by pressing the 0 to 9 keys on the keyboard, where key 0 represents mark 10.

If the Show Marks menu has been selected, each image will have a set of 10 check-boxes displayed adjacent to it in the file pane in both icon and list mode. In addition a set of 10 check-boxes will be shown at the top of the files pane. Clicking any of these will filter the displayed list.

Moving the mouse over any of the check-boxes at the top of the files pane will show mnemonic text for that mark. The text can be modified by right-clicking on the check box.

Marks linked to keywords appear incrementally as image metadata is read. Until a file has been read, its displayed marks may be incomplete. Hiding marks stops pending mark reads. Filtering by marks and explicit mark operations resolve the required values immediately and may take time in large folders.

If the Image Overlay is being displayed, the currently set marks for the image are shown. It is not necessary to include an entry into the overlay template for this to happen.

A keyword can be associated with a single mark by right-clicking on the keyword in the sidebar panel. When a meta-data write operation for a file is triggered either manually or as defined in Metadata , the keyword data indicated by the current set of mark-to-keyword links will be written.

The associations between keywords and marks is preserved when Geeqie is shut down. The current setting of marks can also be optionally saved - the setting is in the Behavior tab of Preferences .

Chapter 6. Plugins

Geeqie provides the facility to execute external commands - any external command may be called via this facility.

Actions may be performed on a single file, or on a set of selected files.

Plugins can do anything from running simple operations like symlink to activating fully featured programs like Gimp. Nor does the action have to be performed on files - you may, for instance, configure a plugin to download images from your digital camera.

To perform these actions, Geeqie uses standard desktop files.

It is not necessary to understand the details of desktop file specifications because the Plugins Configuration dialogue presents you with an easy-to-use template. This dialogue is accessed via Edit $\rightarrow$ Configure Plugins .

Plugins will appear in the menu Plugins , unless specified otherwise in the desktop file.

Commands that are set to display their output in a window will show a dialogue when the command is executed.

The Result dialogue will display the command's text output. If the plugin command will result in multiple runs of the same command for each file, you can use the Stop button to stop Geeqie from executing the command for remaining files.

Once all commands have completed, or if the Stop button was pressed and the pending command has completed, you can use the Close button, or press `Escape` , to close the Result dialogue. The Result dialogue can not be closed while a command is still running.

The supplied standard plugins are described in the Reference section.

Chapter 7. File Management

Geeqie includes a set of utilities to copy, move, rename, and delete images. Cache maintenance utilities are also provided.

> **Note**
> The standard commands for copy, move, rename, delete and create folder can be overridden by Special Plugins

Copy and Move

To copy a file in Geeqie select File → Copy File To… (**Ctrl+C**) and to move a file select File → Move (**Ctrl+M**) . These operations also appear in the context menu of all images and files.

The Copy and Move operations and dialogs behave the same in Geeqie, with the only difference being the final result. The remaining text of this page applies equally to the Copy and Move operations, for the remainder of this page these will be referred to as a Copy/Move operation.

When copying or moving an individual symbolic link, Geeqie preserves the location it points to. Relative links remain relative, with their target path adjusted for the destination folder; absolute links retain their target path. A link to a missing target is preserved as a link. Renaming a link within the same folder retains its original target text. When transferring a whole folder, link target text inside the folder is preserved so that links between files in the transferred tree remain relative to that tree.

Initiating a Copy/Move operation will display a dialog to select the destination folder to Copy/Move the selected file, after the destination folder is selected Geeqie will proceed to Copy/Move the images to the destination. If an error occurs or a file exists with the same name as one of the files being Copy/Moved, a dialog will appear prompting how to handle the situation.

Copy and clipboard commands

Geeqie menus contain several commands whose names include “Copy”. They copy different kinds of data and should not be confused with one another:

Copy File To…	Opens Geeqie's file-operation dialog and copies the selected files to a destination folder. This performs the filesystem operation immediately after the destination and any conflict options are confirmed. It does not use the clipboard. Associated sidecar files are copied with the main file and retain the corresponding destination names, preserving the group. The copy dialog lists these files in its Sidecars column. If only part of an expanded group is explicitly selected, however, the selected members are treated as independent files; unselected members of the group are not copied.
Copy to clipboard	Places the selected file paths on the clipboard as shell-quoted text. Geeqie also supplies file-manager clipboard data marked as a copy operation, so pasting into a compatible file manager copies the files. Pasting into a text editor inserts the paths.
Copy to clipboard (unquoted)	Provides the same file-copy clipboard data, but its plain-text representation contains paths without shell quoting. Paths containing spaces or shell metacharacters may therefore be ambiguous when pasted into a command line.

Cut File to Clipboard	Places the selected paths on the clipboard and marks the file-manager data as a cut operation. No file is moved when this command is selected. A compatible file manager moves the files when the clipboard is pasted.
Copy Image	Places the decoded image pixels on the clipboard for pasting into an image editor or another application that accepts image data. It does not copy the original encoded file, file metadata, or sidecar files. In the file-pane context menu it applies to the file that was right-clicked.

The clipboard used for path and file commands is controlled by the Copy path clipboard selection preference. Copy Image always uses the standard clipboard.

The Copy/Move dialog

This is the dialog that is displayed when initiating a Copy/Move operation. The dialog contains a row of buttons near the top, vertical list of bookmarks along the left side, a vertical list of folders in the current path, a text entry field, and a row of buttons along the bottom.

Home button	Activate this button to change the active path to your home folder.
New Folder button	Activate this to create a new empty folder as a sub folder of the active path. The new folder will appear in the folder list ready for you to type in the desired name.
Show Hidden check box	Enabling this check box will cause hidden folders to appear in the folder list.
Bookmark list	This list on the left side of the window allows you to save links to folders that are frequently used. Activating a bookmark will change the active path to that pointed to by the link. To add items to the bookmark list highlight a folder and press **Ctrl+B** , select “Add bookmark” from the folder's context menu, or drag a folder onto the bookmark list. The bookmark has a context menu that can be displayed by clicking the right mouse button, or pressing the Menu key when a bookmark has the focus. From the context menu you can edit a bookmark, move it up or down in the list, or remove the bookmark from the list. A bookmark can also be moved by pressing Shift + Up or Shift + Down while it has the focus. A bookmark with the name "." is treated as a special case. Each time this bookmark is displayed, its path is loaded with Geeqie's current working folder. This bookmark, like any other, can be deleted and re-created at any time.
Folder list	This is the list that displays the folders contained in the active path. To change the active path to a folder displayed in this list double click the primary mouse button or highlight the folder with the keyboard and press **Enter** or **Spacebar** . This list has a context menu that allows renaming or bookmarking the selected folder. A folder can also be dragged onto the bookmark list to add a bookmark.
Path entry	This text entry displays the active path of the dialog. The path can be directly edited using this entry. Tab completion

	is supported, and the drop down arrow will display a list of recently selected folders. **Tab** completes the path rather than moving focus. Use **Ctrl**+**Tab** to move focus forward out of the path entry, or **Shift**+**Tab** to move backward. Continue keyboard navigation to reach the dialog buttons.
Copy/Move button	The name of this button will differ depending on the selected operation, activating it will start the operation with the folder displayed in the path entry as the destination.
Cancel button	Activate this button to close the dialog, the operation will be aborted.

Press **Return** to activate the default action, either Copy, Move, or With Rename, depending on the operation and preferences. A path completion popup may use Return to accept a completion first. To choose a different action, move focus to its button and press **Space**. Press **Escape** to cancel; if a completion popup is open, dismiss it first.

The error dialog

If a problem occurs because Geeqie is unable to Copy or Move a selected file a dialog will appear to notify the user. The dialog will also ask if the operation should continue if one or more files remain to be Copied or Moved.

The overwrite dialog

If a file with the same name exists in the selected destination a dialog will appear to notify the user and ask how to resolve the conflict. The dialog will also display the source and destination images. The following possible solutions will be offered:

Auto rename	Enabling this check box to have Geeqie select a unique name for the destination file, the proposed new name will appear in the rename entry field.
Rename	Enable this check box to manually enter a new name for the file.
Overwrite	Activate this will overwrite the existing file, or rename it if one of the rename check boxes are enabled.
Overwrite all	Activating this will perform the same operation as the Yes button, however any further file name conflicts will be handled quietly in the same method as is selected for the current file.
Skip	This button will skip doing anything to the file in question, and resume the operation with the next file. The source file will remain untouched.
Skip All	This will perform the same as the Skip button, however any further file name conflicts will be quietly ignored for all remaining files.
Cancel	This will close the dialog and stop the operation, all remaining files will be ignored.

Rename

To rename a file in Geeqie select File → Rename (**Ctrl+R**) . Rename is also available in the context menu of all images and files.

When renaming a file, a dialog will display allowing entry of the new name for the file. The dialog will be different depending on the number of files selected to be renamed. A simple dialog will appear when renaming a single file, however when renaming multiple files the dialog will include a list of the file to be renamed and additional options.

Rename single file dialog

The dialog that appears when renaming a single file includes an image of the file, an entry to type the new name of the file, a Rename button, and a Cancel button. Type the new name into the text entry and press **Enter** or activate the Rename button to rename the file. Press **Escape** or activate the Cancel button to close the dialog and abort renaming the file.

When renaming a file and the new name already exists on the file system a dialog will appear showing the renamed and existing files. To overwrite the existing file press **Enter** or activate the Overwrite button. Press **Escape** or activate the Cancel button to close the dialog abort renaming the file.

Rename multiple files dialog

The dialog that appears when renaming multiple files includes the list of files to be renamed, a preview image of the active file, an entry to enter the new file name, an auto rename check box, a Rename button, and a Cancel button.

Manually enter new file names

To manually enter the new name of each file, the active file will be displayed in the image preview and it's original name will be displayed above the text entry. To rename the file type the new name and press **Enter** or activate the Rename button. When a file is successfully renamed it will be removed from the list and the next one will become the active file. After all files have been renamed, the dialog will close.

Files do not have to be renamed in the order that they appear in the list, highlighting a name in the list will make it become the active file.

When renaming a file and the new name already exists on the file system a dialog will appear showing the renamed and existing files. To overwrite the existing file press **Enter** or activate the Overwrite button. Press **Escape** or activate the Cancel button to return to the original rename dialog, the file that prompted the overwrite dialog will remain active.

Automatic renaming of multiple files

To automatically rename the files in the list, enable the Auto rename check box. The dialog will change to batch mode for numerical naming of files.

The file name list will change to include a preview column that will show the resulting name of each file. The text entry will change to three entries; the first represents the text that appears in the new names before the number, the second will be the start number to use when incrementing the file number, and the third represents the text to append to the name after the

number. The end text should include the file name extension, therefore batch renaming multiple types of files (those with different extensions) all at once is not recommended.

The padding number can be used to change the minimum length of characters to use for the number portion of the new name. When the file number results in fewer characters than the padding size, the result will be padded with leading zeros to fill the remaining space. For example if padding size is 3, a file with the number 24 will appear in the file name as “024”.

The order of the file list can be changed by dragging a name to the desired location within the list. The preview column will update to reflect the new order.

When the preview of new names is satisfactory, press **Enter** or activate the Rename button to rename the files to match the name in the preview column. Press **Escape** or activate the Cancel button to close the dialog and abort the rename operation.

If the resulting new names match any existing files, a warning dialog will notify you. The rename dialog will remain so that you can select a different set of names.

Formatted renaming of multiple files

Formatted rename permits file to be renamed in a pre-determined way whilst still including the original file name.

In the Format box any alpha-numerics can be entered, but the special character ***** will be replaced by the current filename. One or more **#** characters can be included, and will be replaced by an incrementing number.

The current file extension is automatically appended to new new file name.

The order of the file list can be changed by dragging a name to the desired location within the list. The preview column will update to reflect the new order.

When the preview of new names is satisfactory, press Enter or activate the Rename button to rename the files to match the name in the preview column. Press Escape or activate the Cancel button to close the dialog and abort the rename operation.

If the resulting new names match any existing files, a warning dialog will notify you. The rename dialog will remain so that you can select a different set of names.

Delete

To delete a file in Geeqie select File → Delete (**Ctrl+D**) . Delete is also available in the context menu of all images and files. If Enable delete key is enabled in Options, pressing the **Delete** key can also be used to delete files.

The behavior when starting a delete operation depends on the Confirm file delete option in Options. If the option is enabled a dialog will appear to confirm the operation, if it is not enabled the files will be

deleted without further interaction. In all cases if the file is unable to be deleted you will be notified by a dialog.

Confirm delete dialog

The dialog that appears when delete confirmation is enabled will show the file or files to be deleted. To delete the file press **Enter** or activate the Delete button. Pressing **Escape** or activating the Cancel button will close the dialog and abort the operation.

When deleting multiple files, the dialog will include two arrow buttons to step the image preview through the list of files to be deleted. Initially the preview will be blank, activate the right arrow to display the first file.

The dialog will also display the status of the Safe Delete feature. Use this to determine if a file to be deleted will be immediately lost, or if the file can possibly be recovered from the Safe Delete folder.

If a file is unable to be deleted, a dialog will appear to notify you. If multiple files were selected and any remain, the dialog will include a Continue button, activating this button or pressing Enter will delete the remaining files. Pressing **Escape** or activating the Cancel button will close the dialog and the remaining files will not be deleted.

Safe Delete

Geeqie includes a feature to temporarily store deleted files in a selected folder, to use this feature enable Safe Delete on the advanced tab of the preferences dialog.

Cache and Data Maintenance

Thumbnails and other cached data can be maintained from the dialog accessible by selecting Edit → Cache maintenance .

Each displayed cache location has a Cache data size value, calculated in the background. It is the total size of file contents and symbolic-link entries in that directory tree, rather than allocated disk space. Symbolic links are not followed. Thumbnails stored alongside source images elsewhere are not included.

Select Refresh sizes to recalculate the totals. Sizes also refresh when an in-dialog cleanup, thumbnail rendering, or similarity-cache operation finishes. Use Refresh sizes after an external cache-maintenance job. A directory that does not yet exist is identified separately from a size that could not be calculated. Closing the dialog cancels pending size scans.

Geeqie thumbnail and sim. cache

The utilities listed here operate on the Geeqie caching mechanism. This also includes the data cached for the search and find duplicates utilities.

Clean up	Removes thumbnails, sim. files, and data for which the source image is no longer present, or has been modified since the cached data was generated. Valid similarity grids without perceptual hashes are retained and can be upgraded during similarity-cache creation.
Clear cache	Removes all thumbnails, sim. files, and data stored in the designated folder.

Shared thumbnail cache

The utilities listed here operate on the shared thumbnail mechanism.

Clean up	Removes thumbnails for which the source image is no longer present, or has been modified since the cached data was generated. Valid similarity grids without perceptual hashes are retained and can be upgraded during similarity-cache creation.
Clear cache	Removes all thumbnails stored in the designated folder.

Create thumbnails

This utility will render thumbnails using the current thumbnail caching mechanism, as determined in Preferences .

Render	Pre-render thumbnails for a specific folder, the utility has the following options:
Include subfolders	Enable to include all images contained in the subfolders of folder.
Store thumbnails local to source images	The generated thumbnails will be stored local to the source images, if you have the permissions to write to the folder containing the images.

Create file similarity cache

Some file operations, such as search and find duplicates , are speeded up if similarity data files for the images have been created. This utility will create sim. files recursively under the specified folder. When Perceptual hash algorithm is selected in the Duplicates window, it also creates all eight orientation hashes. Existing RGB grids are reused when adding missing hashes.

The sim. files contain some or all of the following fields:

Dimensions	width x height
DateTimeOriginal	Exif.Photo.DateTimeOriginal in time_t format, or -1 if unavailable. Legacy Date fields are ignored and removed when the cache is rewritten.
checksum	MD5sum 32 character ascii text digest
SimilarityGrid	[32 x 32] = 3072 bytes of data (1024 pixels in RGB format, 1 pixel is 24bits)
PerceptualHashV1	Eight 64-bit DCT perceptual hashes, stored as 16-digit hexadecimal values separated by spaces. These cover all right-angle rotations and reflections. This optional field is added when pHash is enabled; the RGB grid is also retained.

Metadata

This utility operates on the data store for Metadata located in the folder:

```
$HOME/.local/share/Geeqie/metadata
```

Clean up	Removes keywords and descriptions for which the source image is no longer present.

Clean and Update Caches

Select Select in the Clean and update caches section, enter or choose a folder, and select Start. Geeqie launches the command line program described below as a separate background job, creating thumbnails and similarity data for the selected folder and all its subfolders. The initial cleanup removes orphaned and outdated entries from Geeqie's cache, not just entries for the selected folder.

Thumbnail caching must be enabled in the saved Preferences. A progress window shows the maintenance stage and current path. Select Stop or close this window to terminate maintenance. The viewer remains available during the background job. The job continues if you close Geeqie; use its progress window to stop it.

Command line program

Geeqie can be run as a command line program: `GQ_CACHE_MAINTENANCE=y[es] geeqie --cache-maintenance=<path to images>`. It will recursively remove orphaned and outdated thumbnails and .sim files, and create thumbnails and similarity data for all images found.

It may also be called from `cron` or `anacron` thus enabling automatic updating of the cached data for all your images.

Scheduling cache maintenance

Run scheduled maintenance as the same user who normally runs Geeqie. Start Geeqie normally at least once and save its configuration before scheduling maintenance. Thumbnail caching must be enabled in Preferences. If you use custom `XDG_CONFIG_HOME`, `XDG_CACHE_HOME`, or `XDG_DATA_HOME` values, set the same values in the maintenance script.

Although maintenance does not open a viewer window, the current GTK4 implementation requires a display connection. For jobs without a desktop session, install `Xvfb`, `xvfb-run` (including its `xauth` dependency), and `dbus-run-session` using your distribution's package manager. The following wrapper supplies a virtual display and a private D-Bus session.

Save this script as `$HOME/.local/bin/geeqie-cache-maintenance`, creating the directory if necessary. Replace `/usr/bin/geeqie` with the absolute path reported by `command -v geeqie`, and replace the image folder with your own. Check the paths to the wrapper programs too. Quoting the image folder allows spaces in its name.

```
#!/bin/sh
# SPDX-License-Identifier: GPL-2.0-or-later
export GQ_CACHE_MAINTENANCE=y
exec /usr/bin/xvfb-run --auto-servernum /usr/bin/dbus-run-session -- /usr/bin/g
```

Make the script executable, create the log directory, and test it manually before adding a schedule:

```
chmod +x "$HOME/.local/bin/geeqie-cache-maintenance"
mkdir -p "$HOME/.local/state/geeqie"
"$HOME/.local/bin/geeqie-cache-maintenance" >> "$HOME/.local/state/geeqie/cache
```

Check the log for configuration errors or a `Caching not enabled` message. The examples below append both standard output and errors to this log; remove or rotate it periodically. Ensure the image folder, including any external or network drive, is available when the job runs.

Using cron

Run `crontab -e` as your normal user and add the following entry to run maintenance daily at 03:00. The cron service must be enabled, and the computer must be running at the scheduled time.

```
0 3 * * * "$HOME/.local/bin/geeqie-cache-maintenance" >> "$HOME/.local/state/ge
```

Using anacron

Anacron can run a daily job that was missed while the computer was switched off. To keep maintenance under your own account, create a personal configuration and timestamp directory:

```
mkdir -p "$HOME/.config" "$HOME/.local/state/geeqie/anacron"
```

Save the following line in `$HOME/.config/geeqie-anacrontab`. The first two fields specify a one-day interval and a five-minute delay; the third is the job identifier.

```
1 5 geeqie-cache-maintenance "$HOME/.local/bin/geeqie-cache-maintenance" >> "$H
```

Anacron must itself be started periodically. For example, add this entry with `crontab -e` to check hourly for a due job. Replace `/usr/sbin/anacron` with the path reported by `command -v anacron`. Use this entry instead of the daily cron entry above.

```
0 * * * * /usr/sbin/anacron -s -t "$HOME/.config/geeqie-anacrontab" -S "$HOME/.
```

Chapter 8. Color Management

TBD.

Color management options

Chapter 9. Image Presentation

Geeqie can be used to display a set of images, showing each image for a period of time before changing to the next image. Images can also be set to fill the entire screen.

Slide Show

The image pane can be set to automatically change the displayed image after a period of time, in Geeqie this is referred to as a slide show.

Starting a slide show

A slide show can be started in a number of ways:

- From the keyboard or mouse View → Toggle slide show (**S**)
- From a context menu - right click an image pane, and select **Start slide show** , or **Stop slide show** ; the text will change depending on the state of the current slide show.

Stopping a slide show

An active slide show can be stopped by using the same functions available to start one, as they work like a toggle switch. If slide show repeat is not enabled in the Preferences dialog , the slide show will stop when the last image in the sequence is displayed.

Any action that changes the image pane to a new image will stop the active slide show, with the exception of the methods discussed in Slide show sequence navigation .

Slide show status

The main window status bar can be used to determine when a slide show is active, **Slide show** will be displayed next to the file count field. An individual Image Window can also display a slide show, however the status will not be immediately visible. To determine the slide show status in an image window, open a context menu by right clicking or using the menu key, the slide show menu item will reflect the slide show status; if **Start slide show** is displayed, no slide show is active, conversely if **Stop slide show** is displayed a slide show is active.

When a Slide show is paused, the main window status bar will display **Paused** instead of **Slide show** . Likewise, the image context menu will display **Pause slide show** or **Continue slide show** , depending on the current state.

Pausing a slide show

An active slide show can be paused in two ways:

- From the keyboard - press the **P** key while a slide show is active.
- From a context menu - right click the image pane displaying the slide show, and select **Pause slide show** , or **Continue slide show** depending on the current slide show state.

Selecting the images to include in a slide show

When a single image is selected, all images in the current folder are included in the slide show. The sequence will begin with the current image; however if random slide shows are enabled, the sequence will begin with a random image from the list.

To only include some images in a slide show, use multiple selection to select the images to include before starting the slide show.

An image collection can be used as the list of images, to do this drag an image from a collection in the files pane onto the image pane. Starting a slide show will then include the images from that collection.

So far, the methods described are for slide shows in a main window, in an Image Window , the slide show will use all images referenced by that window.

Recursive slide show

In the main window, a recursive slide show can be started from a folder's context menu; the folder's right click menu will include two slide show options:

- Slide show: This will start a slide show that includes all images in the selected folder.
- Slide show recursive: The slide show will include all images in the selected folder, and all images in the folders that are contained within the selected folder.

When Repeat is enabled, a recursive slide show rescans the starting folder and its subfolders before each new cycle. Newly added images are included and removed images are omitted. The original sort order is retained, or the updated list is shuffled if Random is enabled. If no images remain, the slide show stops.

Repeating a collection

To start a slide show from a collection displayed as a virtual folder, right-click its entry in the folder pane and select Slide show. With Repeat enabled, the saved collection is reread before each new cycle so changes to its membership are included. This does not scan the folders containing its images.

If the collection has unsaved edits, playback uses its current in-memory contents instead. Reloading affects the slide show playlist without replacing the collection being edited. If the saved file cannot be read, playback retains its previous playlist; if it is successfully read and empty, playback stops. A slide show limited to selected images continues to use that selection.

Slide show sequence navigation

When a slide show is active, functions that normally change the image forward and backward can be used to step through those contained in the slide show.

Configuring a slide show

The options that control the behavior of slide shows are located on the General tab of the options dialog.

Full Screen

Geeqie allows for the image pane to be expanded to fill the entire screen, this is referred to as full screen.

Turning on full screen

Full screen can be activated in a number of ways:

- View → Full screen (**F**)
- From a context menu - right click an image pane, and select **Full screen** .

Turning off full screen

To return to the normal view:

- From the keyboard - press **F** or **Esc** to stop full screen.
- From a context menu - right click the image pane, and select **Exit full screen** .
- From the menu bar - the menu bar may be visible when using multiple monitors, select View → Full screen (**F**)

Configuring full screen

Full screen behavior and location can be adjusted on the Windows tab of the preferences dialog.

Chapter 10. Printing

To print an image select File → Print (**Shift+P**) .

Geeqie first opens a Print options window for image layout and text settings. Select Continue to open the standard desktop print dialog for printer and paper settings. This also works when the desktop print portal is used.

Print options

File → Print in the main window uses the file-list selection. To print every image, use Select All before opening Print. To print just the displayed image, right-click the image and choose Print image…. This is also available in fullscreen and standalone image windows. In split view, it prints the image in the pane that was right-clicked, regardless of other selected files or image panes.

Choose One image per page or Proof sheet. Image size sets the percentage of the available image area for a single image. Proof width and Height set the proof cell dimensions in points (72 points equal one inch). Proof sheets fit as many cells as possible inside the paper margins and continue on additional pages as needed. Oversized cells are reduced to the printable area.

GTK's standard controls provide paper size, orientation, margins, preview, printer selection, and printing to a PDF file. Layout settings are remembered for future print jobs.

Image Text	This text will be attached to each image printed. It may contain any combination of plain text and predefined tags.
Page Text	This text is printed once per page, including on proof sheets. It may contain only plain text. On proof sheets it appears above or below the grid.
Header 1/2 Footer 1/2	These checkboxes define the layout for Image Text and Page Text. Image text may be printed above or below each image, and page text above or below the page content. When both apply to a single image, their order is defined by these checkboxes.

Chapter 11. Preferences

There are two sets of preferences in Geeqie: those set via the main Preferences tabbed dialogue, and those set either via a menu entry or by editing a configuration file.

This chapter explains in detail the main Geeqie preferences dialog, which is accessed via Edit → Preferences (**Ctrl+O**) or on the Toolbar.

General Options

This section describes the options presented under the General Tab of the preferences dialog.

Color scheme

Select Follow desktop, Light, or Dark to control the interface appearance. The default follows the desktop preference. Changes take effect when you confirm the Preferences dialog; Cancel leaves the current scheme unchanged. Custom themes may not provide both light and dark variants.

The command-line option `--color-scheme=system|light|dark` overrides this preference for the running application session without changing the saved setting. For example, **geeqie --color-scheme=dark** requests a dark interface. If Geeqie is already running, the override applies to its existing windows and remains active until it exits. `system` follows the desktop even if the saved preference specifies Light or Dark. Changing the saved preference does not replace an active command-line override.

This setting controls the interface theme. Custom image background colors configured in Image preferences remain separate.

Thumbnails

The files pane loads visible thumbnails first, then prefetches approximately one screen of thumbnails on either side. Loading resumes as you scroll or resize the pane; it does not generate thumbnails for the entire folder. This applies to both list and icon views, including collections and expanded file groups.

Thumbnail memory (MiB) sets the memory budget for retained thumbnail pixels shared by all file panes. The default is 64 MiB. Older thumbnails are released and reloaded from the disk cache or source image when needed. Visible thumbnails and active loads are protected, so they can exceed the budget. This setting does not limit total application memory, image decoding buffers, display textures, or file-list metadata, and does not change the disk cache.

Size	Selects the size of the thumbnails displayed throughout Geeqie, dimensions are width by height in pixels.
Quality	Selects the method to use when scaling an image down for thumbnails:

Nearest	Fastest scaler, but results in poor thumbnail quality.
Tiles	Thumbnail results are very close to bilinear, with better speed.
Bilinear	High quality results, moderately fast.

Cache thumbnails and sim. files

Enable this to save thumbnails and sim. files to disk. Subsequent requests for a thumbnail will be faster, as will searches and find duplicates.

Use Geeqie thumbnail style and cache

Thumbnails are stored in a folder hierarchy that mirrors the location of the source images. Thumbnails have the same name as the original appended by the file extension .png.

The root of the hierarchy is:

`$XDG_CACHE_HOME/geeqie/thumbnails/` or, if $XDG_CACHE_HOME is not defined: `$HOME/.cache/geeqie/thumbnails/`

Store thumbnails local to image folder (non-standard)

When enabled, Geeqie attempts to store cached thumbnails closer to the source image. This way multiple users can benefit from a single cache, thereby reducing wasted disk space.

Thumbnails have the same name as the original appended by the file extension .png.

The resulting location is the source image's folder, in a sub folder with the name `.thumbnails`

When the image source folder cannot be written, Geeqie falls back to saving the thumbnail in the configured Geeqie cache folder.

Use standard thumbnail style and cache, shared with other applications

This will use a thumbnail caching method that is compatible with applications that use the standard thumbnail

specification. When this option is enabled thumbnails will be stored in:

`$XDG_CACHE_HOME/thumbnails/` or, if $XDG_CACHE_HOME is not defined: `$HOME/.cache/thumbnails/`

All thumbnails are stored in the same folder, with computer-generated filenames. Refer to Thumbnails Reference for additional details.

Use Geeqie thumbnail style and custom folder

The folder field selects a parent folder. Geeqie creates a dedicated geeqie-cache subfolder for its thumbnails and .sim files, and marks it as a Geeqie cache. Existing nonempty folders without this marker are not adopted. Select Use Geeqie thumbnail style and cache to restore the default location. A dedicated folder on a larger local or network drive can reduce local disk usage. Existing cache files are not moved. Shared-style thumbnails remain in their standard location, while their .sim files use the configured Geeqie cache folder.

Use EXIF thumbnails when available

Geeqie will extract thumbnail from EXIF data if available, instead of generating one. This will speed up thumbnails generation, but the EXIF thumbnail may be not in sync with the image if it was modified by a tool which did not also update the thumbnail data.

Thumbnail color management

Where possible store thumbnails with an sRGB color profile.

Note

Selecting this option will increase the time taken to generate thumbnails.

You probably will want to deselect the "Use EXIF thumbnails" option.

Star Rating

The characters used to display the Star Rating are defined here. They are defined as a hexadecimal Unicode character. The complete list of Unicode characters can be found in many places on the Internet.

Slide show

Delay between image change
: Specifies how long each image is displayed in a slide show, in seconds. The delay starts after the image has finished loading and rendering, so slow-loading images are not skipped. When a transition is enabled, the delay starts after the transition finishes.

Random
: When enabled, slide show images will appear in random order.

> **Note**
>
> Random images are displayed such that each image appears once per cycle of all images. When the slide show repeat option is enabled, the image order is randomized after completing each cycle.

Repeat
: This will cause the slide show to loop indefinitely, it will continue with the first image after displaying the last image in the slide show list.

Transition
: Choose a transition: None (the default), Cross-fade, Slide in each direction or paired directions, Over and Under in each direction, paired Over directions, and Rotate left, right, or left/right. Paired transitions alternate direction between slides. The outgoing view remains visible until the incoming image is ready. Manual navigation or stopping the slide show cancels the transition. Desktop animation settings are respected.

Transition duration (milliseconds)
: Sets the transition duration, from 50 to 5000 milliseconds (default 300). Image loading and transitions extend the time between image changes. The outgoing view is retained in memory until the transition finishes.

Image loading and caching

Decoded image cache size
: Limit the amount of memory available for caching images.

JPEG decode memory limit (MiB)
: Sets the maximum estimated memory for decoding one JPEG image, from 1 to 65536 MiB (default 1024 MiB). Increase this limit to open very large JPEG images. Rendering, caches, other images, and decoder overhead require additional memory. The limit is also capped by the process address space.

Preload next image
: Enabling this option will cause Geeqie to read the next logical image from disk when idle, it will also retain the previously viewed image in memory. By reading the nearest images into memory, time to display the next image is reduced.

Note

This option will increase Geeqie memory requirements, and may cause performance issues with very large images. If the use of Geeqie results in the system noticeably swapping memory to disk, try disabling this feature.

Refresh on file change

Geeqie will monitor currently active images and folders for changes in their modification time, and update the display if it changes.

Note

Disable this if the system will not go into sleep mode due to occasional disk activity from the time check, or if Geeqie updates too often for folders with continuously changing content.

Menu style

Use a ≡ "hamburger" style menu button instead of the standard menu across the top of the frame.

Note

Geeqie must be restarted for changes to take effect.

Expand menu/toolbar

Expand the menu/toolbar to the full width of the window.

Note

Geeqie must be restarted for changes to take effect.

Hide Selectable Bars

The menu item Hide Selectable Bars will toggle display of the bars selected here. The default keystroke for this action is `control + backtick`

Note

Geeqie must be restarted for changes to take effect.

Show predefined keyword tree

Deselecting this option will hide the list of predefined keywords on the right-hand side of the keywords pane of the info sidebar.

Note

Geeqie must be restarted for the change to take effect.

Timezone Database

The timezone database is used to correct exif time and date for UTC offset and Daylight Saving Time as described here. This option allows you to install or update the database. An Internet connection is required.

On-line help search

An internet search engine may be used to search the help files on Geeqie's website. The string used to conduct the search is defined here. In most cases it will be in one of two formats:

```
https://www.search-engine.com/search?q=site:geeqie.org/help
```

```
https://www.search-engine.com/?q=site:geeqie.org/help'
```

Image Options

This section describes the options presented under the Image Tab of the preferences dialog.

Zoom

Use nearest-neighbor sampling above 200% zoom	Displays crisp pixel boundaries above 200% magnification. At 200% and below, the selected zoom quality is used. Disabled by default. A keyboard shortcut can be assigned to this option in Preferences → Keyboard.
Allow panning beyond image edges	Allows panning until any image edge or corner reaches the center of the viewing area. This makes it easier to inspect image boundaries. Disabled by default.
Quality	Selects the method used to scale the size of an image:

Nearest	Fastest scaler, but results in poor image quality.
Tiles	Results are somewhat close to bilinear, with better speed.
Bilinear	High quality results, moderately fast.

Two pass rendering (apply HQ zoom and color correction in second pass)	Enables Geeqie to first display a scaled image using the *Nearest* zoom quality. After image decoding is complete, the image is scaled again using the selected *Zoom Quality* method, and ICC color correction is applied if color management is enabled. This allows faster display of an image as it is decoded from the source file, but the first pass is not color corrected and the colors may visibly change during the second pass. Disable this option to apply the selected zoom quality and ICC color correction in a single rendering pass. This avoids displaying an uncorrected first pass, but may delay the initial display. Image preloading caches decoded source pixels; it does not pre-render them for the display's color profile.
Zoom increment	Adjusts the step size when zooming in or out on an image.
Zoom style	Selects whether the zoom step size is applied as a geometric (n * step) or arithmetic (n + step) factor.

Fit Image To Window

Allow enlargement of image (max. size in %)	Enable this to allow Geeqie to increase the image size for images that are smaller than the current view area when the

zoom is set to *Fit image to window* . This value sets the maximum expansion permitted in percent i.e. 100% is full-size.

Virtual window size (% of actual window)

This value will set the virtual size of the window when *Fit image to window* is set. Instead of using the actual size of the window, the specified percentage of the window will be used. It allows one to keep a border around the image (values lower than 100%) or to auto zoom the image (values greater than 100%). It affects fullscreen mode too.

Tile Size

Tile size in pixels

This option is only available when GPU acceleration is not selected. It sets the size in pixels that large images are split into during rendering. Setting a larger value will reduce the tiling effect seen as the image is displayed, but will also slightly increase the delay until the first part of the image is seen.

Appearance

Custom border color

Enable this to draw the image background (the area around the image) in the specified color.

Border color

Use this color chooser to define the color to use as image background.

Note

You may use the *Virtual window size* (see above) option to keep a border around the image in fullscreen mode.

Alpha channel color 1/2

These two colors define the checkerboard background used when images with an alpha channel are displayed.

OverlayScreen Display

This section describes the options presented under the OSD Tab of the preferences dialog.

Overlay Screen Display

The contents of the Overlay Screen Display is defined by the Image Overlay Template. This template is easily customised to display a wide range of data.

The format of each entry is:

```
%tag[:max_length][:extra]%
```

Table 11.1. Overlay Screen Display templates

Tag	Replaced by
name	Filename of the picture
path	Full path, including filename, of the picture
collection	Name of the collection
collection_info	Info text for the current image in a collection

number	Current number of image in the list
total	Total number of images
date	File date
size	File size
width	Image width
height	Image height
res	Image dimensions in pixels (width x height)
formatted.Resolution	Image density and units from EXIF metadata, such as 200 x 200 (dot/inch)
keywords	Image keywords from metadata
description	Image description from XMP metadata (Xmp.dc.description). The legacy name comment is also recognized.
imagecomment	JPEG image comment
rating	Image rating
<meta_tag>	The Exif, XMP, or IPTC tag from metadata
lua/<lua_script>/	The output of a Lua script file
lua//<lua_command>	The output of a Lua command

Pixel dimensions and image density are different values. The `%res%` tag displays the image width and height in pixels. To also display the recorded density, use:

```
%res% | %formatted.Resolution%
```

See `formatted.Resolution` for supported density fields, units, and fallback behavior.

As an aide, in addition to standard metadata tags, Geeqie provides a number of pre-formatted tags .

Examples of usage are:

```
%keywords%
%Exif.Photo.DateTimeOriginal%
%lua/jpeg_comment.lua/:12%
%lua//return(os.date())%
```

Refer to Lua Extensions for further information.

The length of displayed data can be limited by using the *:max_length* parameter. The following example will truncate the displayed data to 20 characters and will add 3 dots at the end to denote the truncation.

```
%formatted.Camera:20%
```

If two or more variables are connected with the `|` character, the variables are displayed with a separator. For example::

```
%formatted.ShutterSpeed%|%formatted.ISOSpeedRating%|
%formatted.FocalLength%
```

could show:

```
"1/20s - 400 - 80 mm"
```

Or, if there is no ISO information in the Exif data:

```
"1/200 - 80 mm"
```

If there is no data for a requested tag, the line is not displayed.

The *:extra* parameter may be used to format the output by prepending and appending a text string to the displayed item.

The special character * is used to mark the position of the Tag data item. If no * is present, the extra string is just appended to the standard data displayed. Any "\n" is replaced by a newline on display. Pango mark up is accepted in both left and right parts. If the data item is empty, nothing will be displayed.

Examples:

Table 11.2. Overlay Screen Display template examples

Template	Example display
%name: *\n%	*filename001.jpg*
%size:\n%	123456
%formatted.ISOSpeedRating:ISO *%	ISO 100
%collection:Collection: * \n%	Collection: **my_new_collection**

Font , Text , Background	The font used for the Overlay Screen Display, as well as the font colour and background colour, may be set via these buttons. Colour transparency can also be set.
Help	Show syntax help
Defaults	Restore default image overlay template

Window Options

This section describes the options presented under the Windows Tab of the preferences dialog.

State

Remember session	Restores saved window sizes and maximized states between Geeqie sessions. In GTK4, Geeqie does not restore window coordinates or monitor placement; the window manager or compositor controls where windows appear.
Use saved window positions also for new windows	A new window opened by menu File → New Window (**Ctrl+N**) will also use the saved window size and maximized state. This option applies to additional windows, rather than the initial window restored at startup. Window coordinates and monitor placement are not restored.
Remember window workspace	Remember which workspace a window was displayed in. This is supported on X11 and is separate from monitor placement.
Remember tool state	Maintains the selection set by View → File and Folders → Float File List (**L**)
Remember dialog window positions	Restores saved dialog window sizes between Geeqie sessions. The window manager or compositor controls dialog placement;

	Geeqie does not restore dialog coordinates or monitor placement.
Hide window decorations	By default, windows are decorated with a title bar, resize controls, etc. If this option is selected, some window managers may be able to disable these decorations, creating a borderless window.
Show window IDs	Show the window ID in the titlebar of each window. When multiple Geeqie windows are opened, this option shows a unique identifier for each window. It may be used in conjunction with the command line option: `--id=<ID>`
Use current layout for default	Press the `Set` button to store the current window layout for use as the default for all new windows.

Size

Fit window to image when tools are hidden/floating	The main window will be resized to accommodate each image's size and proportions when the image pane is the only one visible within the window.
Limit size when auto-sizing window	This will restrict the maximum size a window can grow to automatically fit an image. The value represents the percentage of the desktop size in width and height.

Full Screen

Location	Selects the location and position of the full screen window. 'Determined by window manager' will leave the window placement up to the window manager. 'Active screen' places the window on the same screen as the Geeqie main window. 'Active monitor' does the same, but limits the full screen window size to the monitor containing the main window. The remaining choices are dependent on the multiple monitor configuration.
Smooth image flip	This option delays the image change in full screen until the next image is entirely decoded into memory, the result is a smoother transition between images.
Disable screen saver	This option will execute *xscreensaver-command --deactivate* once per minute to avoid the screen saver from activating when Geeqie is displayed full screen.

Keyboard Options

This dialogue enables changes to be be made to the default key short-cut allocated to menu items.

The data displayed may be ordered by clicking at the top of each column.

To change or create a new short-cut, click on the Keystroke Converter box and type the key combination you wish to use. The correct character sequence is automatically loaded into the clipboard. Double click on the Shortcut column of the relevant item. Insert the new text. If multiple shortcuts are used, separate them with a semi-colon. Geeqie must be restarted for the new code to be registered.

Commands preceded by "app." are available in all windows. Commands preceded by "win." are available only in the relevant window.

A text search may be made in any of the columns. Click anywhere in a column and type characters. The search is case-insensitive

File Filters Options

This section describes the options presented under the File Filters Tab of the preferences dialogue.

Filters

Show hidden files or folders	Enables the display of hidden files and folders, also known as dot files.
Use dot prefix as hidden file indicator	Use only the file dot prefix as a method of identifying hidden files. This may speed up file operations. However, the contents of a `.hidden` file will be ignored, as will some other methods of hiding files.
Show parent folder	Enables the display of the parent folder as a double dot in the directories views.
Case sensitive sort (Collection and Search windows, and tab completion)	Files and folders are sorted such that upper case letters occur before lower case letters. Enabled result: A, B, C, a, b, c Disabled result: A, a, B, b, C, c
Disable file extension checks	During rename operations, allows files to be renamed to a different extension with no warning issued.
Disable file filtering	Enable this to allow Geeqie to list all files, even those that do not contain images.

Grouping sidecar extensions

This field defines which extensions will lead to file grouping. Grouped files will appear as a single entry in file views. This is useful, for example, when browsing directories containing two files for the same image (such as raw+jpeg files generated by some cameras).

The group is represented by the first extension in the list for which a file exists.

Each extension should include the starting dot and should be separated by a semi-colon from following entries. It is possible to use wild-cards corresponding to entries in the *File Types Class* below:

`%unknown`	expands to all files listed in class *Unknown* ,
`%image`	expands to all files listed in class *Image* ,
`%raw`	expands to all files listed in class *RAW Image*
`%meta`	expands to all files listed in class *Metadata*

File types

This list contains the image formats that are understood by Geeqie. To disable a specific file type from being displayed, uncheck it's corresponding check-box. Filters can be edited by first selecting the line

containing the item to be changed, then clicking the filter or description field, making any changes, then pressing **Enter** ; press **Escape** to cancel an active edit.

Note

Unlike all other options in the dialogue, filter list changes are applied immediately.

The table consists of:

Enabled	Whether or not the filter is used.
Filter	List of extensions this filter applies to, separated by a `;` . Each extension must be preceded by a dot.
Description	Brief description. This is not used elsewhere.
Class	This may be used to define sets of file types, for use in the *Grouping* function described above. The drop-down list has 7 entries: • Unknown • Image • RAW Image • Metadata • Video • Collection • Document • Archive (.zip, .rar etc.)
Writable	If this option is checked, Geeqie will try to write metadata to the image file itself. This, and the following option, are mutually exclusive.
Sidecar is allowed	If this option is checked, Geeqie will try to write metadata to a sidecar file associated with the image. This, and the previous option, are mutually exclusive.
Add	Adds a new blank line to the filter list, change the parameters to match the desired file type.
Remove	Removes the selected line from the filter list.
Defaults	Resets the filter list to the default values, discarding any customizations made by the user.
Column Sorting	Columns other than Filter can be sorted in ascending or descending order by clicking on the column header.
Filter Search	A search of the file extensions listed in the Filter column can be initiated either by typing `Ctrl + F` or by pressing any alpha-numeric key when the File Types table has focus.

Metadata

This section describes the options for metadata writing.

Refer also to Metadata Processing in the Reference section.

Metadata writing process

When writing metadata, Geeqie will follow these steps. This process will stop when the first successful write occurs. A flowchart showing the sequence is here .

1. Save metadata in image files or sidecar files, according to the XMP standard

   If checked, Geeqie will attempt to write metadata to the file itself, or to an associated XMP sidecar file

   The destination is dependent on the settings in the "Writable" and "Sidecar Is Allowed" columns of the File Filters tab .

2. Save metadata in '.metadata' folder, local to image folder (non-standard)

   If checked, Geeqie will attempt to write to an xmp file in a local .metadata folder

3. Save metadata in Geeqie private directory

   If neither of the above are successful, Geeqie will write to an xmp file in the folder

   `$HOME/.local/share/geeqie/metadata`

When writing to local metadata folders, the directory structure duplicates that of the location of the source files.

Step 1 Options:

If Step 1 above was successful, the following options take effect:

Saving edits preserves unrelated EXIF, XMP and IPTC fields. For example, changing orientation does not convert an existing XMP date-only value into an EXIF timestamp.

- Store metadata also in legacy IPTC tags (converted according to IPTC4XMP standard)

  If checked, Geeqie will also update the corresponding IPTC tags for edited XMP fields, as defined in Standards

  The conversion from XMP tag to IPTC tag is performed by exiv2. A list of the XMP-IPTC conversions is at the end of this table [https://exiv2.org/conversion.html] . Unrelated existing IPTC tags are preserved, including when this option is unchecked.

- Warn if the image files are not writeable

  If checked, open a warning dialogue if the file is not writeable

- Ask before writing to image files

  If checked, open a confirmation dialogue before writing to the file

- Create sidecar files named image.ext.xmp (as opposed to image.xmp)

  If checked, new sidecar files will use the image.ext.xmp naming scheme. This scheme is used by Darktable. Each image has its own sidecar: for example, `photo.jpg.xmp` belongs to `photo.jpg`, and `photo.cr2.xmp` belongs to `photo.cr2`. Grouping the images or changing the group leader does not change these associations. When this option is unchecked, images with the same basename use the shared `photo.xmp` sidecar.

Steps 2 and 3 Option:

If Step 2 or 3 above was successful, the following option takes effect:

- Use GQview legacy metadata format (supports only keywords and descriptions) instead of XMP

  If checked, Geeqie will write metadata in GQview legacy files (file extension .meta) format

Miscellaneous

- Write the same description tags to all grouped sidecars

  If checked the same descriptive tags are written to all files in a group. File grouping is as defined in the File Filters tab of main Preferences. Technical tags (exif) are handled separately for each file.

  The tags written are:

```
"Xmp.dc.title"
"Xmp.photoshop.Urgency"
"Xmp.photoshop.Category",
"Xmp.photoshop.SupplementalCategory",
"Xmp.dc.subject",
"Xmp.iptc.Location",
"Xmp.photoshop.Instruction",
"Xmp.photoshop.DateCreated",
"Xmp.dc.creator",
"Xmp.photoshop.AuthorsPosition",
"Xmp.photoshop.City",
"Xmp.photoshop.State",
"Xmp.iptc.CountryCode",
"Xmp.photoshop.Country",
"Xmp.photoshop.TransmissionReference",
"Xmp.photoshop.Headline",
"Xmp.photoshop.Credit",
"Xmp.photoshop.Source",
"Xmp.dc.rights",
"Xmp.dc.description",
"Xmp.photoshop.CaptionWriter",
"Xmp.xmp.Rating",
```

- Allow keywords to differ only in case

  If checked, permits keywords to be case-sensitive (e.g. *Place* and *place* are two different keywords)

- Write altered image orientation to the metadata

  If checked, the results of orientation commands (Rotate, Mirror and Flip) issued on an image will be written to metadata either automatically as specified below or when manually initiated. This option also affects the Auto rotate image using EXIF information menu item.

  ## Warning

  If this option is not checked, the results of orientation commands will be lost when Geeqie closes.

  Do not change this option while you have temporary rotations applied to images.

Auto-save options

- Write metadata after timeout

  If checked, metadata writes will be made automatically at the defined interval. Warning dialogues will be issued if required.

- Write metadata on image change

  If checked, metadata writes will be made when you move focus to another image.

- Write metadata on directory change

  If checked, metadata writes will be made when you move focus to another folder.

Spelling Checks

If selected, spelling checks will be enabled on the following panes:

- Title
- Description
- Headline

Pre-load metadata

- Read metadata in background

  Using the folder sorting options:

  Exif date original

  Exif date digitized

  Rating

  requires metadata to be read from all files in a folder before the sort action can be made. If a folder contains a large number of file, this can take a noticeable period of time.

  If this option is checked, Geeqie will automatically read the required metatada in the background as soon as a folder is opened. This will reduce the amount of time you have to wait until the sort is completed.

  If you do not use these sort options, leave this option unchecked.

Metadata Write Sequence Flowchart

Keywords

This section describes the keywords list used for autocompletion.

Keyword Search

Pressing the Search button will open a dialog which permits a recursive search to be made for keywords already attached to images. The result of the search is automatically appended to the existing list.

Keyword List

The list shows all keywords currently used for autocompletion. Text may be copy-pasted into the list or deleted from the list.

When the list is saved, duplicates will automatically be removed and the list sorted into alphabetical order.

Color management options

Use of this function requires LCMS to be installed. If you are compiling from sources, LCMS functionality will be available if the development files dependencies are met.

If you are using a pre-compiled distribution, availability depends on the package maintainer.

Here one can defined color profiles to use. Color management can be enabled or disabled from View → Color Management or on the status bar.

Type	Either input color profile or screen color profile.
Menu name	Name of the menu entry corresponding to the color profile.
File	Path to the color profile file.

Stereo image management

TBD

Behavior Options

This section describes the options presented under the Behavior Tab of the preferences dialog.

Delete

Confirm permanent file delete	Prompts for confirmation before deleting files.
Confirm move file to Trash	Prompts for confirmation before moving files to the trash bin.
Confirm permanent folder delete	Enabled by default. Prompts before permanently deleting a folder and all its contents, including subfolders. The folder context menu offers both Move to Trash and Permanently Delete, each using its own confirmation setting. Disabling this option starts permanent deletion without confirmation; errors are still reported.
Confirm move folder to Trash	Enabled by default. Prompts before moving a complete folder tree to Trash as one item. The confirmation shows the number of descendant folders and all files, including hidden files,

sidecars, symbolic links, and files excluded by display filters. Symbolic link targets are never traversed. If contents cannot be counted reliably, an error is shown and deletion stops.

Built-in counting and recursive operations run in the background with a progress dialog and a Cancel button. Cancellation stops further work; files already removed are not restored. Failure messages distinguish incomplete copying from a completed copy followed by failed source removal. They identify the source and destination paths, say whether an incomplete copy was removed or remains, and report how many source entries were removed. A completed copy is retained when source removal fails. Built-in deletion refuses folder trees containing mount points, including bind mounts, before deleting any contents; symbolic links are not followed. Sources that change while being copied are checked before removal and the operation stops if a change is detected. Disabling this option starts trashing without confirmation. Failed trashing never falls back to permanent deletion.

Enable **Delete key**

Disable this to disassociate the delete file function from the **Delete** key.

If *Confirm file delete* , above, is disabled, accidentally hitting the **Delete** can result in files being erroneously deleted without warning. Disabling the **Delete** key will prevent this. The `Ctrl + D` key sequence is not affected.

Use Geeqie trash location

Moves deleted files to a temporary folder, for easy retrieval of deleted files. Items in the temporary folder are retained until the maximum size is reached, at which point deleting additional items will result in removal of the oldest trash entries to reclaim space. Folder sizes include their descendants, and a folder tree is removed as a whole item. Automatic removal displays a desktop notification showing the number of entries permanently deleted and the space reclaimed. An incoming item larger than the limit is refused with an error before removing any existing trash entries. The original item and existing trash entries are preserved.

Note

Deleted files retain their original names in the `files` subfolder. If a name is already present, Geeqie adds a numeric suffix before the file extension, for example `image.2.jpg`, to make it unique without preventing the file type from being recognized.

For each file, the `info` subfolder contains a freedesktop.org-compatible `.trashinfo` file recording its original path and deletion date. A folder tree has one information file for the top-level folder, preserving the names of all its contents.

When viewing the `files` subfolder in Geeqie, right-click a file or folder to copy or move it back to its recorded original location. Copying leaves the trash entry intact; moving removes it after a successful restore. Existing destinations are not overwritten. If

	the original parent no longer exists, use Restore to another folder to choose a destination. Files created by older versions, with a six-digit prefix, remain in the Safe Delete folder and are still included when enforcing the maximum size or clearing the trash.
Use system Trash bin	Moves deleted files to the system-defined trash bin.
Use no trash at all	Selects permanent deletion as the default. The folder context menu still offers Move to Trash; that explicit action uses the last selected trash location without changing this default.
Folder	The location of the folder used by the Safe Delete feature.
Maximum size	The size limit for Geeqie trash, in MiB. The default for new configurations is 5120 MiB (5 GiB). The control accepts limits up to 102400 MiB (100 GiB). Existing saved limits are preserved. Setting this value to 0 (zero) disables the limit.
Clear button	Removes all files contained in the Safe Delete folder.
View button	Changes the Geeqie main window to display the deleted files in the Safe Delete folder.

Behavior

Check subfolders for expansion arrows	This will allow Geeqie to read sub folder content to determine if it should display expander indicators in the folder tree view. **Note** Disable this if Geeqie wakes up auto mounted network shares, or takes too long to initially display the folder view.
In place renaming	When renaming a single file in a folder shown in list view, the rename entry appears directly over the original filename. This applies to the main Rename action (F2 by default) and the files pane context menu. Other views and multiple-file selections use the Rename dialog.
List directory view uses single click to enter	If selected, a single click will enter a directory, rather than the GTK+ default of a double click.
Circular selection lists	If selected, when traversing a list of selected images using Next or Previous Image, when the last or first image is encountered the focus moves to the beginning or end of the list. When deselected, focus will go no further than the start or end of the list.
Auto next folder	If selected, when the last image in the current folder is reached using Next Image, Geeqie will automatically advance to the first image in the next sibling folder. Likewise, when the first image is reached using Previous Image, Geeqie will automatically go to the last image in the previous sibling folder.
Save marks on exit	Save all marks that have been set. Note that marks that are linked to a keyword will always be saved irrespective of this setting.

Saved marks are retained for temporarily unavailable files, such as files on a disconnected drive. They become available again when the files return at the same paths. Clearing all marks also clears retained marks for unavailable files.

Use "With Rename" as default for Copy/Move dialogs
: Move the "With Rename" button to the default position.

Permit duplicates in Collections
: If this option is set, an image can be inserted into a Collection any number of times.

Hide window in fullscreen
: When alt-tabbing, prevent Geeqie window showing twice.

Hide OSD in fullscreen
: Hide Overlay Screen Display in fullscreen mode.

Recent folder list maximum size
: This is the maximum number of entries to be displayed in the recent open folder list drop-down box.

Recent collections list maximum size
: The maximum number of recently opened or saved collections retained between sessions. The default is 100, and the range is 1 to 1000. Open collection combines this history with files in the default collections directory; the limit applies only to history entries. Unavailable collections remain in history and appear grayed out in the list. Existing .gqv files in this history are also updated when images are renamed or moved within Geeqie, even if the collections are saved outside the default collections directory. This setting is independent of the recent folder list and does not control Open recent, which uses GTK's shared recent-files database.

Recent folder-image list maximum size
: The last image viewed in a folder is stored in a list. This value is the maximum number of entries in the list. When a folder is subsequently re-opened, focus moves to the last image viewed in that folder.

Drag'n drop icon size
: This is the width and height of the icon generated from image and displayed when doing drag and drop actions. High resolution screens may require to increase this number.

Drag'n drop default action
: When using drag and drop to copy or move files, the default action is to use a pop-up menu to permit you to select which action is to be taken. This option allows you to set the default action to either Copy or Move.

Note

When using the pop-up menu, the menu will also include an option for a symbolic link and any other plugins that are defined as being a filter. If you set the default action to Copy or Move these additional options will not be available to you.

Copy path clipboard selection
: This option determines which clipboard selection is used for the "Copy path" menu item. The exact method of the paste operation will depend on the operating system, but in general will be:

  Primary
  : Paste using mouse middle-click, or shift-insert.

Clipboard	Paste using mouse right-click and select paste, or control-v.
Both	Paste using either method.

Navigation

Progressive keyboard scrolling	This will cause image panning speed to gradually accelerate when holding down the keyboard arrows.
Keyboard scrolling step multiplier:	Defines the amount to pan an image when using the keyboard arrows.
Mouse wheel scrolls image	The default is for the mouse wheel to change the image, and to scroll the image when **Shift** is used in combination with the wheel. Enabling this swaps the behavior so that operating the wheel will scroll the image, and holding down **Shift** will change the image.
Show bird's-eye view when zoomed in	When enabled, zooming or panning an image that is larger than the image pane temporarily displays an overview of the complete image. The overview indicates the visible region and can be clicked or dragged to pan the image. It hides automatically after a short period of inactivity. This option is disabled by default.
Navigation by left or middle click on image	If selected, mouse left-click will select the next image; mouse middle-click will select the previous image.
Open archive by left click on image	If selected, mouse left-click on an image belonging to the archive class will decompress the file contents and display them in a new Geeqie window.
Play video by left click on image	If selected, mouse left-click on an image belonging to the video class will start a program chosen in a box below.
Mouse button Back/Forward	Some mice have additional buttons designated Back/Forward (mouse buttons 8 and 9). You may allocate any menu item to either of these buttons.

Debugging

Debug level	Displayed when compiled with the option --enable-debug-log. This defines the verbosity of debug info sent to console and log window (0 disables the debug output).
Log Window max. lines	The maximum number of data lines to be displayed. The window will show the most recent data.

Toolbar

This dialogue enables you to change the items displayed on either the Main Toolbar or the Status Toolbar.

The initial display shows the current setup. You may re-order these items by right-clicking on any item.

Additional items may be included by clicking on the "Plus" symbol at the bottom of the tab. A list of all available options is displayed.

Most of the Geeqie menu items are available. The Plugins submenu lists enabled plugins for both the Main Toolbar and the Status Toolbar.

Note

Plugin icons may be theme names or absolute file paths. A generic icon is used when no icon is specified or a theme icon cannot be found. The desktop file format is described in the Standards section

Advanced

External preview extraction

The intention of this feature is to give the user the possibility to display image previews of files that the standard libraries cannot decode.

An example is the .dng files produced by LG V30 cameras. Neither `exiv2` nor `libraw` can extract a preview, but the command line program `dcraw` can.

This feature allows a work-around until the standard libraries provide a solution.

Two command files are required: one to identify which files to process, and one to extract or decode the preview image.

The format for the identification tool is:

```
Parameter 1: (input) full path name to the current image.
Returns: 0 for file match, any other value for no match.
```

The format for the extraction tool is:

```
Parameter 1: (input) full path name to the current image.
Parameter 2: (output) a temporary file name generated by Geeqie.
Returns: not used.
```

The tool should load the output file with the decoded image.

When external previews are enabled, image and read-ahead loading waits briefly after navigation so that rapid navigation can skip intermediate files. External preview commands are serialized across image, read-ahead, and thumbnail requests. Commands for cancelled loads are terminated together with their child processes; scripts should keep their extraction work in the same process group and run it in the foreground. Missing or unreadable output is reported as a loading failure.

This is an example of an identification tool using a shell script:

```
#! /bin/bash

        filename=$(basename -- "$1")
        extension="${filename##*.}"

        shopt -s nocasematch
        if [[ $extension == "DNG" ]]
```

```
        then
        cameramodel=$(exiv2 -K Exif.Image.UniqueCameraModel -Pt "$1" )
        if [[ $cameramodel  == "LG-H930" ]]
        then
        exit 0
        else
        exit 1
        fi
        else
        exit 1
        fi
```

This is an example of an extraction/decode tool using a shell script:

```
#! /bin/bash
          dcraw -e -c   "$1" > "$2"
```

Alternatively:

```
#! /bin/bash
          gm convert "$1" "$2"
```

If the decode tool requires an output file with a particular extension, use this method:

```
#! /bin/bash
          tmpfile=$(mktemp --tmpdir=$tempdir geeqie_tmp_XXXXXX.jpg)
          gm convert "$1" $tmpfile
          mv $tmpfile "$2"
```

Thread Pools

This option will limit the number of threads (cores) that are used when performing a duplicate image search. A value of `0` means use all available threads. This will give the fastest processing time, but will slow other processes including user input response time.

Chapter 12. Additional Preferences

There are two sets of preferences in Geeqie: those set via the main Preferences tabbed dialogue, and those set either via a menu entry or by editing a configuration file.

This chapter explains in detail the Geeqie preferences that are set via a menu entry or by editing a configuration file.

Window Layout

This item is accessed via Edit → Configure this window .

Home Path	The path to be moved to when is clicked.
Show date in directories list view	The folder pane will show dates.
Start-up directory	This may be set for each instance of Geeqie. • No change: Use the Current Working Directory. If called from a system menu rather than the command line, the Home directory will probably be used. • Restore last path: Use the folder in use when Geeqie was previously closed. • Home path: Use the folder specified in the Home Path box.
Layout	The Geeqie main window consists of three main panes: the folder pane , file pane and the image pane . Their position and order can be changed here. The four choices shown represent the basic layout of the main window, the difference being that each has the larger pane in a different location: right, bottom, left, and top. The numbers within each selection represent the order of the pane content. The order in which the panes are filled is determined by the three row list located below the layout selection. The three numbered rows correspond to the three numbered regions of the selected layout. To change the order, drag a row to the desired location, or select a pane and use the Move Up and Move Down buttons. With the list focused, use the arrow keys to select a pane and Ctrl+Up or Ctrl+Down to move it. The moved pane remains selected. **Note** In this dialogue the file and image panes are labeled as such, however for historical reasons the folder pane is labeled as tools.

Plugins Configuration

Plugins Configuration Dialog

This dialog allows user to add new plugins or modify the system ones. It is available in the menu Edit → Configure Plugins .

The Plugins dialog shows list of all relevant plugins i.e. all installed desktop files that are designated either:

- `Categories=Graphics`
- `Categories=X-Geeqie`

The list has the following columns:

Disabled	If the checkbox is ticked, the plugin will not be displayed in Geeqie menus.
Name	Plugin name as specified in desktop file, and is the name displayed in menus.
Hidden	A plugin can be *Hidden* for one of these reasons: • the desktop file contains `Hidden=TRUE` or `NoDisplay=TRUE` • `TryExec` binary was not found • `MimeType` list does not contain images • `MimeType` list is empty and `Categories` does not contain `X-Geeqie`
Desktop file	Name of the desktop file, used as an identifier in Keyboard Options .
Path	Full path to the desktop file. Desktop files in user directories override the system ones with the same name.

Adding new plugin

`New` opens a text editor with a desktop file template. You must amend the line

```
Exec=command %f
```

to contain the command you wish to execute. If it is more than a single command, you must create a script file and call that. If the location of the script file is not in your $PATH environment variable, you must include the full pathname.

Modifying an existing plugin

`Edit` opens a text editor with existing desktop file. For desktop files that are not writable by user, it allows saving to a Geeqie specific directory, where it overrides the system file (but only for Geeqie).

Deleting a plugin

`Delete` can delete user writable desktop files. System desktop files can't be deleted directly, but it is possible to edit them and set `Hidden=TRUE` , see above.

Convert image

Choose Convert image… from the Plugins menu to create JPEG or PNG copies of the selected images. This plugin requires Zenity and ImageMagick (the **magick** or **convert** program).

Choose JPEG or PNG and a destination folder. For JPEG, also choose quality (1–100; default 90). PNG uses lossless compression and preserves transparency; JPEG renders transparent areas white. Originals are preserved. Existing destination names are skipped and reported in the plugin output

window, along with conversion failures. Files with the same base name share a destination name, so only the first successful conversion is saved.

Images are automatically oriented, and metadata, including embedded color profiles, is retained where supported by ImageMagick and the output format. Only the first frame or page is converted. Supported input formats, including RAW formats, depend on the installed ImageMagick delegates and security policy. This operation converts the original file; it does not export Geeqie's displayed adjustments.

The separate Export jpeg plugin extracts an existing embedded preview from a RAW file.

Symlink grouped files

Drop files onto a writable folder in the folder pane, including the parent folder entry, and choose Symlink grouped files. The plugin creates symbolic links for the selected files and their grouped sidecars.

The plugin reports an error if grouped-file or destination lookup fails, a destination already exists, or a link cannot be created. It stops at the first failure. Existing files, directories and symbolic links are not replaced; links already created before a failure remain in place. Enable `X-Geeqie-Verbose=true` in the plugin's desktop file to display its diagnostic output.

Special plugins

A desktop file with one of the following names has a special function. It will replace the corresponding internal command.

```
geeqie-copy-command.desktop
geeqie-move-command.desktop
geeqie-rename-command.desktop
geeqie-delete-command.desktop
geeqie-folder-command.desktop
```

This can be used for example for a custom trash command or for manipulation of files under version control.

Geeqie desktop file keys

A desktop file for use only by Geeqie should have the following entries:

```
Categories=X-Geeqie;
OnlyShowIn=X-Geeqie;
```

Plugins appear in the Plugins menu and in plugin context menus. The legacy `X-Geeqie-Menu-Path` key does not control menu placement in GTK4; omit it from new desktop files.

To let the user choose a destination folder for a plugin, add this entry to its desktop file:

```
X-Geeqie-Filter=true
```

The plugin will appear in the Folder action list of the Sort Manager Sidebar . Clicking a bookmark uses that folder as the destination. When you run the plugin from the Plugins menu, Geeqie asks you to choose a destination folder.

In the plugin script, use `geeqie --get-destination="$file"` to get the destination folder with the file's name appended. For example, if the file is `/photos/image.jpg` and you choose

`/backup`, the command returns `/backup/image.jpg`. It only returns the path; the script must perform the operation.

```
for file in "$@"; do
  destination=$(geeqie --get-destination="$file")
  printf '%s\n' "$destination"
done
```

The same path is available as `$GEEQIE_DESTINATION` for the file currently being processed:

```
printf '%s\n' "$GEEQIE_DESTINATION"
```

Note

A restart of Geeqie is required for this plugin to be registered

If you want to run a plugin in full-screen mode and wish full-screen to be maintained, include the following entry in the desktop file:

```
X-Geeqie-Keep-Fullscreen=true
```

Any terminal output from the plugin command can be displayed with the following command:

```
X-Geeqie-Verbose=true
```

Any terminal output from the plugin command can be displayed, only when multiple files are selected, with the following command:

```
X-Geeqie-Verbose-Multi=true
```

The plugin can be restricted to run on only certain file types, for example:

```
X-Geeqie-File-Extensions=.jpg; .cr2
```

The entries are case insensitive.

The key or key combination to execute this function can be set with:

```
X-Geeqie-Hotkey=
```

Key combinations are of the form:

```
X-Geeqie-Hotkey=<control>y
```

Note

This value may conflict with a setting in Keyboard Options

Legacy Configuration File Options

This section describes the options not accessible by the graphical interface, these options exist to maintain behavior of older releases of Geeqie. To change these settings, edit the Geeqie configuration file .

lazy_image_sync: [true *false*] : This will suppress the image view from changing when selecting a new folder; the current image will remain displayed until the user manually selects another image.

place_dialogs_under_mouse: [true *false*] : Enabling this will cause new dialogs to initially position themselves such that they are under the current mouse position.

scroll_reset_method: value : This will set the method used to set the scroll position, or visible region of a new image, value can be:

0: Move to upper left corner.

1: Center image.

2: Maintain same visible region as previous image, if possible.

Chapter 13. Reference

Command Line Options

Name

Geeqie — GTK based multiformat image viewer

DESCRIPTION

Geeqie is an interactive GTK based image viewer that supports multiple image formats, zooming, panning, thumbnails and sorting images into collections.

Generated for version:

Usage:

geeqie [OPTION…] [path…]

Geeqie is an image viewer.

Version: Geeqie 3.3

Help Options:

-h, **--help**	Show help options
--help-all	Show all help options
--help-gapplication	Show GApplication options

Application Options:

--action=<ACTION>	execute keyboard action (https://www.geeqie.org/help/GuideReferenceRemoteKeyboardActions.html)
--action-list	list available keyboard actions (some are redundant)
-b, **--back**	previous image
--cache-metadata	clean the metadata cache
--cache-render=<folder>	render thumbnails
--cache-render-recurse=<folder>	render thumbnails recursively
--cache-render-shared=<folder>	render thumbnails (see Help)
--cache-render-shared-recurse=<folder>	render thumbnails recursively (see Help)
--cache-shared=*clean*|clear	clear or clean shared thumbnail cache
--cache-thumbs=*clear*|clean	clear or clean thumbnail cache
--close-window	close window
--config-load=<FILE>	load configuration from FILE
-d, **--delay=**<[H:][M:][N][.M]>	set slide show delay to Hrs Mins N.M seconds,
--dupes=<FOLDER>	find duplicates in folder
--dupes-export	export duplicates search result
--dupes-recurse=<FOLDER>	find duplicates in folder recursively

--file=<FILE>|<URL>	open FILE or URL bring Geeqie window to the top
--File=<FILE>|<URL>	open FILE or URL do not bring Geeqie window to the top
--file-extensions	list known file extensions
--first	first image
-f, **--fullscreen**	start / toggle in fullscreen mode
--geometry=<W>x<H>[+<XOFF>+<YOFF>]	set main window location and geometry
--get-collection=<COLLECTION>	get collection content
--get-collection-list	get collection list
--get-destination=<FILE>	get destination path of FILE (https://www.geeqie.org/help/GuidePluginsConfig.html)
--get-file-info	get file info
--get-filelist=*[*<FOLDER>]	get list of files and class
--get-filelist-recurse=*[*<FOLDER>]	get list of files and class recursive
--get-rectangle	get rectangle coordinates
--get-render-intent	get render intent
--get-selection	get list of selected files
--get-sidecars=<FILE>	get list of sidecars of FILE
--get-window-list	get list of windows
--id=<ID>	window id for following commands
--last	last image
-o, **--log-file=**<file>	save log data to file
--lua=<FILE>,<lua script>	run lua script on FILE
--new-window	new window
-n, **--next**	next image
--pixel-info	print pixel info of mouse pointer on current image
--print0	terminate returned data with null character instead of newline
-q, **--quit**	quit
--raise	bring the Geeqie window to the top
--selection-add=*[*<FILE>]	adds the current file (or the specified file) to the current selection
--selection-clear	clears the current selection

--selection-remove=*[*<FILE>]	removes the current file (or the specified file) from the current selection
-w, **--show-log-window**	show log window
--slideshow-recurse=<FOLDER>	start recursive slide show in FOLDER
-s, **--slideshow**	toggle slide show
--tell	print filename [and Collection] of current image
-t, **--tools**	toggle tools
-v, **--version**	print version info
--view=<FILE>	open FILE in new window

All other command line parameters are used as plain files if they exist, or a URL or a folder. The name of a collection, with or without either path or extension (.gqv) may be used.

If more than one folder is on the command line, only the last will be used.

If more than one file is on the command line:

If they are in the same folder, that folder will be opened and those files will be selected. If they are not in the same folder, a new Collection containing those files will be opened.

To run Geeqie as a new instance, use: GQ_NEW_INSTANCE=y[es] geeqie Normally a single set of configuration files is used for all instances. However, the environment variables XDG_CONFIG_HOME, XDG_CACHE_HOME, XDG_DATA_HOME can be used to modify this behavior on an individual basis e.g. XDG_CONFIG_HOME=/tmp/a XDG_CACHE_HOME=/tmp/b GQ_NEW_INSTANCE=y geeqie

To run or stop Geeqie in cache maintenance (non-GUI) mode use: GQ_CACHE_MAINTENANCE=y[es] geeqie **--help** Note that bash command line completion does not work in this mode.

User manual: https://www.geeqie.org/help/GuideIndex.html

: https://www.geeqie.org/help-pdf/help.pdf

BUGS

Please send bug reports and feedback to https://github.com/BestImageViewer/geeqie/issues

COPYRIGHT

Copyright (C) 1999-2004 by John Ellis. Copyright (C) 2004-2026 by The Geeqie Team. Use this software at your own risk! This software released under the GNU General Public License. Please read the COPYING file for more information.

Use `--color-scheme=system|light|dark` to override the interface color scheme for the running session. This does not change the saved Color scheme preference.

Independent Instances

Normally, launching Geeqie again sends the command to the running application instance. Opening another window does not create an independent instance. **Ctrl**+**Q** quits that instance and closes all its windows; **Ctrl**+**W** closes only the current window, quitting the instance if it is the last main window.

To start an independent application instance, use:

```
GQ_NEW_INSTANCE=y geeqie /path/to/folder
```

Quitting this instance does not close windows belonging to another instance. By default, independent instances share configuration and state files. Each instance saves configuration on exit, so the last one to exit can overwrite settings saved by another instance.

To keep a separate persistent configuration, set a different configuration root:

```
XDG_CONFIG_HOME="$HOME/.config/geeqie-work" GQ_NEW_INSTANCE=y geeqie /path/to/f
```

This instance uses a separate configuration under `$HOME/.config/geeqie-work/geeqie`. The `XDG_CACHE_HOME` and `XDG_DATA_HOME` environment variables can also be set to separate cache and data locations. Separate configurations do not isolate changes made to image files or their metadata.

Remote Keyboard Actions

The command:

```
geeqie --action=<action>
```

can be used to simulate keyboard actions. An example is:

```
geeqie --action="ZoomIn"
```

The actions available are shown in the Edit/Preferences/Keyboard dialog. The two end columns are relevant. The final component of the Accel column is the action code that should be used. The Tooltip column provides a description of the action. Both columns can be sorted by clicking the header.

The available action codes may also be listed by:

```
geeqie --action-list
```

Thumbnails

Note

The storage layout below describes Geeqie's private thumbnail cache. The shared thumbnail cache format is described in Thumbnail Standards .

When Geeqie detects that a tracked source file has changed or been replaced, it discards its cached thumbnails and reloads metadata, including orientation. Detection includes file identity and subsecond modification and status-change timestamps where the platform supports them. Cached thumbnails are retained when a source becomes unavailable, allowing missing collection members to keep their previews.

Shared thumbnails written by Geeqie also record the source file size. A cached thumbnail with a different recorded size is rejected even when its modification time matches. Older thumbnails without

a size field remain supported. These checks cannot detect every replacement made while a file is not being tracked, particularly when cache-validation values are unchanged. Use Rebuild thumbnails in the files pane context menu to regenerate previews for selected files.

Format

Thumbnails are stored in PNG image format. The thumbnail name is the name of the source image with ".png" appended.

The modification time (mtime) of the thumbnail is set to match the source file. Thumbnails are regenerated when the timestamps of the thumbnail and source file do not match.

Location

Thumbnails are stored in a location specified in Thumbnail Preferences .

The directory structure of the thumbnail cache duplicates that of the location of the source files.

Size

Geeqie allows the following sizes for thumbnails:

24x24, 32x32, 48x48, 64x64, 96x72, 96x96, 129x96, 128x128, 160x120, 160x160, 192x144, 192x192, 256x192, 256x256, custom

Custom size has a maximum of 512x512.

The thumbnail is scaled to fit within the preferred size maintaining the aspect ratio. Thumbnails are not cached for images that are equal to or smaller than the preferred thumbnail size.

When a cached thumbnail's width and height do not match the preferred size, the thumbnail is regenerated.

Metadata Processing

This section describes the metadata reading and writing process.

Metadata templates

Select files in the files pane, then choose Edit → Metadata Templates… to apply description, headline, keywords, copyright, creator, and credit together. The dialog applies to the selection captured when it was opened.

Choose one action for each field: Leave unchanged, Replace, Append, or Clear. New templates start with every field unchanged. Replace overwrites existing values. Text append adds the value after a newline; applying the template again repeats the appended text. List append adds values without duplicates. Replace and Append ignore blank values. Clear explicitly removes existing metadata. Text entry is available only for Replace and Append. Keywords can be separated by commas or newlines.

Choose New template to start with blank fields. Enter a name under Save as and choose Save template to store the values and per-field actions. Saving with an existing name replaces that template; use another name to create a copy. Choose a saved template from the list to load it, or choose Delete template to remove it. Editing or applying values does not automatically save the template.

Expand Add field to add more metadata fields. Choose a suggested field, or enter a label and a simple XMP metadata key such as `Xmp.photoshop.City`. Keys can be copied from the Exif window. Select List of values (one per line) for list fields, then choose Add field. Duplicate keys are rejected. Custom fields support the same four actions and blank-value behavior as the standard fields, and their labels, keys, values, list settings, and actions are saved with the template.

Remove field removes a custom field from the current template without changing file metadata. Save the template to keep the change. Loading a template restores its own custom fields; New template starts with the standard fields.

Choose Apply to selection to queue the changes using the same metadata save and grouped-file settings as the info sidebar. Use Save metadata to write queued changes manually. Templates are stored as readable INI files in the `metadata-templates` directory inside the Geeqie configuration directory. Filenames use the template name with unsuitable characters replaced by underscores. Numeric suffixes distinguish filenames that collide, including differences only in letter case. Saving an exact existing template name updates that template. Templates saved with the previous hashed filenames remain usable.

Metadata handling

Read algorithm

1. Read Exif, IPTC and XMP from the file.
2. Display this "raw" data in the Exif Window .
3. Read XMP sidecar or Geeqie private metadata file. This data will override the XMP data read from the file.
4. Synchronise XMP data with Exif and IPTC data. This process corresponds to the section "Reconciling metadata properties" as described in the XMP Specification .
   - detect which of Exif and Xmp.exif is newer and copy the data in the appropriate direction
   - detect which of IPTC and XMP is newer and copy the data in the appropriate direction
5. Use this "processed" data within Geeqie, with the exception of in the Exif Window

Metadata modification

- Metadata is written to a single file or group of files as specified in the Metadata tab of main Preferences.
- Modifying the keywords, description, orientation, etc. does not ask for confirmation, but immediately modifies the metadata in memory and adds the file to "write queue"
- Writing of the queued changes is triggered either manually or as defined in Metadata. It is possible to review the changes and exclude selected files.

Write algorithm

1. Exif and IPTC are updated from XMP (for example *Exif.Image.Orientation* is updated from *Xmp.exif.orientation*). See XMP and IPTC4XMP specifications for the list of corresponding tags.
2. Try to write metadata in the order specified in the Metadata writing process section of Preferences.
3. After the first successful write attempt delete old metadata files in all possible locations.

Options that influence the process

File types with allowed sidecars are listed in the File Filters Options tab of the main Preferences.

Lua Extensions

Lua scripts allow the functionality of Geeqie to be extended. Lua scripts may only be used in conjunction with the Overlay Screen Display , List panes on the Info Sidebar, and the

```
geeqie --lua:
```

command line option.

Some knowledge of the Lua programming language is required.

Requirements

Use of Lua within Geeqie requires Lua to be installed. If you are compiling from sources, Lua functionality will be available if the development files dependencies are met.

If you are using a pre-compiled distribution, availability depends on the package maintainer.

How to use Lua

Lua scripts must be stored in a single folder as defined in Configuration Files and Locations .

A link to a Lua script must be inserted into the overlay template. Refer to the Overlay Screen Display section of Window Options.

The full extent of the Lua language is available.

Geeqie Lua built-in functions

The following functions are built in to Geeqie:

Function	**Returns**
Image:get_path()	The full path of the file, including filename and extension
Image:get_name()	The full filename including extension
Image:get_extension	The file extension including preceding dot
Image:get_date()	The file date in Unix timestamp format.
Image:get_size()	The file size in bytes
Image:get_marks()	An integer representing the marks set for the file
Image:get_exif()	A data structure containing the entire exif data
<exif_str>:get_datum("<exif_tag>")	A single exif tag extracted from a structure output by the above command

The keyword "Image" refers to the file currently being displayed by Geeqie.

Examples

The following example, which outputs the jpeg comment of a file, demonstrates the use of a built-in function and how to call a system command from a Lua script:

```
path=Image:get_path()
commentfile=io.popen("exiv2 -p c \"" .. path .. "\"" )
comment = commentfile:read("*a")
commentfile:close()
return (comment)
```

Note that it is necessary to have escape characters surrounding path and filenames.

The following example demonstrates how to extract exif data from a file:

```
--Retrieve the DateTimeDigitized exif tag
exif_structure = Image:get_exif();
DateTimeDigitized =
     exif_structure:get_datum("Exif.Photo.DateTimeDigitized");
return (os.date(DateTimeDigitized))
```

Warning

Warning

Lua is a powerful programming language. Errors in script files, besides having undesirable side-effects, may cause Geeqie to crash.

Lua API

The Lua API document [./lua-api/html/lua_8cc.html]

Configuration Files and Locations

The following data lists the locations Geeqie uses for various actions. The uppercase symbols are environment variables. If they are not set on your system the fallback locations are listed in parentheses.

Geeqie will first attempt to load a configuration file from:

`/etc/geeqie/geeqierc.xml`

It will then continue with the following locations.

Most of Geeqie's configuration files are contained in the folder, and sub-folders of:

`$XDG_CONFIG_HOME/geeqie/`

`($HOME/.config/geeqie/)`

Geeqie's standard configuration file is:

`.../geeqierc.xml`

An alternative configuration file may be used by executing:

`geeqie --config-load:<filename>`

Saved window layout files are in the folder:

`.../layouts`

Geeqie-created desktop files used by Plugins are in the folder:

`.../applications`

Lua script files for Lua Extensions are in the folder:

`.../lua`

Historic data such as last several folders visited, bookmarks, and recently used collections, as well as default print settings are contained in this text file:

`.../history`

The `[collection_history]` section records recently opened or saved collection files. Open collection combines these entries with files from the default collections directory. Unavailable paths are retained and shown as disabled entries. Open recent uses GTK's shared recent-files database separately. The obsolete `[recent]` section is discarded when the history file is loaded.

Keyboard shortcut maps are contained in this text file:

`.../accels`

The location for Collections is in the folder:

`$XDG_DATA_HOME/geeqie/collections`

`($HOME/.local/share/geeqie/collections)`

Thumbnails are stored in a location specified in Thumbnail Preferences

Metadata is stored either in the image file or in the location specified in Metadata

Similarity files are stored in the folder:

`($HOME/.cache/geeqie/)`

The safe delete folder is specified in the Safe Delete tab of main Preferences

File date types

Geeqie understands 4 types of file dates:

- Standard date (mtime) - The date the file contents were last modified, or when the file was created if since unchanged.
- Status changed (ctime) - The date the file status was last changed. This includes change of owner, change of read/write permission and change of contents.
- Exif date original (Exif.Photo.DateTimeOriginal) - For images from a digital camera, this is the time the photo was taken.
- Exif date digitized (Exif.Photo.DateTimeDigitized) - For scanned images, this the time the scan was made. For images from a digital camera, it may be identical to the above date.

XMP, Exif and IPTC

The tables below show the XMP, Exif and IPTC tags used by Geeqie.

Geeqie pre-formatted tags

Table 13.1. Pre-formatted tags

Geeqie Tag	Exif tags used	Description
formatted.Camera	Exif.Image.Make Exif.Image.Model Exif.Image.Software	Camera and software name
formatted.DateTime	Exif.Photo.DateTimeOriginal Exif.Photo.SubSecTimeOriginal Exif.Image.DateTime Exif.Photo.SubSecTime	Image Date
formatted.MediaDate	Exif.Photo.DateTimeOriginal Xmp.video.DateUTC Xmp.video.TrackCreateDate Xmp.video.TrackModifyDate File modification date	Media date
formatted.ShutterSpeed	Exif.Photo.ExposureTime Exif.Photo.ShutterSpeedValue	Shutter speed in seconds (e.g. 1/60s)
formatted.Aperture	Exif.Photo.Fnumber Exif.Photo.ApertureValue	Aperture (e.g. f5.6)
formatted.ExposureBias	Exif.Photo.ExposureBiasValue	Exposure bias (e.g. +0.3)
formatted.ISOSpeedRating	Exif.Photo.ISOSpeedRatings	ISO sensitivity (e.g. 100)
formatted.FocalLength	Exif.Photo.FocalLength	Focal length
formatted.FocalLength35mmFilm	Exif.Photo.FocalLengthIn35mmFilm Exif.Photo.FocalLength Exif.Photo.FocalPlaneXResolution Exif.Photo.FocalPlaneYResolution Exif.Photo.FocalPlaneResolutionUnit Exif.Photo.PixelXDimension Exif.Photo.PixelYDimension	Focal length 35mm
formatted.SubjectDistance	Exif.Photo.SubjectDistance	Subject distance
formatted.Flash	Exif.Photo.Flash	Flash
formatted.Resolution	Exif.Image.XResolution Exif.Image.YResolution Exif.Image.ResolutionUnit	Recorded horizontal and vertical image density, for example `300 x 300 (dot/inch)`. This is separate from the image dimensions in pixels and the monitor's physical pixel density.

Geeqie Tag	Exif tags used	Description
		Valid EXIF density values take precedence. EXIF density retains its recorded unit, which may be per inch or per centimeter; a missing unit is shown as unknown. If either EXIF density value is missing or invalid, Geeqie uses JPEG JFIF density or PNG pHYs density when available. JFIF density retains its recorded unit; PNG pixels per meter are converted to pixels per inch. Values that specify only pixel aspect ratio are not displayed. If no usable density is available, the tag is empty.
formatted.ColorProfile	Exif.Image.InterColorProfile Exif.Photo.ColorSpace Exif.Iop.InteroperabilityIndex	Color profile
formatted.GPSAltitude	Exif.GPSInfo.GPSAltitudeRef Exif.GPSInfo.GPSAltitude	Altitude above sea level
formatted.GPSPosition	Exif.GPSInfo.GPSLatitude Exif.GPSInfo.GPSLatitudeRef Exif.GPSInfo.GPSLongitude Exif.GPSInfo.GPSLongitudeRef	Latitude, Longitude
formatted.localtime	Exif.GPSInfo.GPSLatitude Exif.GPSInfo.GPSLatitudeRef Exif.GPSInfo.GPSLongitude Exif.GPSInfo.GPSLongitudeRef Exif.GPSInfo.GPSDateStamp Exif.GPSInfo.GPSTimeStamp	Local time corrected for UTC offset, DST correction [a]
formatted.timezone	Exif.GPSInfo.GPSLatitude Exif.GPSInfo.GPSLatitudeRef Exif.GPSInfo.GPSLongitude Exif.GPSInfo.GPSLongitudeRef	Timezone indicated by lat/long [b]
formatted.countryname	Exif.GPSInfo.GPSLatitude Exif.GPSInfo.GPSLatitudeRef Exif.GPSInfo.GPSLongitude	ISO 3166 country name indicated by lat/long

Geeqie Tag	Exif tags used	Description
	Exif.GPSInfo.GPSLongitudeRef	
formatted.countrycode	Exif.GPSInfo.GPSLatitude Exif.GPSInfo.GPSLatitudeRef Exif.GPSInfo.GPSLongitude Exif.GPSInfo.GPSLongitudeRef	ISO 3166 two-letter abbreviated country name indicated by lat/long
formatted.star_rating	Xmp.xmp.Rating	Rating shown as a set of # characters

[a] Refer to UTC and DST section.

[b] Refer to UTC and DST section.

Info sidebar panes

Table 13.2. Fixed panes

Tag	Pane title
Xmp.dc.description	Description
Xmp.dc.subject	Keywords
Xmp.dc.title	Title
Xmp.xmp.Rating	Rating
Xmp.photoshop.Headline	Headline

Table 13.3. Location and GPS pane

Tag	Field
formatted.GPSPosition	GPS position
formatted.GPSAltitude	GPS altitiude
formatted.timezone	Time zone
Xmp.photoshop.Country	Country
Xmp.iptc.CountryCode	Country Code
Xmp.photoshop.State	State
Xmp.photoshop.City	City
Xmp.iptc.Location	Location

Table 13.4. Copyright pane

Tag	Field
Xmp.dc.creator	Creator
Xmp.dc.contributor	Contributor
Xmp.dc.rights	Rights

Table 13.5. Exif pane

Tag	Field
formatted.Camera	Camera
formatted.DateTime	Date

Tag	Field
formatted.MediaDate	Media date
formatted.localtime	Local time
formatted.ShutterSpeed	Shutter speed
formatted.Aperture	Aperture
formatted.ExposureBias	Exposure bias
formatted.ISOSpeedRating	ISO sensitivity
formatted.FocalLength	Focal length
formatted.FocalLength35mmFilm	Focal length 35mm
formatted.SubjectDistance	Subject distance
formatted.Flash	Flash
formatted.Resolution	Resolution
formatted.ColorProfile	Color profile
Exif.Photo.ExposureProgram	Exposure Program
Exif.Photo.MeteringMode	Metering Mode
Exif.Photo.LightSource	Light Source

Exif IPTC alternative keys

Table 13.6. Conversion table

Tag	Field
Xmp.tiff.Orientation	Exif.Image.Orientation
Xmp.dc.title	Iptc.Application2.ObjectName
Xmp.photoshop.Urgency	Iptc.Application2.Urgency
Xmp.photoshop.Category	Iptc.Application2.Category
Xmp.photoshop.SupplementalCategory	Iptc.Application2.SuppCategory
Xmp.dc.subject	Iptc.Application2.Keywords
Xmp.iptc.Location	Iptc.Application2.LocationName
Xmp.photoshop.Instruction	Iptc.Application2.SpecialInstructions
Xmp.photoshop.DateCreated	Iptc.Application2.DateCreated
Xmp.dc.creator	Iptc.Application2.Byline
Xmp.photoshop.AuthorsPosition	Iptc.Application2.BylineTitle
Xmp.photoshop.City	Iptc.Application2.City
Xmp.photoshop.State	Iptc.Application2.ProvinceState
Xmp.iptc.CountryCode	Iptc.Application2.CountryCode
Xmp.photoshop.Country	Iptc.Application2.CountryName
Xmp.photoshop.TransmissionReference	Iptc.Application2.TransmissionReference
Xmp.photoshop.Headline	Iptc.Application2.Headline
Xmp.photoshop.Credit	Iptc.Application2.Credit
Xmp.photoshop.Source	Iptc.Application2.Source
Xmp.dc.rights	Iptc.Application2.Copyright
Xmp.dc.description	Iptc.Application2.Caption

Tag	Field
Xmp.photoshop.CaptionWriter	Iptc.Application2.Writer

Collection file format

A collection is a UTF-8 text file with the `.gqv` extension. Scripts can generate collections directly using the same format as files saved by Geeqie. A minimal example is:

```
#Geeqie collection
"/home/user/Pictures/first.jpg"
"/home/user/Pictures/Family holiday/second.jpg"
#end
```

Write one image path per line, enclosed in ordinary double quotation marks. Spaces inside a quoted path are preserved. The `#created with Geeqie version` and `#geometry:` comments found in collections saved by Geeqie are optional; scripts do not need to generate them.

Use absolute paths, or relative paths beginning with `./` or `../`. Relative paths are resolved against the folder containing the collection file, not the terminal's working directory. For example, a collection stored in `/home/user/Pictures/example.gqv` can contain:

```
#Geeqie collection
"./first.jpg"
"./Family holiday/second.jpg"
"../Scans/third.jpg"
#end
```

Geeqie recognizes these relative paths and retains relative-path storage when the collection is saved. See Collections for the Save As option controlling relative paths.

Paths are literal: shell expressions such as `~`, `$HOME`, and wildcards are not expanded. A script must expand them before writing the file. Backslashes are not an escape mechanism, and filenames containing double quotation marks cannot be represented reliably by this format.

The collection may be stored outside Geeqie's collections folder. Open it using File → Open collection..., or supply its path on the command line:

```
geeqie /home/user/Pictures/example.gqv
```

It is possible to include informational text along with an image. The format is:

```
#Geeqie collection
#i Some informational text for the following file
"<filepath1>"
#end
```

This text can be displayed from the files pane when viewing a Collections.

To add or edit this text in Geeqie, display the collection in the files pane and select Edit info text from the Info text submenu of a file's context menu. The text is stored when the collection is saved.

Supported File Formats

3FR ANI ARW AVIF BMP CR2 CR3 CRW CUR DDS DJVU DNG ERF EXR GIF GQV HEIC HEIF ICO JP2 JPE JPEG JPG JPS JXL KDC MEF MOS MPO MRW NEF NPY NRW ORF PBM

PDF PEF PGM PNG PNM PPM PSD QIF QTIF RAF RAW RW2 SCR SR2 SRF SVG SVGZ TGA TIF TIFF WEBP XBM XPM

Images in archive files (.ZIP, .TAR etc) can be displayed.

Animated GIF and WEBP files are supported.

Refer to Additional pixbuf loaders for additional information.

Preview and thumbnails of video clips can be displayed. Clips can be run via a defined external program - see Play video by left click on image .

RAW image previews

Geeqie normally displays an embedded preview from a camera RAW file, rather than developing the RAW sensor data. For main-image display, the Exiv2 preview loader selects the largest available embedded preview. LibRaw support also extracts a preview; it does not enable RAW development.

An embedded preview can have a lower resolution than the RAW sensor image. For example, some Sony ARW files do not contain a full-resolution preview. Its colors and detail can also differ from the output of a RAW converter. A zoom level of 100% refers to the displayed preview, not to the resolution of the RAW sensor data.

To display a full-resolution rendering, develop the RAW file with an external RAW converter and open its output in Geeqie. Alternatively, configure External preview extraction to run a converter that develops the RAW data into an image format Geeqie can display. Extracting an embedded preview alone cannot recover detail absent from that preview.

Similarity Algorithms

This function is intended to find images with similar color content. For example when an image was saved at different compression levels or dimensions (scaled down/up) the contents are similar, but these files do not match by file size, dimensions, or checksum.

A 32 x 32 array is created for each image. Imagine the image cut into 1024 rectangles, 32 across and 32 down.

For each array element, the average value of all the red and the green and the blue pixels is computed and stored in the array. Therefore the array represents the average color of each corresponding part of the image.

This data is stored in a file with the same name as the image and with the extension .sim. It is stored in the same location as thumbnails. If many images are to be compared, run-time is reduced by having these .sim files already created. This can be done via Edit/Cache Maintenance or by the command line instruction: `GQ_CACHE_MAINTENANCE=y geeqie --cache-maintenance=<path>`

Standard Algorithm

To compare two images, each array element of each image is compared in turn. The computed value is the percent match of all elements of the two images. For this, simple comparisons are used - basically the value is an average of the corresponding array differences.

The value computed is in the range 0% to 100%.

100% for exact matches (an image is compared to itself)
0% for exact opposite images (compare an all black to an all white image)

Generally only a match of >85% is significant at all, and >95% is useful to find images that have been re-saved to other formats, dimensions, or compression.

If the Ignore Orientation checkbox on the Duplicates window is selected, images are also checked for 90°, 180°, 270°, rotations and mirror and flip. This will increase run-time.

Alternate Algorithm: Perceptual Hash (pHash)

Select Perceptual hash algorithm from the algorithm dropdown in the Duplicates window. The button shows Perceptual or Standard, and changing the selection restarts comparisons in open Duplicates windows. The selection also applies to image searches and similarity cache creation. This replaces the previous alternate algorithm.

Geeqie converts the 32 x 32 RGB grid to grayscale and computes a discrete cosine transform (DCT). The low-frequency 8 x 8 block describes coarse image structure. The constant brightness (DC) coefficient is excluded, and the remaining 63 coefficients are compared with their median to produce a 64-bit hash with one unused bit. This is a Geeqie-specific DCT pHash; hashes from other applications may use different preprocessing and are not interchangeable.

When pHash is selected, similarity cache creation stores eight hashes in each .sim file: one for every combination of right-angle rotation and reflection. These are derived from one DCT using coefficient transposition and sign changes. The RGB grid is retained, so switching back to the standard algorithm does not require rebuilding it. Existing caches without hashes are upgraded from their grids during cache maintenance or a similarity search, without decoding the source images again. Hashes already in a cache are retained when pHash is disabled.

Two images are compared by counting differing hash bits (Hamming distance). The similarity percentage is `100 x (1 - distance / 63)`. When Ignore Orientation is selected in the Duplicates window, the best match between both images' eight orientations is used; otherwise only the original orientation is compared. Considering both sets makes the score independent of image order and initial orientation, and requires up to 64 short hash comparisons per image pair. Exact hash matches score 100%, but this does not imply identical files or pixels.

The high (95%), medium (90%), and low (85%) thresholds allow at most 3, 6, and 9 differing bits respectively. These scores are not equivalent to the standard algorithm's percentages. Use the custom threshold to adjust results for your images. pHash ignores color and is tolerant of uniform brightness and contrast changes; unrelated images with similar coarse structure, including uniformly colored images, can match. Cropping and arbitrary-angle rotations are not explicitly handled.

Cached hashes make each comparison much cheaper than comparing RGB grids. Searches still compare image pairs, so large collections can take significant time. Enable pHash before an overnight cache-maintenance run to prepare all eight hashes in advance. The command line cache-maintenance program uses the saved algorithm selection, so save the configuration before starting it.

Additional pixbuf loaders

Geeqie can display files for which there are existing pixbuf loaders. In addition to the standard ones included with most distributions, the following loaders may also be available to you.

However, in some cases you are required to compile and install them yourself.

JP2/JPC/JPX/J2K/JPF

If your distribution has the libpixbufloader-jasper.so loader, these file formats will also be displayed.

Standard Plugins

The following plugins are supplied as standard with Geeqie.

Apply the orientation to image content

This command affects only jpeg, tiff and png files. The image file on disk is rotated to the orientation as specified by its Exif Orientation tag.

Warning

External commands exiftran, exiv2 and mogrify, are used to implement this command.

This item is displayed in the **Edit/Orientation** menu.

Camera import

Imports images from an attached camera to the current folder. If a file with the same name already exists it will not be overwritten, and the new image will not be downloaded.

File names on digital cameras are not necessarily unique. In some cases the same file name can exist in more than one sub-folder on the camera. As existing images will not be overwritten, subsequent files of the same name will not be downloaded. In this case a dialog will be shown to warn you of this.

You may use the gphoto2 option `--filename` option to ensure that all filenames are unique when downloading. Refer to the gphoto2 documentation. [http://www.gphoto.org/doc/manual/ref-gphoto2-cli.html]

This item is displayed in the **Plugins** menu.

Exif datetime to file

Sets the file datetime (mtime) to the Exif.Image.DateTimeOriginal value. If the exif tag does not exist, no action is taken.

This may be useful if you wish to sort on Exif.Image.DateTimeOriginal. This requires that the metadata for all files in the folder must be read each time. This is time consuming. If this operation is carried out when images are imported from the camera, future sorting can utilize file datetime which will be much faster.

This item is displayed in the **Plugins** menu.

Export jpeg

Extracts an embedded jpeg from a raw image to a file in the `/tmp` folder. If you wish to to keep this image, it must be copied or moved.

If `jpgicc` is installed, the image will be colour-corrected for the currently selected rendering intent.

This item is displayed in the **Plugins** menu.

Open With

Call the standard GTK application chooser for a single file.

This item is also displayed in the **File** menu.

Random image

Selects and displays a random image from a list of all images in Collections and the currently displayed folder.

This item is displayed in the **Plugins** menu.

Resize image

Reduce the size of an image.

Table 13.7. Resize image options

Option	Meaning
Required size kB	desired output file size in kilobytes
Tolerance %	tolerance or allowed size of result greater than desired size expressed as percent of desired
Max. iterations	maximum number of iterations to stop
Copy if unchanged	copy to output when not downsizing and no image format change
Strip metadata	strip all meta data
Show computation	show the values of each iteration
Open output file in Geeqie	on successful conversion set Geeqie focus to the file

This plugin uses the Downsize script. The original may be found here. [http://www.fmwconcepts.com/imagemagick/downsize/index.php] It is subject to the following copyright:

```
Copyright © Fred Weinhaus

My scripts are available free of charge for non-commercial (non-profit) us

For use of my scripts in commercial (for-profit) environments
or non-free applications,
please contact me (Fred Weinhaus) for licensing arrangements.
My email address is fmw at alink dot net.

If you:
1) redistribute,
2) incorporate any of these scripts into other free applications or
3) reprogram them in another scripting language, then you must contact me
especially if the result might be used in a commercial or for-profit envi

Usage, whether stated or not in the script, is restricted to the above li
arrangements. It is also subject, in a subordinate manner, to the ImageMa
can be found at: http://www.imagemagick.org/script/license.php
```

Symlink

Creates a symbolic link on the selected files. If a file has sidecars, they will also be symlinked.

This item is displayed in the **File** menu.

Tethered photography

If a camera is connected to the computer via USB, photos taken are immediately downloaded to the current folder and displayed on the computer.

If a file already exists with the same name, it is not overwritten but the download will not take place.

Auto-rename can be achieved with the --filename option of gphoto2.

In file `$HOME/.config/gphoto2/settings` , set the option: `gphoto2=filename=<see the gphoto2 manual>`

This item is displayed in the **Plugins** menu.

UTC and Daylight Saving Time (DST)

Geeqie can display the local time at which a photo was taken, adjusted for UTC offset and Daylight Saving Time.

If the image exif data contains the required parameters (see Pre-formatted tags), Geeqie will use the latitude and longitude to determine which timezone the image was taken in.

The UTC offset and Daylight Saving Time Correction for that timezone, combined with the GPS UTC value, will then be used to compute the correct local time.

This value may be displayed in either the Info Sidebar or Overlay Screen Display by using the parameter `formatted.localtime`

The timezone for the image may also be displayed by using the parameter `formatted.timezone` .

Decoding Latitude and Longitude

This section is relevant to geocode searches and the map display of Geeqie.

The result of some internet or other searches for placenames can contain a latitude and longitude embedded in a text string. For example an openstreetmap.org search can give a URL such as:

```
https://www.openstreetmap.org/search?
query=51.5542%2C-0.1816#map=12/51.5542/-0.1818
```

If you paste such a string into the search box, the latitude/longitude can be automatically extracted and used as the origin of the search.

You may also drag-and-drop a URL of this type onto the map to cause the map to be re-centered.

This feature uses the file:

```
/usr/local/lib/geeqie/geocode-parameters.awk
```

You may create your own decodes by using the above file as an example, and storing the file in:

```
~/.config/geeqie/applications/geocode-parameters.awk
```

If a successful decode was not made from the first file, your local version will then be used.

Standards

Desktop files: https://specifications.freedesktop.org/desktop-entry-spec/desktop-entry-spec-latest.html

Desktop files: https://standards.freedesktop.org/desktop-entry-spec/ [https://specifications.freedesktop.org/desktop-entry-spec/latest/]

XMP: https://developer.adobe.com/xmp/docs/XMPSpecifications

IPTC4XMP: http://xml.coverpages.org/IPTC-CoreSchema200503-ImplGuideLines3.pdf

Pango mark up: https://docs.gtk.org/Pango/pango_markup.html

Thumbnails: https://www.freedesktop.org/wiki/Specifications/thumbnails/

Perl Compatible Regular Expressions

The Filename and Description sections on the search window use Perl Compatible Regular Expressions [https://en.wikipedia.org/wiki/Perl_Compatible_Regular_Expressions] . In general use there are a number of differences to the wildcard expansion used on the command line:

- Use "." and not "?" for a single character.
- Use "abc.*ghk" and not "abc*ghk" for multiple characters
- Use "\." to represent the dot in a file extension
- Use "(?i)" to make the following characters case-insensitive

Chapter 14. Useful Software

This is a list of other software that users might find useful in conjunction with Geeqie.

Entangle

Tethered photography: https://entangle-photo.org/

Photini

Digital photograph metadata editor: https://photini.readthedocs.io/en/latest/

Unison

Unison allows two replicas of a collection of files and directories to be stored on different hosts (or different disks on the same host), modified separately, and then brought up to date by propagating the changes in each replica to the other. It is available in some distributions. This is the project site: https://github.com/bcpierce00/unison/

Keyword Merge

This program will merge the keyword tree of one Geeqie configuration file into another. The keyword trees are simply concatenated. When Geeqie loads the resulting configuration file, any duplicates are discarded: keyword_merge.sh [https://raw.githubusercontent.com/BestImageViewer/geeqie/master/tools/keyword_merge.sh]

Keyword Marks

This program will list the current keyword-mark links that have been set in the Info Sidebar. The Geeqie configuration file is scanned to produce this list, therefore changes to keyword-marks will not be listed until Geeqie is closed and the config. file updated: keyword-mark.sh [https://raw.githubusercontent.com/BestImageViewer/geeqie/master/tools/keyword-mark.sh]

Install Geeqie from sources on Debian

This script will clone the Geeqie repository, fulfill all dependencies, and install Geeqie on a Debian based system: geeqie-install-debian.sh [https://raw.githubusercontent.com/BestImageViewer/geeqie/master/geeqie-install-debian.sh]

Generic install script

This script will install Geeqie to a defined location, and will run under any system. However, it is left to you to make sure dependencies are fulfilled: build-geeqie [https://raw.githubusercontent.com/pixlsus/Scripts/master/build-geeqie]

Recently used Collections

This script will list recently used Collections. This may be of use to someone who has Collections stored outside the default Collections folder: recent-collections.sh [https://raw.githubusercontent.com/pixlsus/Scripts/master/scripts/recent-collections.sh]

Decoder for .sim files

This program is unlikely to be of interest to most users. It shows a summary of the contents of a .sim file. (Requires perl to be installed): decode_sim [https://raw.githubusercontent.com/BestImageViewer/geeqie/master/tools/decode-sim]

Chapter 15. Frequently Asked Questions

Move to Trash failed

How do I fix move to trash failed?

It appears that some distributions will only trash files that are in the same filesytem as where the Trash bin is located.

The location of the Trash bin for some systems is `$HOME/.local/share/Trash` .

If you attempt to trash a file not in this filesystem, an attempt is made to find directory `<top of filesystem>/.Trash-<uid>` where `<uid>` is your user id (in a terminal type `id` to display it). If this directory does not exist, then the action will fail.

You may create this directory and set write access to it as follows:

`cd` to the directory where the file to be deleted is

`df --human-readable .` (Note the dot!) The "Mounted on" column shows the top of the filesystem.

```
cd <the top of the filesystem shown above>
```

```
sudo mkdir .Trash-<uid>
```

```
sudo chown <your user name> .Trash-<uid>
```

```
sudo chmod u+rwx .Trash-<uid>
```

If you are running Geeqie as a Snap, you may need to use Geeqie's own trash system. See Safe Delete

Image viewing

Limiting enlargement of images

How do I keep images smaller than the visible area from enlarging when the zoom is set to “fit to window”?

In the Preferences Image dialog , disable “Allow enlargement of image for zoom to fit”.

Command line

How do I open an image in an existing Geeqie application ?

Use command line options to send commands to an existing Geeqie instance, for a list of commands supported by the remote switch run:

```
geeqie --help
```

Setting the slide show delay

How do I set the slide show delay when starting Geeqie in slide show from the command line ?

The delay can not be set when used with the standard --slideshow command line switch, as --slideshow uses the delay stored in the configuration file.

For this, use the remote command --slideshow to start the slide show after setting the delay with --delay=. The order of the command line is important. For example for a 12 second slide show delay use:

```
geeqie --delay=12 --slideshow
```

Note that if geeqie is running prior to this command, the slide show should also be stopped prior to changing the delay:

```
geeqie --slideshow --delay=12 --slideshow
```

Miscellaneous

How can I save and restore multiple Geeqie windows

The option Exit program when this window is closed provides this facility.

I want to be able to edit images with Geeqie

Geeqie is an image *viewer* . There is, and never will be, a plan to add editing features. I recommend gimp for image editing.

Exif Auto Rotation

How do the Exif auto-rotation commands work?

Image auto-rotation works in either of two modes, set by Write altered image orientation to the metadata option.

Write altered image orientation to the metadata - not set:

If you change the orientation of an image using any of the available commands, the image will always be displayed in the modified orientation. This new orientation is stored in temporary data structures associated with each image file. When Geeqie closes, these changes will be lost.

Warning

It is not possible to determine if the image as displayed is the original image or one which has been temporarily rotated.

If the option Auto rotate image using EXIF information is selected, any images which have Exif orientation data will also be shown rotated. However, any rotation you have selected manually will always over-ride the embedded Exif rotation value.

Write altered image orientation to the metadata - set:

If you change the orientation of an image using any of the available commands, this new orientation is stored in a metadata write queue.

If the option Auto rotate image using EXIF information is selected, any images which have Exif orientation data will be shown rotated, including any in the metadata write queue.

However if the option Auto rotate image using EXIF information is not selected, no images will be auto-rotated. If you have manually rotated any images, the metadata write icon on the status bar will alert you to that fact. Clicking the icon will give a list of the modified files.

Dark Mode

How to set Geeqie to use a dark mode

If Geeqie does not automatically change mode when a style is changed, use the following as an example to start Geeqie:

```
GTK_THEME=Adwaita:dark geeqie
```

Chapter 16. Licence

The Geeqie program is released under the "GNU General Public License", the complete GNU GPL text can be found in the COPYING file included with each Geeqie release.

This Geeqie documentation is released under the GNU Free Documentation License.

Permission is granted to copy, distribute and/or modify this document under the terms of the GNU Free Documentation License, Version 1.2 or any later version published by the Free Software Foundation; with no Invariant Sections, no Front-Cover Texts, and no Back-Cover Texts.

Chapter 17. Credits

Geeqie is maintained by The Geeqie Team (Colin Clark, Klaus Ethgen, Laurent Monin, Petr Ostadal, Vladimir Nadvornik and others).

This software is a fork of GQview project by John Ellis.

The ever growing list of contributors and language translators can be found in the README file included with the latest release of the program.

The Credits can also be viewed by selecting the Credits button located in the About dialog *(open Help Menu, select About, click Credits... button)*.

Chapter 18. GNU Free Documentation License

GNU GENERAL PUBLIC LICENSE
Version 2, June 1991

Copyright (C) 1989, 1991 Free Software Foundation, Inc.,
51 Franklin Street, Fifth Floor, Boston, MA 02110-1301 USA
Everyone is permitted to copy and distribute verbatim copies
of this license document, but changing it is not allowed.

Preamble

The licenses for most software are designed to take away your freedom to share and change it. By contrast, the GNU General Public License is intended to guarantee your freedom to share and change free software--to make sure the software is free for all its users. This General Public License applies to most of the Free Software Foundation's software and to any other program whose authors commit to using it. (Some other Free Software Foundation software is covered by the GNU Lesser General Public License instead.) You can apply it to your programs, too.

When we speak of free software, we are referring to freedom, not price. Our General Public Licenses are designed to make sure that you have the freedom to distribute copies of free software (and charge for this service if you wish), that you receive source code or can get it if you want it, that you can change the software or use pieces of it in new free programs; and that you know you can do these things.

To protect your rights, we need to make restrictions that forbid anyone to deny you these rights or to ask you to surrender the rights. These restrictions translate to certain responsibilities for you if you distribute copies of the software, or if you modify it.

For example, if you distribute copies of such a program, whether gratis or for a fee, you must give the recipients all the rights that you have. You must make sure that they, too, receive or can get the source code. And you must show them these terms so they know their rights.

We protect your rights with two steps: (1) copyright the software, and (2) offer you this license which gives you legal permission to copy, distribute and/or modify the software.

Also, for each author's protection and ours, we want to make certain that everyone understands that there is no warranty for this free software. If the software is modified by someone else and passed on, we want its recipients to know that what they have is not the original, so that any problems introduced by others will not reflect on the original authors' reputations.

Finally, any free program is threatened constantly by software

patents. We wish to avoid the danger that redistributors of a free program will individually obtain patent licenses, in effect making the program proprietary. To prevent this, we have made it clear that any patent must be licensed for everyone's free use or not licensed at all.

  The precise terms and conditions for copying, distribution and modification follow.

GNU GENERAL PUBLIC LICENSE
TERMS AND CONDITIONS FOR COPYING, DISTRIBUTION AND MODIFICATION

  0. This License applies to any program or other work which contains a notice placed by the copyright holder saying it may be distributed under the terms of this General Public License. The "Program", below, refers to any such program or work, and a "work based on the Program" means either the Program or any derivative work under copyright law: that is to say, a work containing the Program or a portion of it, either verbatim or with modifications and/or translated into another language. (Hereinafter, translation is included without limitation in the term "modification".) Each licensee is addressed as "you".

Activities other than copying, distribution and modification are not covered by this License; they are outside its scope. The act of running the Program is not restricted, and the output from the Program is covered only if its contents constitute a work based on the Program (independent of having been made by running the Program). Whether that is true depends on what the Program does.

  1. You may copy and distribute verbatim copies of the Program's source code as you receive it, in any medium, provided that you conspicuously and appropriately publish on each copy an appropriate copyright notice and disclaimer of warranty; keep intact all the notices that refer to this License and to the absence of any warranty; and give any other recipients of the Program a copy of this License along with the Program.

You may charge a fee for the physical act of transferring a copy, and you may at your option offer warranty protection in exchange for a fee.

  2. You may modify your copy or copies of the Program or any portion of it, thus forming a work based on the Program, and copy and distribute such modifications or work under the terms of Section 1 above, provided that you also meet all of these conditions:

    a) You must cause the modified files to carry prominent notices stating that you changed the files and the date of any change.

    b) You must cause any work that you distribute or publish, that in whole or in part contains or is derived from the Program or any part thereof, to be licensed as a whole at no charge to all third parties under the terms of this License.

    c) If the modified program normally reads commands interactively when run, you must cause it, when started running for such interactive use in the most ordinary way, to print or display an announcement including an appropriate copyright notice and a notice that there is no warranty (or else, saying that you provide a warranty) and that users may redistribute the program under

these conditions, and telling the user how to view a copy of this License. (Exception: if the Program itself is interactive but does not normally print such an announcement, your work based on the Program is not required to print an announcement.)

These requirements apply to the modified work as a whole. If identifiable sections of that work are not derived from the Program, and can be reasonably considered independent and separate works in themselves, then this License, and its terms, do not apply to those sections when you distribute them as separate works. But when you distribute the same sections as part of a whole which is a work based on the Program, the distribution of the whole must be on the terms of this License, whose permissions for other licensees extend to the entire whole, and thus to each and every part regardless of who wrote it.

Thus, it is not the intent of this section to claim rights or contest your rights to work written entirely by you; rather, the intent is to exercise the right to control the distribution of derivative or collective works based on the Program.

In addition, mere aggregation of another work not based on the Program with the Program (or with a work based on the Program) on a volume of a storage or distribution medium does not bring the other work under the scope of this License.

3. You may copy and distribute the Program (or a work based on it, under Section 2) in object code or executable form under the terms of Sections 1 and 2 above provided that you also do one of the following:

a) Accompany it with the complete corresponding machine-readable source code, which must be distributed under the terms of Sections 1 and 2 above on a medium customarily used for software interchange; or,

b) Accompany it with a written offer, valid for at least three years, to give any third party, for a charge no more than your cost of physically performing source distribution, a complete machine-readable copy of the corresponding source code, to be distributed under the terms of Sections 1 and 2 above on a medium customarily used for software interchange; or,

c) Accompany it with the information you received as to the offer to distribute corresponding source code. (This alternative is allowed only for noncommercial distribution and only if you received the program in object code or executable form with such an offer, in accord with Subsection b above.)

The source code for a work means the preferred form of the work for making modifications to it. For an executable work, complete source code means all the source code for all modules it contains, plus any associated interface definition files, plus the scripts used to control compilation and installation of the executable. However, as a special exception, the source code distributed need not include anything that is normally distributed (in either source or binary form) with the major components (compiler, kernel, and so on) of the operating system on which the executable runs, unless that component itself accompanies the executable.

If distribution of executable or object code is made by offering

access to copy from a designated place, then offering equivalent access to copy the source code from the same place counts as distribution of the source code, even though third parties are not compelled to copy the source along with the object code.

4. You may not copy, modify, sublicense, or distribute the Program except as expressly provided under this License. Any attempt otherwise to copy, modify, sublicense or distribute the Program is void, and will automatically terminate your rights under this License. However, parties who have received copies, or rights, from you under this License will not have their licenses terminated so long as such parties remain in full compliance.

5. You are not required to accept this License, since you have not signed it. However, nothing else grants you permission to modify or distribute the Program or its derivative works. These actions are prohibited by law if you do not accept this License. Therefore, by modifying or distributing the Program (or any work based on the Program), you indicate your acceptance of this License to do so, and all its terms and conditions for copying, distributing or modifying the Program or works based on it.

6. Each time you redistribute the Program (or any work based on the Program), the recipient automatically receives a license from the original licensor to copy, distribute or modify the Program subject to these terms and conditions. You may not impose any further restrictions on the recipients' exercise of the rights granted herein. You are not responsible for enforcing compliance by third parties to this License.

7. If, as a consequence of a court judgment or allegation of patent infringement or for any other reason (not limited to patent issues), conditions are imposed on you (whether by court order, agreement or otherwise) that contradict the conditions of this License, they do not excuse you from the conditions of this License. If you cannot distribute so as to satisfy simultaneously your obligations under this License and any other pertinent obligations, then as a consequence you may not distribute the Program at all. For example, if a patent license would not permit royalty-free redistribution of the Program by all those who receive copies directly or indirectly through you, then the only way you could satisfy both it and this License would be to refrain entirely from distribution of the Program.

If any portion of this section is held invalid or unenforceable under any particular circumstance, the balance of the section is intended to apply and the section as a whole is intended to apply in other circumstances.

It is not the purpose of this section to induce you to infringe any patents or other property right claims or to contest validity of any such claims; this section has the sole purpose of protecting the integrity of the free software distribution system, which is implemented by public license practices. Many people have made generous contributions to the wide range of software distributed through that system in reliance on consistent application of that system; it is up to the author/donor to decide if he or she is willing to distribute software through any other system and a licensee cannot impose that choice.

This section is intended to make thoroughly clear what is believed to be a consequence of the rest of this License.

8. If the distribution and/or use of the Program is restricted in certain countries either by patents or by copyrighted interfaces, the original copyright holder who places the Program under this License may add an explicit geographical distribution limitation excluding those countries, so that distribution is permitted only in or among countries not thus excluded. In such case, this License incorporates the limitation as if written in the body of this License.

9. The Free Software Foundation may publish revised and/or new versions of the General Public License from time to time. Such new versions will be similar in spirit to the present version, but may differ in detail to address new problems or concerns.

Each version is given a distinguishing version number. If the Program specifies a version number of this License which applies to it and "any later version", you have the option of following the terms and conditions either of that version or of any later version published by the Free Software Foundation. If the Program does not specify a version number of this License, you may choose any version ever published by the Free Software Foundation.

10. If you wish to incorporate parts of the Program into other free programs whose distribution conditions are different, write to the author to ask for permission. For software which is copyrighted by the Free Software Foundation, write to the Free Software Foundation; we sometimes make exceptions for this. Our decision will be guided by the two goals of preserving the free status of all derivatives of our free software and of promoting the sharing and reuse of software generally.

NO WARRANTY

11. BECAUSE THE PROGRAM IS LICENSED FREE OF CHARGE, THERE IS NO WARRANTY FOR THE PROGRAM, TO THE EXTENT PERMITTED BY APPLICABLE LAW. EXCEPT WHEN OTHERWISE STATED IN WRITING THE COPYRIGHT HOLDERS AND/OR OTHER PARTIES PROVIDE THE PROGRAM "AS IS" WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. THE ENTIRE RISK AS TO THE QUALITY AND PERFORMANCE OF THE PROGRAM IS WITH YOU. SHOULD THE PROGRAM PROVE DEFECTIVE, YOU ASSUME THE COST OF ALL NECESSARY SERVICING, REPAIR OR CORRECTION.

12. IN NO EVENT UNLESS REQUIRED BY APPLICABLE LAW OR AGREED TO IN WRITING WILL ANY COPYRIGHT HOLDER, OR ANY OTHER PARTY WHO MAY MODIFY AND/OR REDISTRIBUTE THE PROGRAM AS PERMITTED ABOVE, BE LIABLE TO YOU FOR DAMAGES, INCLUDING ANY GENERAL, SPECIAL, INCIDENTAL OR CONSEQUENTIAL DAMAGES ARISING OUT OF THE USE OR INABILITY TO USE THE PROGRAM (INCLUDING BUT NOT LIMITED TO LOSS OF DATA OR DATA BEING RENDERED INACCURATE OR LOSSES SUSTAINED BY YOU OR THIRD PARTIES OR A FAILURE OF THE PROGRAM TO OPERATE WITH ANY OTHER PROGRAMS), EVEN IF SUCH HOLDER OR OTHER PARTY HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.

END OF TERMS AND CONDITIONS

How to Apply These Terms to Your New Programs

If you develop a new program, and you want it to be of the greatest possible use to the public, the best way to achieve this is to make it free software which everyone can redistribute and change under these terms.

To do so, attach the following notices to the program. It is safest to attach them to the start of each source file to most effectively convey the exclusion of warranty; and each file should have at least the "copyright" line and a pointer to where the full notice is found.

```
    one line to give the program's name and a brief idea of what it does.
    Copyright (C) year  name of author

    This program is free software; you can redistribute it and/or modify
    it under the terms of the GNU General Public License as published by
    the Free Software Foundation; either version 2 of the License, or
    (at your option) any later version.

    This program is distributed in the hope that it will be useful,
    but WITHOUT ANY WARRANTY; without even the implied warranty of
    MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the
    GNU General Public License for more details.

    You should have received a copy of the GNU General Public License along
    with this program; if not, write to the Free Software Foundation, Inc.,
    51 Franklin Street, Fifth Floor, Boston, MA 02110-1301 USA.
```

Also add information on how to contact you by electronic and paper mail.

If the program is interactive, make it output a short notice like this when it starts in an interactive mode:

```
    Gnomovision version 69, Copyright (C) year name of author
    Gnomovision comes with ABSOLUTELY NO WARRANTY; for details type `show w'.
    This is free software, and you are welcome to redistribute it
    under certain conditions; type `show c' for details.
```

The hypothetical commands `show w' and `show c' should show the appropriate parts of the General Public License. Of course, the commands you use may be called something other than `show w' and `show c'; they could even be mouse-clicks or menu items--whatever suits your program.

You should also get your employer (if you work as a programmer) or your school, if any, to sign a "copyright disclaimer" for the program, if necessary. Here is a sample; alter the names:

```
  Yoyodyne, Inc., hereby disclaims all copyright interest in the program
  `Gnomovision' (which makes passes at compilers) written by James Hacker.

  signature of Ty Coon, 1 April 1989
  Ty Coon, President of Vice
```

This General Public License does not permit incorporating your program into proprietary programs. If your program is a subroutine library, you may consider it more useful to permit linking proprietary applications with the library. If this is what you want to do, use the GNU Lesser General Public License instead of this License.